Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3130 FOUNDATION DESIGN OF CONVEYOR 3130-CNV-001

DESIGN CALCULATION

AS BUILT

MD-216-3130-EG-CV-CAL-1001

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Project Country	KINGDOM OF SAUDI ARABIA

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning a
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for Concrete design calculation of Conveyor Gallery 3130-CNV-001.
This conveyor is between TP-3 and Tertiary Crusher House and belt width of the conveyor is 1200 mm.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Conveyor 3130-CNV-001. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column and beam reinforcement design have been performed in Staad-Pro Software. The foundation of Ground Portal GR-1 to GR-4; TR-4 & VGTU have been performed in MS Excel.
Foundation of TR-1, TR-2 & TR-3 has been designed by FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1m & 3m depth below FGL (ie. RL 911.500m & RL 909.500m respectively) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

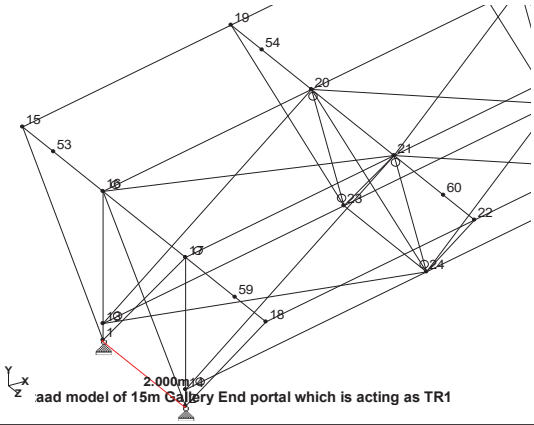
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3130-EG-ST-CAL-1001

5.1 Support Reaction from TR-1

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DL	0.0	83.6	-11.4
	2 LL	-0.4	31.8	-1.5
	3 ML - CONVEYOR MATERIAL	-0.3	18.5	0.0
	4 WL (+Z)	-1.5	-26.0	-15.3
	5 WL (-Z)	1.5	26.0	15.3
	6 TL1	0.2	0.1	6.3
	7 TL2	-0.2	-0.1	-6.3
	8 EARTHQUAKE LOAD (+X)	-0.8	-0.1	0.0
	9 EARTHQUAKE LOAD (+Z)	0.0	-0.8	-0.4
2	1 DL	0.0	20.2	7.4
	2 LL	0.4	28.2	1.5
	3 ML - CONVEYOR MATERIAL	0.3	18.0	-0.1
	4 WL (+Z)	1.5	26.0	-11.9
	5 WL (-Z)	-1.5	-26.0	11.9
	6 TL1	-0.2	-0.1	-6.3
	7 TL2	0.2	0.1	6.3
	8 EARTHQUAKE LOAD (+X)	-0.8	-0.3	0.1
	9 EARTHQUAKE LOAD (+Z)	0.0	0.8	-0.3



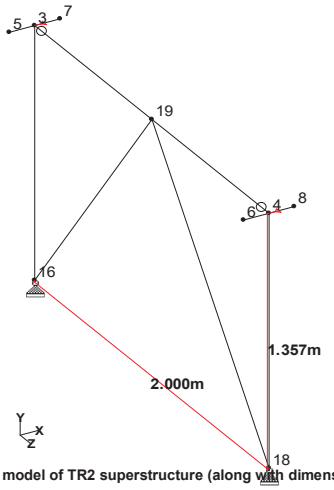
aad model of 15m Gallery End portal which is acting as TR1

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5.2 Support Reaction from TR-2

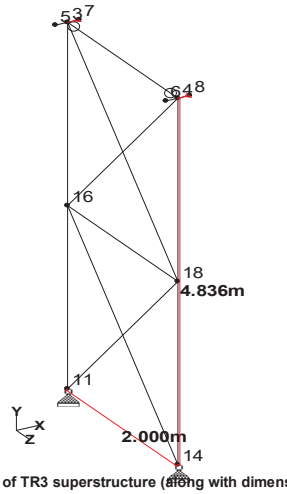
Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
16	1 DEAD LOAD	1.2	182.8	-4.7
	2 LIVE LOAD	0.0	62.3	0.0
	3 MATERIAL LOAD	0.6	42.8	0.0
	4 WL (+Z)	-0.8	-97.8	-31.3
	5 WL (-Z)	0.8	97.8	31.3
	6 WL (+X)	-0.8	0.0	0.0
	7 WL (-X)	0.8	0.0	0.0
	8 TL1	0.0	0.0	0.0
	9 TL2	0.0	0.0	0.0
	10 EL (+X)	-0.1	-0.3	-0.1
	11 EL (+Z)	0.0	-6.5	-3.4
18	1 DEAD LOAD	1.9	64.5	-4.9
	2 LIVE LOAD	1.1	69.7	0.0
	3 MATERIAL LOAD	0.5	43.0	0.0
	4 WL (+Z)	0.8	97.8	-31.3
	5 WL (-Z)	-0.8	-97.8	31.3
	6 WL (+X)	-0.8	0.0	0.0
	7 WL (-X)	0.8	0.0	0.0
	8 TL1	0.0	0.0	0.0
	9 TL2	0.0	0.0	0.0
	10 EL (+X)	0.0	0.1	-0.1
	11 EL (+Z)	0.0	6.5	-3.4



Staad model of TR2 superstructure (along with dimension)

5.3 Support Reaction from TR-3

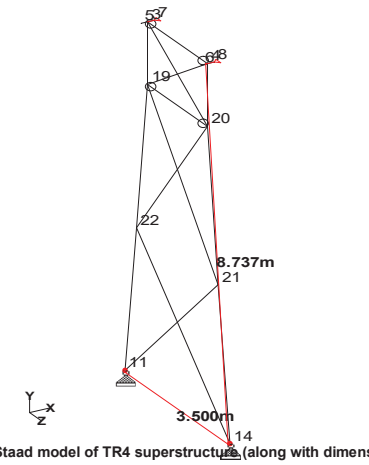
Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
11	1 DEAD LOAD	-0.4	172.3	1.4
	2 LIVE LOAD	0.0	61.1	3.4
	3 MATERIAL LOAD	-0.2	42.4	2.2
	4 WL (+Z)	0.1	-211.7	-35.2
	5 WL (-Z)	-0.1	211.7	35.2
	6 WL (+X)	-2.9	0.0	0.0
	7 WL (-X)	2.9	0.0	0.0
	8 TL1	0.0	-0.2	18.9
	9 TL2	0.0	0.2	-18.9
	10 EL (+X)	0.0	0.5	0.0
	11 EL (+Z)	0.0	-13.2	-2.3
14	1 DEAD LOAD	-0.5	82.1	-11.0
	2 LIVE LOAD	-0.3	70.9	-3.3
	3 MATERIAL LOAD	-0.2	43.4	-2.1
	4 WL (+Z)	-0.1	211.7	-35.1
	5 WL (-Z)	0.1	-211.7	35.1
	6 WL (+X)	-2.9	0.0	0.0
	7 WL (-X)	2.9	0.0	0.0
	8 TL1	0.0	0.2	-18.9
	9 TL2	0.0	-0.2	18.9
	10 EL (+X)	0.0	0.5	0.0
	11 EL (+Z)	0.0	13.2	-2.2



Staad model of TR3 superstructure (along with dimension)

5.4 Support Reaction from TR-4

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
11	1 DEAD LOAD	0.042	132.305	9.049
	2 LIVE LOAD	-0.01	61.654	6.866
	3 MATERIAL LOAD	-0.008	36.458	4.182
	4 WL (+Z)	0.026	-197.421	-40.019
	5 WL (-Z)	-0.026	197.421	40.019
	6 WL (+X)	-7.004	0	0
	7 WL (-X)	7.004	0	0
	8 TL1	-0.002	0.022	-5.56
	9 TL2	0.002	-0.022	5.56
	10 EL (+X)	-0.002	0.048	0
	11 EL (+Z)	-0.004	-9.338	-1.658
14	1 DEAD LOAD	-0.044	95.934	-17.049
	2 LIVE LOAD	0.01	58.346	-6.866
	3 MATERIAL LOAD	0.008	36.542	-4.182
	4 WL (+Z)	-0.026	197.421	-41.033
	5 WL (-Z)	0.026	-197.421	41.033
	6 WL (+X)	-6.999	0	0
	7 WL (-X)	6.999	0	0
	8 TL1	0.002	-0.022	5.559
	9 TL2	-0.002	0.022	-5.559
	10 EL (+X)	0.001	-0.048	0
	11 EL (+Z)	0.004	9.338	-1.707



Staad model of TR4 superstructure (along with dimension)

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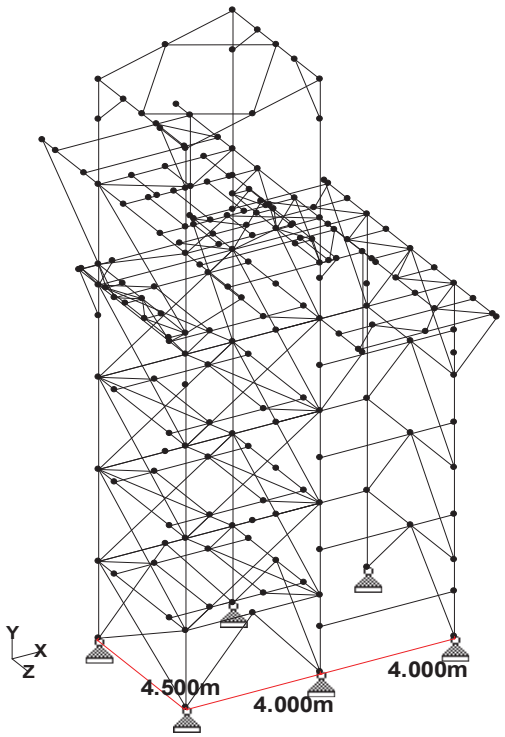
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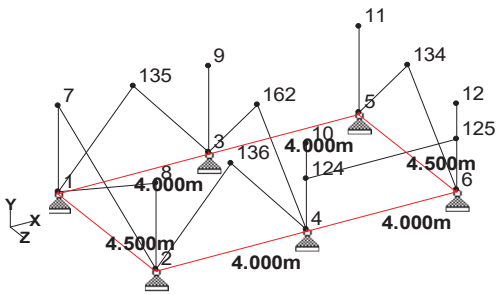
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5.5 Support Reaction from VGTU

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	17.30	292.38	12.76
	2 LIVE LOAD	1.14	208.12	11.28
	3 MATERIAL LOAD	0.01	17.63	1.12
	4 WL (+Z)	1.59	-162.61	-33.39
	5 WL (-Z)	-1.60	162.61	33.39
	6 WL (+X)	-50.72	-263.59	-11.42
	7 WL (-X)	50.72	263.59	11.42
	8 TL1	1.46	0.02	37.28
	9 TL2	-1.46	-0.02	-37.28
	10 EL (+X)	-2.80	-16.85	-0.61
	11 EL (+Z)	0.24	-13.45	-2.44
2	1 DEAD LOAD	18.26	245.67	-16.00
	2 LIVE LOAD	3.27	179.66	-11.54
	3 MATERIAL LOAD	0.11	17.28	-0.92
	4 WL (+Z)	-1.60	162.61	-33.60
	5 WL (-Z)	1.60	-162.61	33.60
	6 WL (+X)	-50.44	-265.50	12.00
	7 WL (-X)	50.44	265.50	-12.00
	8 TL1	1.43	-0.02	-37.31
	9 TL2	-1.43	0.02	37.31
	10 EL (+X)	-2.43	-16.64	0.94
	11 EL (+Z)	-0.24	13.45	-2.46
3	1 DEAD LOAD	14.63	119.94	-2.10
	2 LIVE LOAD	-3.31	279.97	5.28
	3 MATERIAL LOAD	-0.11	24.85	-0.13
	4 WL (+Z)	2.34	-162.04	-28.54
	5 WL (-Z)	-2.34	162.04	28.54
	6 WL (+X)	-50.77	261.31	-0.31
	7 WL (-X)	50.77	-261.31	0.31
	8 TL1	-1.33	-0.06	0.64
	9 TL2	1.33	0.06	-0.64
	10 EL (+X)	-2.80	16.52	-0.17
	11 EL (+Z)	0.30	-13.38	-1.96
4	1 DEAD LOAD	15.82	88.54	-5.97
	2 LIVE LOAD	-1.12	292.85	-6.91
	3 MATERIAL LOAD	-0.01	26.05	-0.23
	4 WL (+Z)	-2.34	162.04	-28.54
	5 WL (-Z)	2.34	-162.04	28.54
	6 WL (+X)	-51.02	267.78	-1.16
	7 WL (-X)	51.02	-267.78	1.16
	8 TL1	-1.18	0.06	-0.63
	9 TL2	1.18	-0.06	0.63
	10 EL (+X)	-2.43	16.84	-0.23
	11 EL (+Z)	-0.30	13.38	-1.96
5	1 DEAD LOAD	-0.01	30.14	1.20
	2 LIVE LOAD	0.01	69.16	0.95
	3 MATERIAL LOAD	0.00	0.44	0.09
	4 WL (+Z)	0.00	-52.84	-15.97
	5 WL (-Z)	0.00	52.84	15.97
	6 WL (+X)	-1.10	2.28	0.45
	7 WL (-X)	1.10	-2.28	-0.45
	8 TL1	-0.10	0.03	0.50
	9 TL2	0.10	-0.03	-0.50
	10 EL (+X)	0.00	-0.12	-0.02
	11 EL (+Z)	0.00	-3.26	-0.68
6	1 DEAD LOAD	0.00	22.22	0.49
	2 LIVE LOAD	0.01	54.46	0.95
	3 MATERIAL LOAD	0.00	-0.44	0.09
	4 WL (+Z)	0.00	52.84	-15.97
	5 WL (-Z)	0.00	-52.84	15.97
	6 WL (+X)	-0.57	-2.27	0.45
	7 WL (-X)	0.57	2.27	-0.45
	8 TL1	-0.30	-0.03	-0.49
	9 TL2	0.30	0.03	0.49
	10 EL (+X)	0.00	0.12	-0.02
	11 EL (+Z)	0.00	3.26	-0.68



Staad model of VGTU superstructure (along witgh dimension)



Node number and orientation of VGTU superstructure

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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

5.6 LOAD COMBINATIONS

5.6.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75 (LL+ML)	
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75 (LL+ML)	
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75 (LL+ML)	
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75 (LL+ML)	
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75 (LL+ML)	
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75 (LL+ML)	
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)	
LOAD COMB 314 0.6 DL + 1.0 WL(+X)	
LOAD COMB 315 0.6 DL + 1.0 WL(-X)	
LOAD COMB 316 0.6 DL + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD COMB 317 0.6 DL + 0.7 EL(-X)	
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)	
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)	

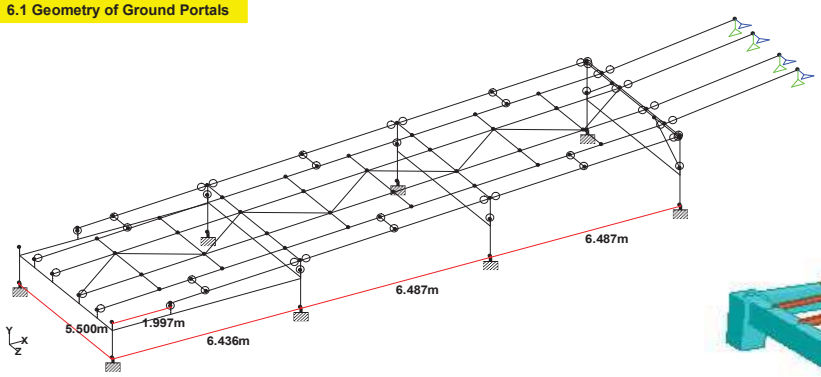
5.6.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5 (LL+ML)	
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5 (LL+ML)	
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5 (LL+ML)	(Eq. 2.3.2-5 of SBC 301)
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5 (LL+ML)	
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5 (LL+ML)	
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5 (LL+ML)	
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)	
LOAD COMB 414 0.9 DL + 1.6 WL(+X)	
LOAD COMB 415 0.9 DL + 1.6 WL(-X)	
LOAD COMB 416 0.9 DL + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD COMB 417 0.9 DL + 1.0 EL(-X)	
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)	
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)	

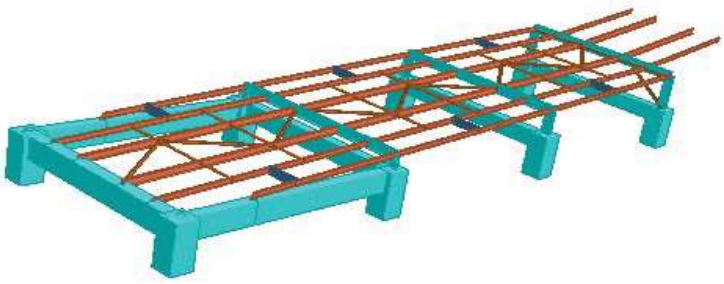
Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

6 GEOMETRY

C01 **6.1 Geometry of Ground Portals**



C01 **Staad model of Ground Portals (along witgh dimension)**

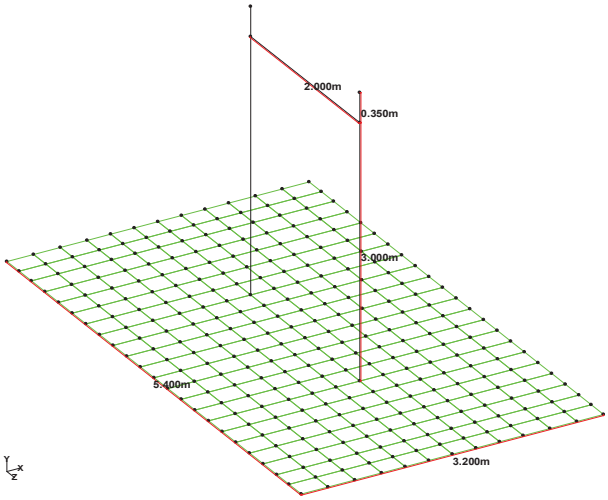


Isometric view of Ground Portals (From Staad)

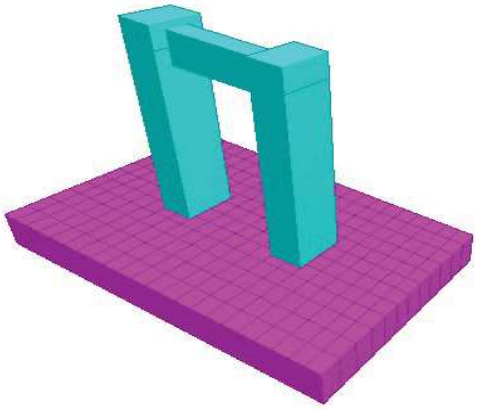
Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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6.2 Geometry of TR-3 Foundation



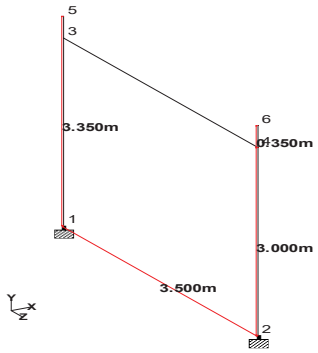
Staad Model of TR3 Foundation mkd F2 (Along with dimension)



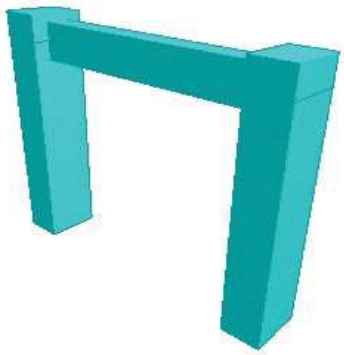
Isometric view of TR3 Foundation mkd F2

Same size as for TR-3 footing has been provided for TR-1 & TR-2 as height of these trestles are less than TR-3. Moreover wind load are also less for those trestles.

6.3 Geometry of TR-4 Foundation

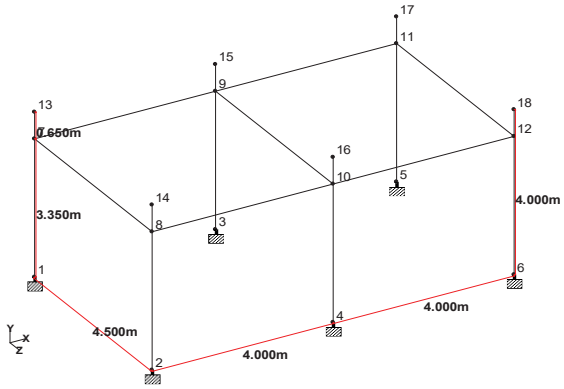


Staad Model of TR4 Foundation mkd F3 (Along with dimension)

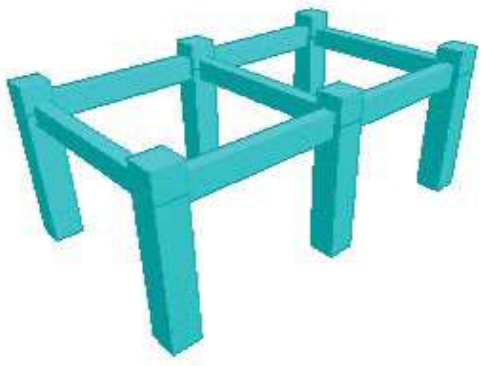


Isometric view of of TR4 Foundation mkd F3

C01 **6.4 Geometry of VGTU Foundation**



C01 Staad Model of VGTU Foundation (Along with dimension)



Isometric view of VGTU foundation (From Staad)

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7. DESIGN OF CONCRETE COMPONENTS

All foundations have been designed using worksheet in accordance to SBC-304.
All beams & columns have been designed in Staad-Pro by ACI 318M.
For Design Results, Refer Attachemnt 3 to 6 - Staad Output file.

7.1 Design Summary of Column

7.1.1 Column Reinforcement Summary

Column mark	Width (Along conv)	Depth (Across conv)	Reinforcement	Stirrup	Remarks	Numbers
P1	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP1	2
P2	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP2 & GP3	4
P2A	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP4	2
P3	600	600	12 No.- D20	3 Nos D10 @ 200c/c	TR1 & TR2	4
P4	700	500	12 No.- D20	3 Nos D10 @ 200c/c	TR3 & TR4	4
P5	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU Grid A-1, A-2, B-1 & B-2	4
P6	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU Grid C-1 & C-2	2
P7	450	450	8 No.- D20	2 Nos D10 @ 200c/c	VGTU Post	2

7.2 Design Summary of Beam

7.2.1 Beam Reinforcement Summary

Beam mark	Width (mm)	Depth (mm)	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	350	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	1
C01 TB1A	300	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	2
TB2	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB3	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	4
TB4	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB5	300	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	4

7.3 Design of Foundation

7.3.1 Foundation marking, Max. Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (Along conv) (mm)	Width (Across conv) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
GP1	F1A	1500	2000	300	114	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.22**	2
C01 GP2, GP3 & GP4	F1	1500	2000	300	114	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.22**	6
TR1, TR2 & TR3 *(Combined footing)*	F2	3200	5400	550	111	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.20**	3
TR4	F3	2500	2500	450	141	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.25**	2
C01 VGTU Grid A-1, A-2; C-1 & C-2	F3	2500	2500	450	205	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.37**	4
C01 VGTU Grid B-1 & B-2	F4	3000	2000	450	153	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.28**	2

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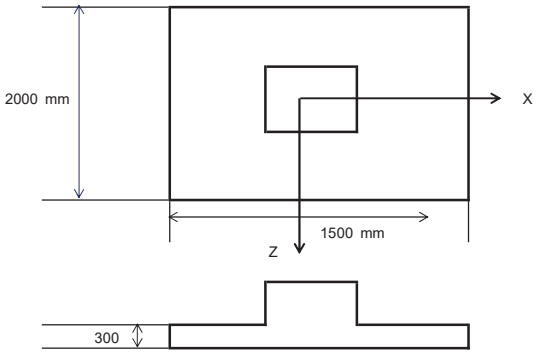
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Approved	PRI	PRI	PRI

7.3.2 Design of foundation mkd. F1A & F1 for supporting trestle mkd. GP1; GP2, GP3 & GP4.

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	83 & 84	
Bottom of foundation	B.O.F	=	-1.500	m	Weight of soil	=	33.26	KN
Staad support reaction level		=	-1.500	m	Weight of foundation	=	21.60	KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00	KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0	KN
Length of footing	Lz	=	2.00	m	Pressure on top for tension	=	18.3	KN/m^2
Width of footing	Lx	=	1.50	m				
Thickness of footing	D	=	0.30	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.000	m				
Section Modulus of Pedestal	Zxx	=	1.00	m^3				
Section Modulus of Pedestal	Zzz	=	0.75	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	518	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	75	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	16	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	16.4	65.953	-12.0	-16.85	-0.5	-1.7	120.8	59.34	21.21	1	0.11
302 1.0 (DL + TL1) +	16.9	111.853	-24.2	-52.69	-0.233	-2.1	166.7	111.09	0.05	1	0.21
303 1.0 (DL + TL1) +	16.9	111.853	-24.2	-52.69	-0.233	-2.1	166.7	111.09	0.05	1	0.21
304 1.0 DL+ 0.75 W	17.2	96.696	-16.4	-38.30	-0.359	-2.4	151.6	92.00	9.04	1	0.18
305 1.0 DL+ 0.75 W	16.3	104.06	-25.9	-49.17	-0.257	-1.6	158.9	104.31	1.64	1	0.20
306 0.6 DL + 1.0 WL	10.4	34.663	-0.9	-2.87	-0.4	-1.5	89.5	34.71	24.97	1	0.07
307 0.6 DL + 1.0 WL	9.2	44.481	-13.5	-17.35	-0.253	-0.5	99.3	51.12	15.11	1	0.10
301 1.0 DL	15.8	67.819	15.2	20.45	0.49	-1.2	122.7	62.97	18.82	1	0.12
302 1.0 (DL + TL1) +	16.2	113.55	26.6	55.40	0.184	-1.6	168.4	113.63	-1.35	1	0.22
303 1.0 (DL + TL1) +	16.2	113.55	26.6	55.40	0.184	-1.6	168.4	113.63	-1.35	1	0.22
304 1.0 DL+ 0.75 W	15.6	105.796	28.4	52.06	0.2	-1.1	160.7	107.08	0.03	1	0.21
305 1.0 DL+ 0.75 W	16.6	98.439	19.0	41.27	0.31	-1.9	153.3	94.85	7.35	1	0.18
306 0.6 DL + 1.0 WL	8.9	45.596	15.4	19.46	0.228	-0.2	100.5	53.25	13.72	1	0.10
307 0.6 DL + 1.0 WL	10.1	35.787	2.8	5.08	0.4	-1.2	90.7	36.94	23.49	1	0.07

Max P_{max}	=	113.63	KN/m^2
Max P_{min}	=	-1.35	KN/m^2
Resultant F.O.S	=	0.22	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	22.9	92.3	-16.8	-23.59	-0.7	-2.33	57.47	4.09
402 1.4 (DL + TL1) +	23.7	170.4	-37.6	-84.53	-0.236	-3.10	145.45	-31.88
403 1.4 (DL + TL1) +	23.7	170.4	-37.6	-84.53	-0.236	-3.10	145.45	-31.88
404 1.2 DL+ 1.6 WL	20.9	94.2	-10.4	-26.55	-0.598	-3.03	62.01	0.82
405 1.2 DL+ 1.6 WL	18.9	109.9	-30.6	-49.73	-0.382	-1.41	88.26	-14.96
406 0.9 DL + 1.6 WL	15.7	51.5	-0.7	-3.57	-0.589	-2.31	23.82	10.52
407 0.9 DL + 1.6 WL	13.7	67.2	-20.9	-26.75	-0.372	-0.68	50.07	-5.26
401 1.4 DL	22.2	94.9	21.2	28.63	0.686	-1.71	62.56	0.74
402 1.4 (DL + TL1) +	22.8	172.7	40.6	88.05	0.167	-2.29	148.67	-33.55
403 1.4 (DL + TL1) +	22.8	172.7	40.6	88.05	0.167	-2.29	148.67	-33.55
404 1.2 DL+ 1.6 WL	18.2	112.1	34.0	53.53	0.33	-0.83	91.99	-17.26
405 1.2 DL+ 1.6 WL	20.2	96.4	13.8	30.51	0.541	-2.45	65.90	-1.64
406 0.9 DL + 1.6 WL	13.3	68.9	23.7	29.91	0.335	-0.29	53.26	-7.34
407 0.9 DL + 1.6 WL	15.2	53.2	3.6	6.89	0.546	-1.91	27.17	8.29

Max Net factored B.P = 148.67 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.45 m

Moment about Z: Muz = 15.05 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 184.4 mm^2

Length of cantilever along Z = 0.7 m

Moment about X: Mux = 36.43 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 275.84 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 487.2 mm^2

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 217 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 390.6 mm^2

A_{req} = 184.4 mm^2

Required Spacing = **514.8** mm

Effective Depth Along X = 201 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 361.8 mm^2

A_{req} = 487.2 mm^2

Required Spacing = **412.7** mm

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.45 m

Moment about Z: Muz = 1.85 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 327.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 22.4 mm^2

Length of cantilever along Z = 0.7 m

Moment about X: Mux = 4.48 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 292.56 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 57.3 mm^2

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of

or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 197.1 mm^2

A_{req} = 22.4 mm^2

Required Spacing = **573.8** mm

Effective Depth Along Z = 207 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of

or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

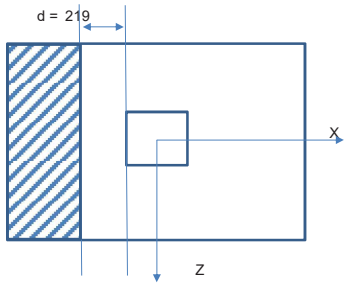
A_{stmin} = 186.3 mm^2

A_{req} = 57.3 mm^2

Required Spacing = **607.1** mm

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

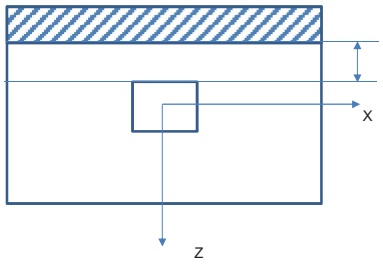
Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	23-03-2020	25-04-2020	01.11.2023	Date	23-03-2020	25-04-2020	01.11.2023
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Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Along X Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0046		
Effective Depth Along X	d	=	219	mm	
Shear Force at d distance	V_u	=	34.34	KN	
Allowable shear strength	φVc	=	168.29	KN	O.K
Utility ratio		=	0.20		

L	=	0.231	m	Least of	
i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$		Table 22.5.5.1 SBC 304
	λ	=	1	Normal concrete	
	Mu	=	3.97	KNm	
	φ	=	0.75	for shear	
	φVc	=	179.8	KN	
ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$		
	φVc	=	168.3	KN	Table 22.5.5.1 SBC 304
iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$		
		=	281.8	KN	

ONE WAY SHEAR CKECK



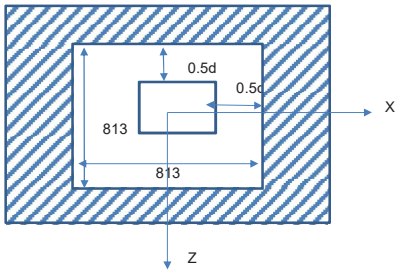
Along Z Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0049		
Effective Depth Along Z	d	=	207	mm	
Shear Force at d distance	V_u	=	73.30	KN	
Allowable shear strength	φVc	=	157.72	KN	O.K
Utility ratio		=	0.46		

L	=	0.493	m	Least of	
i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$		Table 22.5.5.1 SBC 304
	λ	=	1	Normal concrete	
	Mu	=	18.07	KNm	
	φ	=	0.75	for shear	
	φVc	=	157.7	KN	
ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$		
	φVc	=	159.8	KN	
iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$		
	φVc	=	266.4	KN	

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	3	m2
Punching area		=	0.66	m2
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	3252.0	mm

Allowable shear stress — least of — Table 22.6.5.2 SBC 304

v_c	$0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.27 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.25	m	
Depth at section	d	=	213	mm	
Shear Force at section	Vu	=	347.8	KN	
Allowable shear strength	φVc	=	1014.24	KN	O.K
Utility ratio			0.34		

Vc	=	$v_c b_0 d$	
	=	1352.32	KN
φ	=	0.75	for shear
φV_n	=	φVc	
φVc	=	1014.2	KN

Global Project ID	900707
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Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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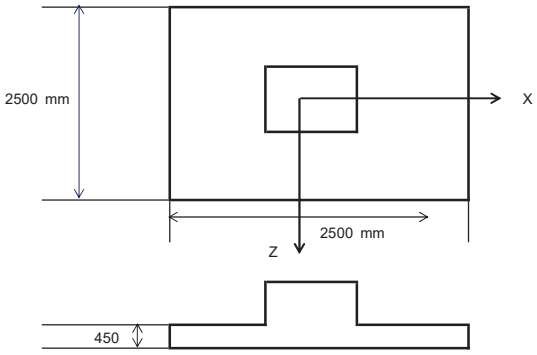
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Date	23-03-2020	25-04-2020	01.11.2023
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Approved	PRI	PRI	PRI

7.3.3 Design of foundation F3 for supporting trestle mkd. TR4

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-3.500	m
Staad support reaction level		=	-3.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	2.50	m
Width of footing	Lx	=	2.50	m
Thickness of footing	D	=	0.45	m
Depth of pedestal	Bx	=	0.70	m
Width of pedestal	Bz	=	0.50	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	2.60	m^3
Section Modulus of Pedestal	Zzz	=	2.60	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	554	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no		=	1	
Weight of soil		=	270.81	KN
Weight of foundation		=	67.50	KN
Uplift		=	0.00	KN
Surcharge		=	0	KN
Pressure on top for tension		=	54.1	KN/m^2

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	0.0	163.187	4.2	8.23	0.0	0.1	501.5	83.44	77.04	1	0.15
302 1.0 (DL + TL1) +	0.0	261.313	4.9	8.78	0.008	0.1	599.6	99.33	92.55	1	0.18
303 1.0 (DL + TL1) +	0.0	261.27	6.4	9.92	0.009	0.1	599.6	99.77	92.10	1	0.18
304 1.0 DL+ 0.75 W	0.0	67.755	35.7	74.28	0.017	0.1	406.1	93.54	36.40	1	0.17
305 1.0 DL+ 0.75 W	0.0	405.776	-25.2	-56.14	0.003	0.0	744.1	140.62	97.49	1	0.25
306 1.0 DL+ 0.75 W	5.2	236.766	5.3	9.07	0.0	-17.5	575.1	102.22	81.80	1	0.18
307 1.0 DL+ 0.75 W	-5.3	236.766	5.3	9.07	0.011	17.7	575.1	102.28	81.75	1	0.18
308 1.0 DL + 0.525	0.0	236.791	5.3	9.07	0.01	0.1	575.1	95.53	88.51	1	0.17
309 1.0 DL + 0.525	0.0	236.74	5.3	9.07	0.01	0.1	575.1	95.52	88.50	1	0.17
310 1.0 DL + 0.525	0.0	231.257	6.2	10.97	0.009	0.1	569.6	95.37	86.89	1	0.17
311 1.0 DL + 0.525	0.0	242.274	4.4	7.17	0.0	0.1	580.6	95.68	90.11	1	0.17
312 0.6 DL + 1.0 WL	0.0	-127.435	43.1	91.88	0.018	0.1	210.9	69.07	-1.59	1	0.12
313 0.6 DL + 1.0 WL	0.0	323.26	-38.1	-82.00	0	0.0	661.6	137.34	74.36	1	0.25
314 0.6 DL + 1.0 WL	7.0	97.912	2.5	4.94	0.0	-23.4	436.2	80.68	58.91	1	0.15
315 0.6 DL + 1.0 WL	-7.0	97.912	2.5	4.94	0.01	23.5	436.2	80.73	58.86	1	0.15
316 0.6 DL + 0.7 EL	0.0	97.946	2.5	4.94	0.008	0.1	436.3	71.72	67.88	1	0.13
317 0.6 DL + 0.7 EL	0.0	97.879	2.5	4.94	0.009	0.1	436.2	71.71	67.87	1	0.13
318 0.6 DL + 0.7 EL	0.0	90.568	3.7	7.47	0.008	0.1	428.9	71.51	65.73	1	0.13
319 0.6 DL + 0.7 EL	0.0	105.257	1.3	2.41	0.0	0.1	443.6	71.92	70.02	1	0.13

Max P_{max} = 140.62 KN/m^2
Max P_{min} = -1.59 KN/m^2
Resultant F.O.S = 0.25

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	0.0	228.5	5.9	11.52	0.0	0.15	41.04	32.07
402 1.4 (DL + TL1) +	0.0	395.3	7.3	12.63	0.009	0.06	68.12	58.37
403 1.4 (DL + TL1) +	0.0	395.2	9.3	14.22	0.011	0.08	68.72	57.74
404 1.2 DL+ 1.6 WL	-0.1	-115.7	70.7	149.54	0.029	0.21	39.00	-76.01
405 1.2 DL+ 1.6 WL	0.0	605.4	-59.2	-128.67	0	0.00	146.28	47.46
406 1.2 DL+ 1.6 WL	11.2	244.9	5.8	10.44	0.013	-37.43	57.56	20.80
407 1.2 DL+ 1.6 WL	-11.2	244.9	5.8	10.44	0.016	37.64	57.64	20.72
408 1.2 DL + 1.0 EL	0.0	244.9	5.8	10.44	0.014	0.10	43.23	35.14
409 1.2 DL + 1.0 EL	0.0	244.8	5.8	10.44	0.015	0.11	43.22	35.12
410 1.2 DL + 1.0 EL	0.0	234.4	7.5	14.06	0.013	0.10	42.94	32.07
411 1.2 DL + 1.0 EL	0.0	255.4	4.1	6.82	0.016	0.12	43.52	38.20
412 0.9 DL + 1.6 WL	-0.1	-213.7	68.7	146.52	0.027	0.20	22.15	-90.53
413 0.9 DL + 1.6 WL	0.0	507.4	-61.2	-131.70	-0.001	-0.01	131.76	30.61
414 0.9 DL + 1.6 WL	11.2	146.9	3.8	7.41	0.012	-37.44	40.72	6.28
415 0.9 DL + 1.6 WL	-11.2	146.9	3.8	7.41	0.015	37.64	40.80	6.20
416 0.9 DL + 1.0 EL	0.0	146.9	3.8	7.41	0.013	0.09	26.39	20.63
417 0.9 DL + 1.0 EL	0.0	146.8	3.8	7.41	0.014	0.10	26.38	20.61
418 0.9 DL + 1.0 EL	0.0	136.4	5.5	11.03	0.012	0.09	26.09	17.55
419 0.9 DL + 1.0 EL	0.0	157.4	2.1	3.79	0.015	0.11	26.67	23.68

Max Net factored B.P = 146.28 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.9 m

Moment about Z: Muz = 59.24 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 919.61 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 430.1 mm^2

Effective Depth Along X = 367 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 660.6 mm^2

A_{req} = 430.1 mm^2

Required Spacing = **304.4** mm

Length of cantilever along Z = 1 m

Moment about X: Mux = 73.14 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 841.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 556.8 mm^2

Effective Depth Along X = 351 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 631.8 mm^2

A_{req} = 556.8 mm^2

Required Spacing = **318.2** mm

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.9 m

Moment about Z: Muz = 21.92 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 929.66 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 157.4 mm^2

Effective Depth Along X = 369 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 332.1 mm^2

A_{req} = 157.4 mm^2

Required Spacing = **340.6** mm

Length of cantilever along Z = 1 m

Moment about X: Mux = 27.06 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 870.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 201.1 mm^2

Effective Depth Along Z = 357 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 321.3 mm^2

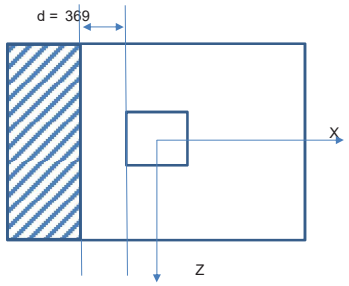
A_{req} = 201.1 mm^2

Required Spacing = **352.0** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



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Along X Direction Cantilever

L	=	0.531	m		
Reinforcement ratio ρ_w	=	0.0027			
Effective Depth Along X d	=	369	mm		
Shear Force at d distance V_u	=	77.67	KN		
Allowable shear strength φVc	=	274.78	KN	O.K	
Utility ratio	=	0.28			

Least of

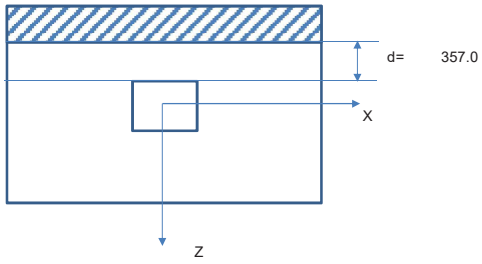
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 20.62 KNm
φ = 0.75 for shear
φVc = 279.8 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 274.8 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
= 474.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.643	m		
Reinforcement ratio ρ_w	=	0.0028			
Effective Depth Along Z d	=	357	mm		
Shear Force at d distance V_u	=	94.06	KN		
Allowable shear strength φVc	=	266.26	KN	O.K	
Utility ratio	=	0.35			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

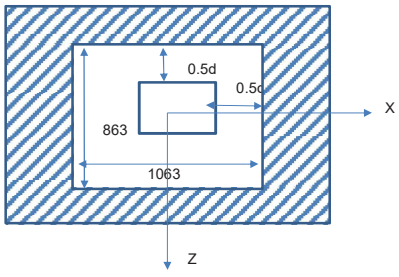
λ = 1 Normal concrete
Mu = 30.24 KNm
φ = 0.75 for shear
φVc = 267.7 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 266.3 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
φVc = 459.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 6.25 m2
Punching area = 0.92 m2
λ = 1
β = 1.4
αs = 40
b0 = 3852.0 mm

Allowable shear stress least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 2.44 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.83 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face u	=	3.85	m		
Depth at section d	=	363	mm		
Shear Force at section Vu	=	780.1	KN		
Allowable shear strength φVc	=	2047.40	KN	O.K	
Utility ratio		0.38			

Vc = $v_c b_0 d$
= 2729.86 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 2047.4 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	23-03-2020	25-04-2020	01.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
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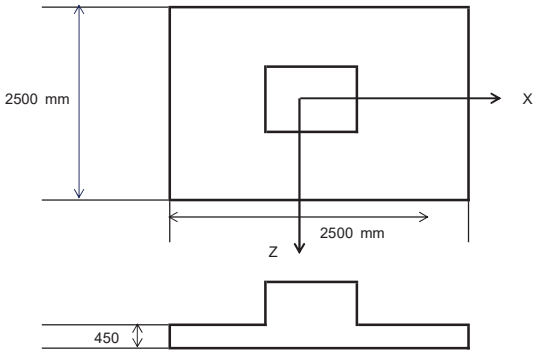
Outotec / L&T	0	1	2
Date	23-03-2020	25-04-2020	01.11.2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
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Approved	PRI	PRI	PRI

7.3.4 Design of foundation F3 for supporting VGTU pedestals along grid A-1 & A-2.

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-3.500	m
Staad support reaction level		=	-3.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	2.50	m
Width of footing	Lx	=	2.50	m
Thickness of footing	D	=	0.45	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.60	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	2.60	m^3
Section Modulus of Pedestal	Zzz	=	2.60	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	554	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no		=	1	
Weight of soil		=	270.35	KN
Weight of foundation		=	67.50	KN
Uplift		=	0.00	KN
Surcharge		=	0	KN
Pressure on top for tension		=	54.1	KN/m^2

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-9.8	347.557	2.9	5.55	-0.4	24.3	685.4	121.14	98.19	1	0.22
302 1.0 (DL + TL1) +	-9.5	573.678	14.4	16.20	-0.818	23.8	911.5	161.21	130.48	1	0.29
303 1.0 (DL + TL1) +	-10.0	573.536	-2.6	0.64	-0.497	24.1	911.4	155.34	136.30	1	0.28
304 1.0 DL+ 0.75 W	-10.5	382.126	29.9	75.70	0.132	26.1	720.0	154.27	76.12	1	0.28
305 1.0 DL+ 0.75 W	-8.9	652.064	-19.7	-60.29	-1.315	22.1	989.9	190.01	126.76	1	0.34
306 1.0 DL+ 0.75 W	9.1	303.291	3.0	5.54	-0.7	-28.0	641.1	115.47	89.69	1	0.21
307 1.0 DL+ 0.75 W	-28.5	730.899	7.2	9.87	-0.525	76.2	1068.8	204.04	137.96	1	0.37
308 1.0 DL + 0.525	-9.0	507.7	4.9	7.44	-0.624	22.1	845.6	146.63	123.94	1	0.26
309 1.0 DL + 0.525	-10.4	526.489	5.3	7.97	-0.558	26.0	864.3	151.35	125.24	1	0.27
310 1.0 DL + 0.525	-9.8	509.413	6.4	11.17	-0.537	24.3	847.3	149.16	121.96	1	0.27
311 1.0 DL + 0.525	-9.7	524.777	3.8	4.24	-0.6	23.9	862.6	148.82	127.22	1	0.27
312 0.6 DL + 1.0 WL	-6.9	28.576	34.8	93.99	0.729	17.3	366.4	101.35	15.91	1	0.18
313 0.6 DL + 1.0 WL	-4.8	388.493	-31.3	-87.33	-1.2	11.9	726.3	154.33	78.10	1	0.28
314 0.6 DL + 1.0 WL	19.2	-76.537	-1.1	0.45	-0.3	-54.9	261.3	63.05	20.57	1	0.11
315 0.6 DL + 1.0 WL	-30.9	493.606	4.5	6.21	-0.147	84.1	831.5	167.70	98.37	1	0.30
316 0.6 DL + 0.7 EL	-4.9	196.009	1.5	2.98	-0.28	12.0	533.9	91.16	79.68	1	0.16
317 0.6 DL + 0.7 EL	-6.8	221.06	1.9	3.68	-0.191	17.2	558.9	97.45	81.40	1	0.18
318 0.6 DL + 0.7 EL	-6.0	198.292	3.4	7.95	-0.163	14.8	536.1	94.54	77.03	1	0.17
319 0.6 DL + 0.7 EL	-5.8	218.777	0.0	-1.29	-0.3	14.3	556.6	95.07	83.06	1	0.17

Max P_{max} = 204.04 KN/m^2

Max P_{min} = 15.91 KN/m^2

Resultant F.O.S = 0.37

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-13.7	486.6	4.0	7.77	-0.6	34.06	93.91	61.79
402 1.4 (DL + TL1) +	-13.2	871.0	21.0	23.54	-1.225	33.24	161.16	117.55
403 1.4 (DL + TL1) +	-13.9	870.8	-2.8	1.76	-0.775	33.70	152.94	125.71
404 1.2 DL+ 1.6 WL	-13.4	242.2	57.8	153.15	0.939	33.29	110.34	-32.85
405 1.2 DL+ 1.6 WL	-10.0	818.0	-48.0	-136.96	-2.147	24.75	192.98	68.79
406 1.2 DL+ 1.6 WL	28.4	74.0	0.4	3.48	-0.746	-82.12	44.71	-21.03
407 1.2 DL+ 1.6 WL	-51.8	986.2	9.5	12.71	-0.461	140.16	216.49	99.09
408 1.2 DL + 1.0 EL	-10.4	512.2	4.6	7.59	-0.667	25.27	94.57	69.33
409 1.2 DL + 1.0 EL	-13.0	548.0	5.2	8.60	-0.541	32.77	103.56	71.80
410 1.2 DL + 1.0 EL	-11.8	515.5	7.3	14.70	-0.5	29.38	99.40	65.55
411 1.2 DL + 1.0 EL	-11.6	544.7	2.5	1.49	-0.708	28.66	98.74	75.58
412 0.9 DL + 1.6 WL	-10.5	24.9	55.5	150.05	1.19	26.16	71.64	-63.69
413 0.9 DL + 1.6 WL	-7.1	600.7	-50.3	-140.06	-1.896	17.63	156.67	35.57
414 0.9 DL + 1.6 WL	31.3	-143.3	-2.0	0.38	-0.496	-89.24	11.48	-57.34
415 0.9 DL + 1.6 WL	-48.9	768.9	7.1	9.61	-0.211	133.03	177.80	68.25
416 0.9 DL + 1.0 EL	-7.5	294.9	2.3	4.49	-0.416	18.15	55.88	38.49
417 0.9 DL + 1.0 EL	-10.1	330.7	2.9	5.50	-0.29	25.64	64.87	40.96
418 0.9 DL + 1.0 EL	-8.9	298.2	5.0	11.60	-0.25	22.25	60.70	34.71
419 0.9 DL + 1.0 EL	-8.7	327.4	0.2	-1.61	-0.457	21.54	61.28	43.50

Max Net factored B.P = 216.49 KN/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	23-03-2020	25-04-2020	01.11.2023	Date	23-03-2020	25-04-2020	01.11.2023
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 97.69 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 919.61 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 713.1 mm²

Effective Depth Along X = 367 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 660.6 mm²

A_{req} = 713.1 mm²

Required Spacing = **281.9** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 97.69 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 841.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 746.6 mm²

Effective Depth Along X = 351 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 631.8 mm²

A_{req} = 746.6 mm²

Required Spacing = **269.3** mm

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Ast prov = 1005.3 mm²

Top Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 929.66 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm²

Effective Depth Along X = 369 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 332.1 mm²

A_{req} = 0.0 mm²

Required Spacing = **340.6** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 870.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm²

Effective Depth Along Z = 357 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 321.3 mm²

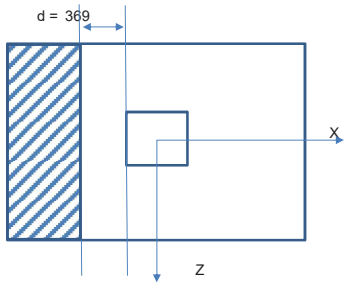
A_{req} = 0.0 mm²

Required Spacing = **352.0** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Ast prov = 565.5 mm²

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	23-03-2020	25-04-2020	01.11.2023	Date	23-03-2020	25-04-2020	01.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
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Along X Direction Cantilever

L	=	0.581	m		
Reinforcement ratio ρ_w	=	0.0027			
Effective Depth Along X d	=	369	mm		
Shear Force at d distance V_u	=	125.78	KN		
Allowable shear strength φVc	=	274.78	KN	O.K	
Utility ratio	=	0.46			

Least of

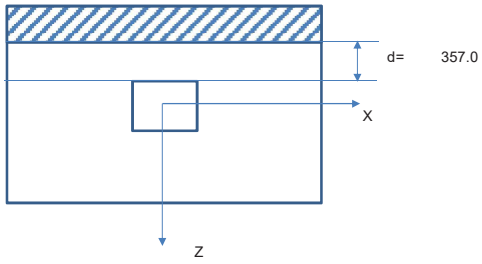
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 36.54 KNm
φ = 0.75 for shear
φVc = 278.2 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 274.8 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
= 474.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.593	m		
Reinforcement ratio ρ_w	=	0.0028			
Effective Depth Along Z d	=	357	mm		
Shear Force at d distance V_u	=	128.38	KN		
Allowable shear strength φVc	=	266.26	KN	O.K	
Utility ratio	=	0.48			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

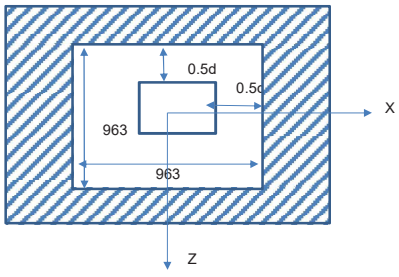
λ = 1 Normal concrete
Mu = 38.06 KNm
φ = 0.75 for shear
φVc = 268.9 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 266.3 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
φVc = 459.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 6.25 m2
Punching area = 0.93 m2
λ = 1
β = 1
αs = 40
b0 = 3852.0 mm

Allowable shear stress least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.83 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face u	=	3.85	m		
Depth at section d	=	363	mm		
Shear Force at section Vu	=	1152.3	KN		
Allowable shear strength φVc	=	2047.40	KN	O.K	
Utility ratio		0.56			

Vc = $v_c b_0 d$ = 2729.86 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 2047.4 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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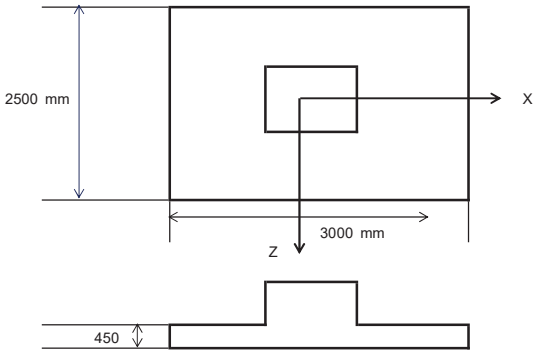
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Prepared By	APL	APL	APL
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Approved	PRI	PRI	PRI

7.3.5 Design of foundation F4 for supporting VGTU pedestals along grid B-1 & B-2.

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-3.500	m
Staad support reaction level		=	-3.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	2.50	m
Width of footing	Lx	=	3.00	m
Thickness of footing	D	=	0.45	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.60	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	3.13	m^3
Section Modulus of Pedestal	Zzz	=	3.75	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	554	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no		=	4	
Weight of soil		=	327.73	KN
Weight of foundation		=	81.00	KN
Uplift		=	0.00	KN
Surcharge		=	0	KN
Pressure on top for tension		=	54.5	KN/m^2

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-13.5	148.838	5.0	10.48	0.3	28.5	557.6	85.29	63.39	1	0.15
302 1.0 (DL + TL1) +	-14.8	465.018	3.8	10.47	0.526	31.1	873.7	128.15	104.85	1	0.23
303 1.0 (DL + TL1) +	-14.4	465.198	4.6	11.34	0.268	30.7	873.9	128.33	104.72	1	0.23
304 1.0 DL+ 0.75 W	-12.9	518.646	25.1	67.90	2.122	27.7	927.4	152.76	94.54	1	0.28
305 1.0 DL+ 0.75 W	-15.7	253.435	-16.3	-46.30	-1.373	32.9	662.2	111.88	64.70	1	0.20
306 1.0 DL+ 0.75 W	15.8	586.166	4.7	11.78	0.4	-32.9	994.9	145.19	120.11	1	0.26
307 1.0 DL+ 0.75 W	-44.4	185.915	4.0	9.82	0.397	93.5	594.6	107.35	51.22	1	0.19
308 1.0 DL + 0.525	-13.3	394.839	4.5	10.99	0.354	28.2	803.6	118.17	96.11	1	0.21
309 1.0 DL + 0.525	-15.3	377.242	4.3	10.60	0.395	32.4	786.0	116.83	92.76	1	0.21
310 1.0 DL + 0.525	-14.2	393.566	5.4	13.50	0.5	30.1	802.3	119.31	94.64	1	0.22
311 1.0 DL + 0.525	-14.4	378.515	3.4	8.10	0.2	30.5	787.2	115.70	94.23	1	0.21
312 0.6 DL + 1.0 WL	-6.2	266.109	30.6	82.42	2.514	13.6	674.8	119.98	59.98	1	0.22
313 0.6 DL + 1.0 WL	-9.9	-87.504	-24.6	-69.84	-2.146	20.6	321.2	70.67	14.99	1	0.13
314 0.6 DL + 1.0 WL	32.0	356.137	3.5	7.60	0.2	-67.1	764.9	122.32	81.65	1	0.22
315 0.6 DL + 1.0 WL	-48.2	-177.532	2.5	4.98	0.214	101.3	231.2	59.44	2.21	1	0.11
316 0.6 DL + 0.7 EL	-6.7	101.034	3.1	6.55	0.157	14.3	509.8	73.87	62.07	1	0.13
317 0.6 DL + 0.7 EL	-9.4	77.571	2.9	6.03	0.211	19.9	486.3	72.08	57.60	1	0.13
318 0.6 DL + 0.7 EL	-7.9	99.337	4.3	9.89	0.352	16.8	508.1	75.38	60.10	1	0.14
319 0.6 DL + 0.7 EL	-8.2	79.269	1.7	2.69	0.0	17.4	488.0	70.57	59.56	1	0.13

Max P_{max}	=	152.76	KN/m^2
Max P_{min}	=	2.21	KN/m^2
Resultant F.O.S	=	0.28	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-18.8	208.4	7.0	14.68	0.4	39.88	43.11	12.45
402 1.4 (DL + TL1) +	-21.1	745.9	5.0	14.78	0.764	44.28	115.99	82.91
403 1.4 (DL + TL1) +	-20.5	746.2	6.2	16.00	0.402	43.66	116.25	82.72
404 1.2 DL+ 1.6 WL	-13.7	619.6	49.8	134.60	4.142	29.80	133.64	31.60
405 1.2 DL+ 1.6 WL	-19.7	53.9	-38.6	-109.02	-3.316	40.98	53.00	-38.64
406 1.2 DL+ 1.6 WL	47.5	763.7	6.4	14.89	0.366	-99.38	133.09	70.56
407 1.2 DL+ 1.6 WL	-80.9	-90.2	4.8	10.69	0.461	170.16	36.77	-60.82
408 1.2 DL + 1.0 EL	-14.8	353.5	5.7	13.16	0.374	31.34	59.70	34.56
409 1.2 DL + 1.0 EL	-18.6	320.0	5.5	12.42	0.452	39.43	57.15	28.18
410 1.2 DL + 1.0 EL	-16.5	351.1	7.5	17.94	0.653	34.93	61.87	31.75
411 1.2 DL + 1.0 EL	-17.0	322.4	3.7	7.64	0.173	35.84	54.99	30.98
412 0.9 DL + 1.6 WL	-9.1	416.8	48.7	131.25	4.005	20.05	102.92	8.23
413 0.9 DL + 1.6 WL	-15.1	-148.9	-39.7	-112.38	-3.453	31.23	24.43	-64.15
414 0.9 DL + 1.6 WL	52.1	560.9	5.3	11.53	0.229	-109.13	107.58	41.99
415 0.9 DL + 1.6 WL	-76.3	-293.0	3.7	7.34	0.323	160.41	6.06	-84.19
416 0.9 DL + 1.0 EL	-10.2	150.7	4.6	9.81	0.237	21.60	28.99	11.20
417 0.9 DL + 1.0 EL	-14.0	117.2	4.4	9.06	0.315	29.68	26.44	4.81
418 0.9 DL + 1.0 EL	-11.9	148.3	6.4	14.58	0.516	25.18	31.15	8.39
419 0.9 DL + 1.0 EL	-12.3	119.6	2.6	4.29	0.036	26.09	24.28	7.62

Max Net factored B.P = 133.64 KN/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
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Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 96.22 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 919.61 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 702.2 mm^2

Effective Depth Along X = 367 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 660.6 mm^2

A_{req} = 702.2 mm^2

Required Spacing = **286.3** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 60.30 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 841.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 458.2 mm^2

Effective Depth Along X = 351 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 631.8 mm^2

A_{req} = 458.2 mm^2

Required Spacing = **318.2** mm

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 929.66 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 369 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 332.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **340.6** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 870.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 357 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 321.3 mm^2

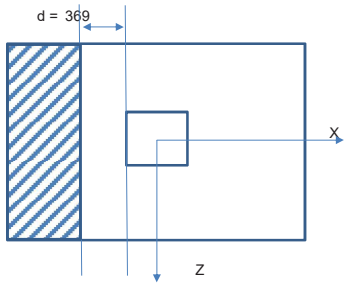
A_{req} = 0.0 mm^2

Required Spacing = **352.0** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	23-03-2020	25-04-2020	01.11.2023	Date	23-03-2020	25-04-2020	01.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Along X Direction Cantilever

L	=	0.831	m		
Reinforcement ratio ρ_w	=	0.0027			
Effective Depth Along X d	=	369	mm		
Shear Force at d distance V_u	=	111.05	KN		
Allowable shear strength φVc	=	273.35	KN	O.K	
Utility ratio	=	0.41			

Least of

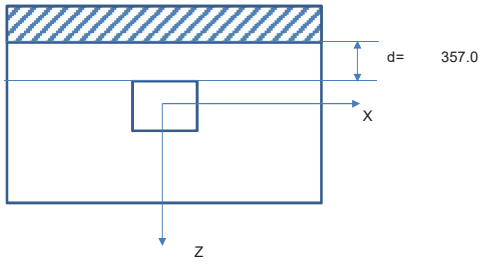
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 46.14 KNm
φ = 0.75 for shear
φVc = 273.3 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 274.8 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
= 474.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.593	m		
Reinforcement ratio ρ_w	=	0.0028			
Effective Depth Along Z d	=	357	mm		
Shear Force at d distance V_u	=	79.25	KN		
Allowable shear strength φVc	=	266.26	KN	O.K	
Utility ratio	=	0.30			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*

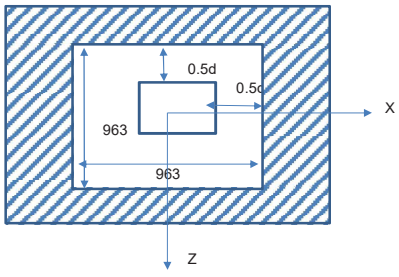
λ = 1 Normal concrete
Mu = 23.50 KNm
φ = 0.75 for shear
φVc = 268.9 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 266.3 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
φVc = 459.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 7.5 m2
Punching area = 0.93 m2
λ = 1
β = 1
αs = 40
b0 = 3852.0 mm

Allowable shear stress least of *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.83 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face u	=	3.85	m		
Depth at section d	=	363	mm		
Shear Force at section Vu	=	878.3	KN		
Allowable shear strength φVc	=	2047.40	KN	O.K	
Utility ratio		0.43			

Vc = $v_c b_0 d$
= 2729.86 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 2047.4 KN

Global Project ID	900707
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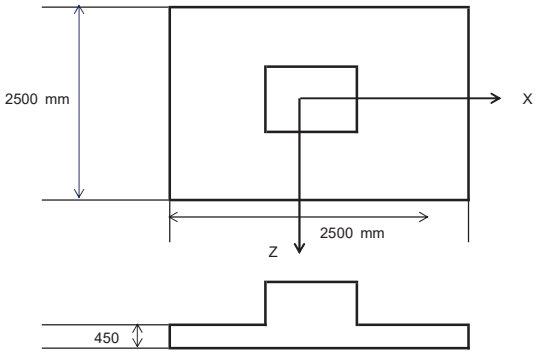
Ma'aden Revision No:	A01	C01	D01
Date	23-03-2020	25-04-2020	01.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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Date	23-03-2020	25-04-2020	01.11.2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
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Approved	PRI	PRI	PRI

7.3.6 Design of foundation F3 for supporting VGTU pedestals along grid C-1 & C-2.

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-3.500	m
Staad support reaction level		=	-3.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	2.50	m
Width of footing	Lx	=	2.50	m
Thickness of footing	D	=	0.45	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.60	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	2.60	m^3
Section Modulus of Pedestal	Zzz	=	2.60	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	554	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	75	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad Node no	=	6		
Weight of soil	=	270.35	KN	*inputs
Weight of foundation	=	67.50	KN	
Uplift	=	0.00	KN	
Surcharge	=	0	KN	
Pressure on top for tension	=	54.1	KN/m^2	



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-10.1	65.236	1.9	1.50	0.6	24.5	403.1	74.48	54.51	1	0.13
302 1.0 (DL + TL1) +	-9.8	120.092	1.0	-0.84	0.76	25.4	457.9	83.33	63.21	1	0.15
303 1.0 (DL + TL1) +	-9.8	119.969	1.2	-0.57	0.788	25.4	457.8	83.21	63.29	1	0.15
304 1.0 DL+ 0.75 W	-8.5	154.413	14.3	35.55	2.21	22.6	492.3	101.07	56.45	1	0.18
305 1.0 DL+ 0.75 W	-11.3	58.252	-11.7	-35.86	-0.756	27.7	396.1	87.79	38.96	1	0.16
306 1.0 DL+ 0.75 W	17.9	121.39	1.1	-0.79	0.8	-34.4	459.2	87.00	59.96	1	0.16
307 1.0 DL+ 0.75 W	-37.7	91.274	1.5	0.48	0.62	84.7	429.1	101.37	35.95	1	0.18
308 1.0 DL + 0.525	-8.9	106.993	1.3	-0.09	0.728	23.1	444.8	80.09	62.26	1	0.14
309 1.0 DL + 0.525	-10.8	105.671	1.3	-0.23	0.726	27.2	443.5	81.48	60.45	1	0.15
310 1.0 DL + 0.525	-9.8	108.371	1.7	1.02	0.836	24.9	446.2	81.36	61.44	1	0.15
311 1.0 DL + 0.525	-10.0	104.293	0.9	-1.34	0.6	25.4	442.1	81.00	60.49	1	0.15
312 0.6 DL + 1.0 WL	-4.2	103.249	18.5	48.50	2.33	11.3	441.1	93.52	47.63	1	0.17
313 0.6 DL + 1.0 WL	-8.0	-24.965	-16.2	-46.71	-1.625	18.1	312.9	74.97	25.16	1	0.14
314 0.6 DL + 1.0 WL	31.0	59.219	0.8	0.05	0.5	-64.7	397.1	88.41	38.66	1	0.16
315 0.6 DL + 1.0 WL	-43.2	19.064	1.5	1.74	0.209	94.1	356.9	93.92	20.29	1	0.17
316 0.6 DL + 0.7 EL	-4.8	40.024	1.2	0.99	0.354	12.0	377.9	65.45	55.47	1	0.12
317 0.6 DL + 0.7 EL	-7.3	38.26	1.1	0.80	0.351	17.4	376.1	67.17	53.19	1	0.12
318 0.6 DL + 0.7 EL	-5.9	41.86	1.7	2.47	0.498	14.4	379.7	67.23	54.28	1	0.12
319 0.6 DL + 0.7 EL	-6.2	36.423	0.6	-0.68	0.2	15.0	374.3	65.91	53.86	1	0.12

Max P_{max}	=	101.37	KN/m^2
Max P_{min}	=	20.29	KN/m^2
Resultant F.O.S	=	0.18	

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-14.2	91.3	2.7	2.09	0.8	34.30	28.59	0.64
402 1.4 (DL + TL1) +	-13.7	184.6	1.1	-1.84	1.12	35.75	43.97	15.09
403 1.4 (DL + TL1) +	-13.6	184.4	1.5	-1.47	1.159	35.77	43.80	15.21
404 1.2 DL+ 1.6 WL	-9.0	208.3	29.6	76.86	3.962	24.31	72.17	-5.53
405 1.2 DL+ 1.6 WL	-15.0	3.1	-25.8	-75.47	-2.366	35.34	43.05	-42.06
406 1.2 DL+ 1.6 WL	47.3	137.8	1.4	-0.66	1.027	-97.25	59.65	-15.55
407 1.2 DL+ 1.6 WL	-71.3	73.6	2.4	2.04	0.569	156.91	72.81	-49.27
408 1.2 DL + 1.0 EL	-10.2	106.9	1.9	0.83	0.8	25.98	27.40	6.82
409 1.2 DL + 1.0 EL	-13.8	104.4	1.8	0.56	0.796	33.68	29.85	3.56
410 1.2 DL + 1.0 EL	-11.8	109.6	2.7	2.94	1.005	29.39	29.95	5.11
411 1.2 DL + 1.0 EL	-12.2	101.8	1.1	-1.56	0.591	30.27	28.51	4.07
412 0.9 DL + 1.6 WL	-6.1	161.3	29.4	77.51	3.692	16.53	61.92	-10.31
413 0.9 DL + 1.6 WL	-12.1	-43.9	-26.0	-74.82	-2.635	27.57	32.30	-46.33
414 0.9 DL + 1.6 WL	50.2	90.8	1.2	-0.01	0.757	-105.03	54.87	-25.80
415 0.9 DL + 1.6 WL	-68.4	26.6	2.2	2.70	0.3	149.13	62.56	-54.05
416 0.9 DL + 1.0 EL	-7.3	60.0	1.8	1.48	0.531	18.20	17.15	2.04
417 0.9 DL + 1.0 EL	-10.9	57.5	1.7	1.21	0.526	25.90	19.60	-1.22
418 0.9 DL + 1.0 EL	-8.9	62.6	2.5	3.60	0.736	21.61	19.69	0.34
419 0.9 DL + 1.0 EL	-9.4	54.8	0.9	-0.90	0.321	22.49	17.76	-0.21

Max Net factored B.P = 72.81 KN/m^2

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Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 32.85 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 919.61 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 237.6 mm^2

Effective Depth Along X = 367 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 660.6 mm^2

A_{req} = 237.6 mm^2

Required Spacing = **304.4** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 32.85 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 841.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 248.6 mm^2

Effective Depth Along X = 351 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 631.8 mm^2

A_{req} = 248.6 mm^2

Required Spacing = **318.2** mm

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 929.66 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 369 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of

or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 332.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **340.6** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 870.17 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 357 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

greater of

or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 321.3 mm^2

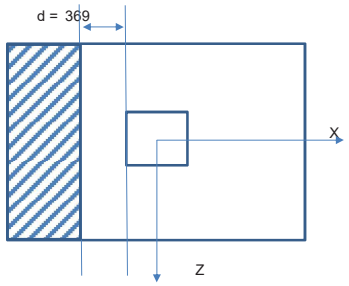
A_{req} = 0.0 mm^2

Required Spacing = **352.0** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



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Along X Direction Cantilever

L	=	0.581	m		
Reinforcement ratio ρ_w	=	0.0027			
Effective Depth Along X d	=	369	mm		
Shear Force at d distance V_u	=	42.30	KN		
Allowable shear strength φVc	=	274.78	KN	O.K	
Utility ratio	=	0.15			

Least of

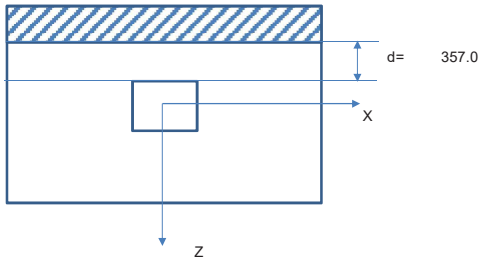
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 12.29 KNm
φ = 0.75 for shear
φVc = 278.2 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$
φVc = 274.8 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$
= 474.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.593	m		
Reinforcement ratio ρ_w	=	0.0028			
Effective Depth Along Z d	=	357	mm		
Shear Force at d distance V_u	=	43.17	KN		
Allowable shear strength φVc	=	266.26	KN	O.K	
Utility ratio	=	0.16			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ — *Table 22.5.5.1 SBC 304*

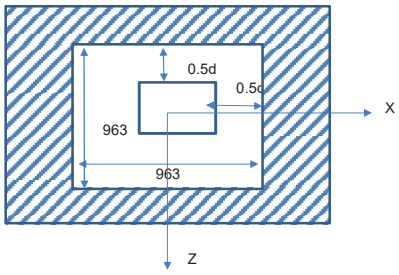
λ = 1 Normal concrete
Mu = 12.80 KNm
φ = 0.75 for shear
φVc = 268.9 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$
φVc = 266.3 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$
φVc = 459.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 6.25 m2
Punching area = 0.93 m2
λ = 1
β = 1
αs = 40
b0 = 3852.0 mm

Allowable shear stress least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1 + \frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2 + \frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.83 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face u	=	3.85	m		
Depth at section d	=	363	mm		
Shear Force at section Vu	=	387.5	KN		
Allowable shear strength φVc	=	2047.40	KN	O.K	
Utility ratio		0.19			

Vc = $v_c b_0 d$
= 2729.86 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 2047.4 KN

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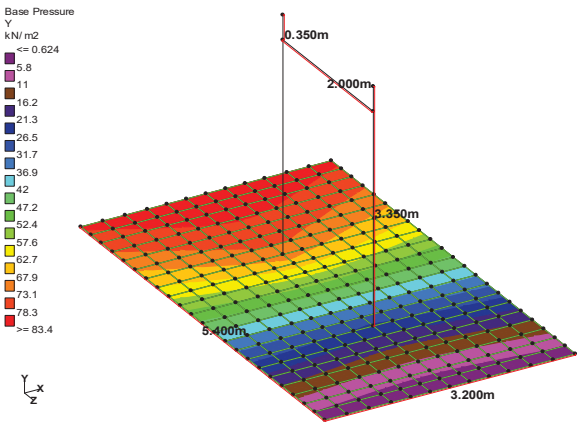
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7.3.7 Design of combined footing for Trestle mkd. TR1, TR2 & TR3 and foundation mkd. as F2

For Calculation of Base pressure and design of beam, column and raft, Refer STAAD File: 3130-CNV-001_TR3 FDN MKD F2

Staad Supoort Nodes are 5 & 6



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.15	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.200	m	
Width of footing	Lx	=	5.400	m	
Thickness of footing	D	=	0.55	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	26.1	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	110.8	KN/m^2	Load Case 305
Minimum Base Pressure		=	0.6	KN/m^2	Load Case 313

2) Design of foundation Raft :

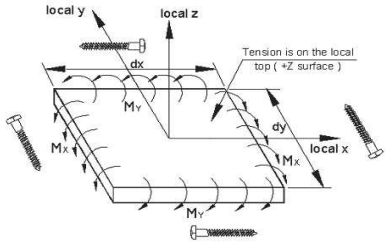
Slab Thickness =	550	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	27.617	kN-m	Corresponding Mxy	0.757	kN-m	*(Plate No. - 24 , L/C No. - 413)*
Min. Mx	-65.478	kN-m	Corresponding Mxy	11.433	kN-m	*(Plate No. - 115 , L/C No. - 405)*
Max. My	68.074	kN-m	Corresponding Mxy	1.606	kN-m	*(Plate No. - 25 , L/C No. - 413)*
Min. My	-108.882	kN-m	Corresponding Mxy	4.209	kN-m	*(Plate No. - 114 , L/C No. - 405)*

FOR TOP FACE REINFORCEMRNT :



Mx1 =	Mx + abs(Mxy) =	27.617 + 0.757 =	28.374 kN-m
My1 =	My + abs(Mxy) =	68.074 + 1.606 =	69.680 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	27.617 + 0.008 =	27.625 kN-m
My2 =	My+abs(Mxy^2/Mx) =	68.074 + 0.062 =	68.136 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	28.374 kN-m
Myd =	69.68 kN-m

ALONG LOCAL X - DIRECTION :

Mu =	28.374 kN-m						
Φ*Kn =	Mu/bd^2				deff. =	469	mm
Mu/bd^2 =	28.374X10^6/(1000X469^2) =	0.129	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0003	Reqd. Ast =	0.0003X1000X469/100 =	1.41	cm^2		
Provided	12 mm @	200	C/C	(Provide Ast =	5.65	cm^2	
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0012			

ALONG LOCAL Y - DIRECTION :

Mu =	69.68 kN-m						
Φ*Kn =	Mu/bd^2				deff. =	457	mm
Mu/bd^2 =	69.68X10^6/(1000X457^2) =	0.317	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0008	Reqd. Ast =	0.0008X1000X469/100 =	3.66	cm^2		
Provided	12 mm @	200	C/C	(Provide Ast =	5.65	cm^2	
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0012			

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FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -65.478 - 11.433 = -76.911 kN-m
My1 = My - abs(Mxy) = -108.882 - 4.209 = -113.091 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -65.478 - 1.201 = -66.679 kN-m
My2 = My-abs(Mxy^2/Mx) = -108.882 - 0.271 = -108.611 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -76.911 kN-m
Myd = -113.091 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 76.911 kN-m
Φ*Kn = Mu/bd^2 deff. = 467 mm
Mu/bd^2 = 76.911X10^6/(1000X467^2) = 0.353 N/mm^2 < 6.9 N/mm^2
Pt = 0.0009 Reqd. Ast = 0.0009X1000X467/100 = 4.2 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 **O.K.**
Total Provide Pt = 0.0022

ALONG LOCAL Y - DIRECTION :

Mu = 113.091 kN-m
Φ*Kn = Mu/bd^2 deff. = 451 mm
Mu/bd^2 = 113.091X10^6/(1000X451^2) = 0.556 N/mm^2 < 6.9 N/mm^2
Pt = 0.0015 Reqd. Ast = 0.0015X1000X451/100 = 6.77 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 **O.K.**
Total Provide Pt = 0.0022

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.158 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 247 , L/C No. - 405)*
Vu = 73.79 kN

Provided Pt = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X73.786X10^3X467/76.911/10^6)X1000X467/10^3
= 449.875 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X467/10^3
= 459.515 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X467/10^3
= 801.215 kN
HENCE, Vc = 449.875 kN
Φt*Vc = 0.75X449.875 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 337.406 kN **> 73.786T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.221 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 146 , L/C No. - 405)*
Vu = 99.671 kN

Pt provided = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X99.671X10^3X451/113.091/10^3)X1000X451/10^3
= 433.609 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X451/10^3
= 443.772 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X451/10^3
= 773.764 kN
HENCE, Vc = 433.609 kN
Φt*Vc = 0.75X433.609 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 325.207 kN **> 99.671T , Hence O.K.**

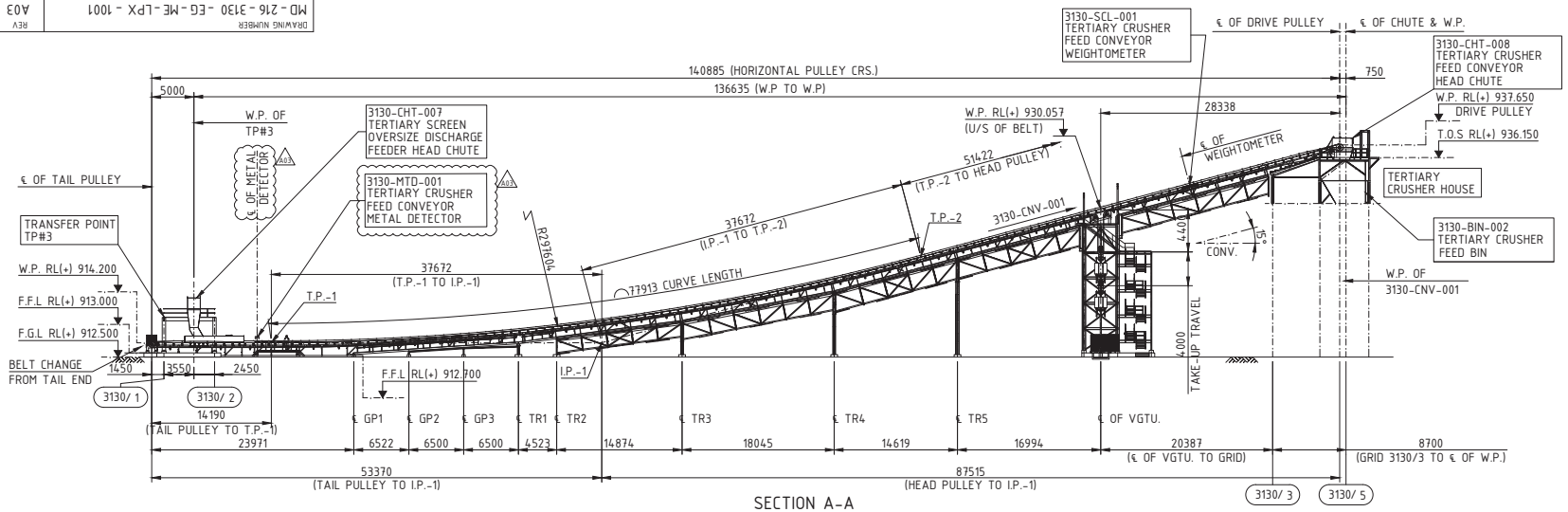
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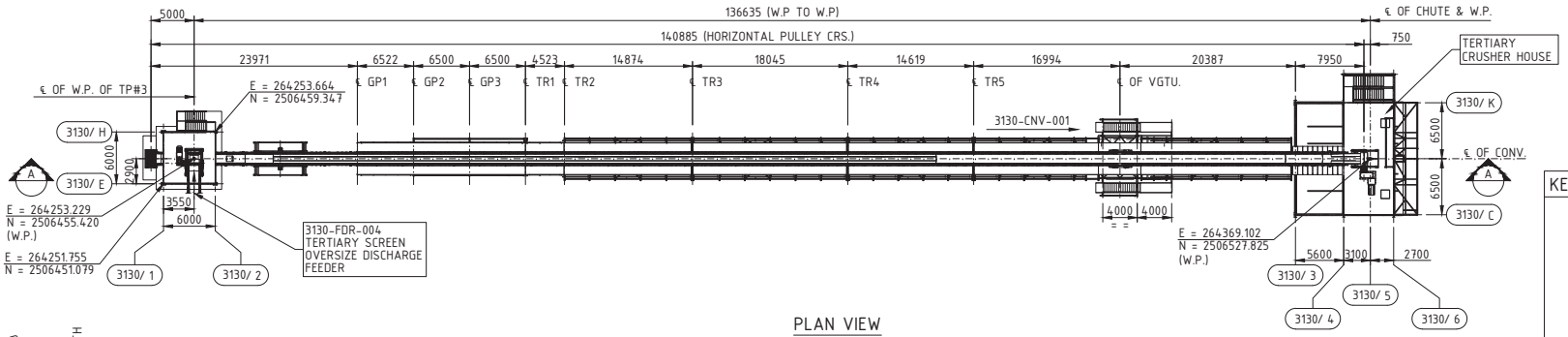
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Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

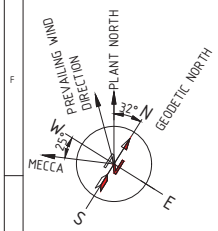
DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1001 REV A03



SECTION A-A
SCALE 1:300



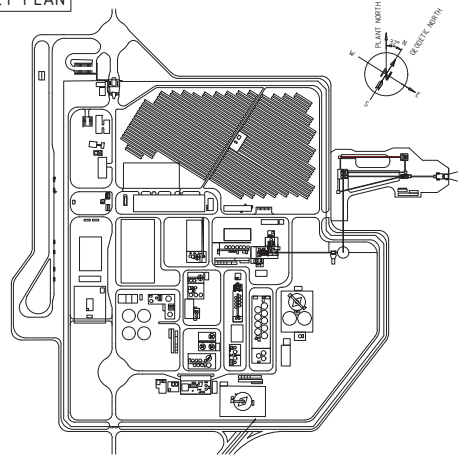
PLAN VIEW
SCALE 1:300



NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+) 913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A03	14/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
				COMMENTS DT. 08.04.2020 & REVISED AS MARKED ⚠3					
MD-216-3130-EG-ME-ELV-1003	CONV. 3130-CNV-001 -ELEVATION	A02	18/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3130-EG-ME-ELV-1002	CONV. 3130-CNV-001 -ELEVATION			COMMENTS DT. 04.03.20 & REVISED AS MARKED ⚠2					
MD-216-3130-EG-ME-ELV-1001	CONV. 3130-CNV-001 -ELEVATION	A01	06/02/2020	ISSUED FOR APPROVAL					

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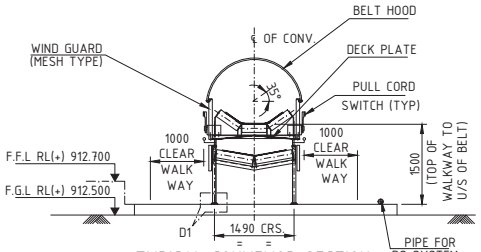
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31326 — DOCUMENT ID: OU100669067 — REVISION: 2

DRAWN	AFO	03/02/2020
CHECKED	BRD	04/02/2020
ENGINEER	BD	04/02/2020
APPROVED	BGR	06/02/2020
EMPLOYER		
SCALE	1:300	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
CONV. 3130-CNV-001 - PLAN

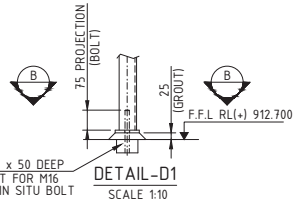
DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1001 — REV: A03

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1001 | REV: A03



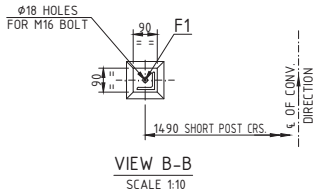
TYPICAL CONVEYOR SECTION ON GROUND (1200 B/W)

SCALE 1:50



DETAIL-D1

SCALE 1:10



VIEW B-B

SCALE 1:10

CONVEYOR CHARACTERISTICS

	PARTICULARS	SPECIFICATION	QUANTITY
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	VGTU. / 4200 mm	
	TAKE-UP LOAD	~4680 kg.(APPROXIMATE) RCC BLOCKS, 40 kg EACH	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	INTERNAL V-PLOUGH	1 NO.
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS)	@ 50 m INTERVAL	3 PAIRS
	PULL CORD SWITCH (HS)	@ 30 m INTERVAL	5 PAIRS
	ZERO SPEED SWITCH (SS)	AT TAIL PULLEY	1 NO.
	BELT RIP SWITCH (YS)	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm thk. CONTINUOUS	1 LOT.
	BELT HOOD	GALVANIZED CORRUGATED SHEET	1 LOT.

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
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N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
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STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

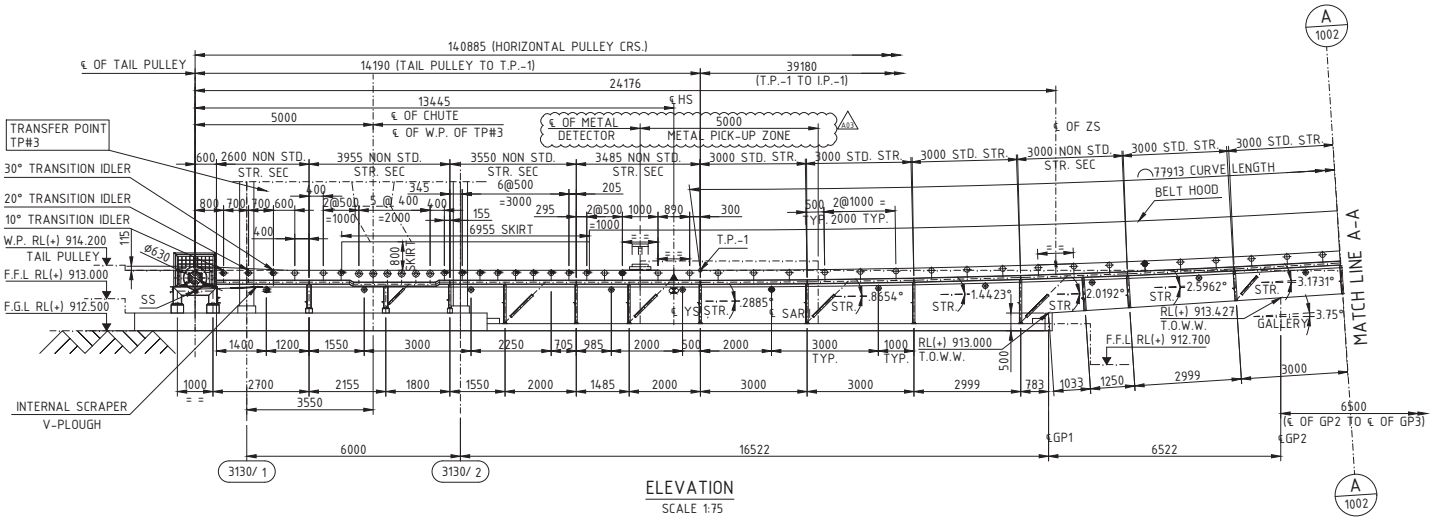
FOR CONV. 3130-CNV-001

POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
	DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F1	(+)0.50	(+)0.55	(±)0.15	(±)0.08

SIGN CONVENTION

UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design



ELEVATION

SCALE 1:75

CONVEYOR CHARACTERISTICS

	PARTICULARS		SPECIFICATION	QUANTITY
CONVEYOR	EQUIPMENT TAG NO.		3130-CNV-001	1 NO.
	EQUIPMENT DESCRIPTION		TERTIARY CRUSHER FEED CONVEYOR	
	LOCATION		TP-3 TO TERTIARY CRUSHER HOUSE	
	CAPACITY RATED / DESIGN	(t/h)	782 / 870	
	BELT WIDTH	(mm)	1200	
	BELT SPEED	(m/sec)	1.673	
	TRIPPER FIXED / MOVABLE		N.A.	
	CONVEYOR LENGTH / LIFT	(mm)	140885 / 23450	
MATERIAL	MATERIAL		GOLD ORE : FRESH ORE / OXIDE ORE	
	BULK DENSITY	(t/m³)	1.6	
	MAX. LUMP SIZE	(mm)	(-)40	
	MOISTURE CONTENT	(%)	2	
	SURCHARGE ANGLE	(deg.)	20	
BELTING	MAX. RUNNING TENSION	(T1 kN)	78.879	
	SLACK SIDE TENSION	(T2 kN)	30.089	
	TAIL TENSION	(T0 kN)	31.013	
	BELT DESIGNATION		EP-1000 / 4 (HEAVY DUTY)	
	TOP / BOTTOM COVER THICKNESS	(mm)	6 / 3	
	COVER GRADE / FABRIC		DIN-X / EP	
PULLEYS	DISCHARGE WITH DRIVE	(MKD. x DIA.)	D-5 x 800 mm	1 NO.
	DISCHARGE WITHOUT DRIVE	(MKD. x DIA.)	N.A.	
	SNUB	(MKD. x DIA.)	ND-7 x 500 mm	1 NO.
	TAIL	(MKD. x DIA.)	ND-6 x 630 mm	1 NO.
	TAKE-UP	(MKD. x DIA.)	ND-6 x 630 mm	1 NO.
	DEFLECTOR	(MKD. x DIA.)	DR-3 x 273 mm	1 NO.
	TRIPPER DISCHARGE	(MKD. x DIA.)	N.A.	
	TRIPPER BEND	(MKD. x DIA.)	N.A.	
	BEND	(MKD. x DIA.)	ND-6 x 630 mm	2 NOS.
	LAGGING	(mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK	DISCHARGE WITH DRIVE		TBA	2 SETS
	DISCHARGE WITHOUT DRIVE		N.A.	
	SNUB		TBA	2 SETS
	TAIL		TBA	2 SETS
	TAKE-UP		N.A.	
	DEFLECTOR		TBA	2 SETS
	BEND		TBA	4 SETS
	TRIPPER DISCHARGE		N.A.	
	TRIPPER BEND		N.A.	
IDLER	TROUGHING	(DIA. x deg.)	152.4 x 35°	110 NOS.
	TROUGHING RETRACTABLE	(DIA. x deg.)	152.4 x 45°	10 NOS.
	TROUGHING TRAINER	(DIA. x deg.)	152.4 x 35°	9 NOS.
	RETURN	(DIA. x deg.)	152.4 x 10°	36 NOS.
	SELF ALIGNING RETURN TRAINER INVERTED IDLER	(DIA. x deg.)	152.4 x 10°	5 NOS.
	IMPACT RETRACTABLE	(DIA. x deg.)	190 x 45°	6 NOS.
	TRANSITION (ADJUSTABLE)	(DIA. x deg.)	152.4 x 10°/20°/30°	(2+2+1) NOS.
	RETURN SELF CLEANING	(DIA. x deg.)	190 x 0°	2 NOS.
	NON MAGNETIC TROUGHING	(DIA. x deg.)	152.4 x 35°	2 NOS.
	WEIGHING IDLER	(DIA. x deg.)	152.4 x 35°	4 NOS.
	SPECIAL MACHINED IDLER	(DIA. x deg.)	152.4 x 35°	8 NOS.
	RETURN FLAT IDLER	(DIA. x deg.)	152.4 x 0°	3 NOS.
DRIVE	MOTOR	RATING kW/rpm (SYN)	132 / 1800	1 NO.
		FRAME SIZE	TBA	
		TYPE	SQUIRREL CAGE	
		SUPPLY	400V, 60Hz	
	REDUCER	TYPE	HELICAL	1 NO.
		MODEL NO./COOLING/RATIO	TBA	
		RATIO	45	
		HOLD BACK	YES	
	BRAKE	TYPE	ELECTRO HYDRAULIC THRUSTER	1 NO.
		MODEL NO.	TBA	
		DRAM DIAMETER	TBA	
	COUPLING LOW SPEED	TYPE	FLUID COUPLING	1 NO.
		MODEL NO.	TBA	
	COUPLING HIGH SPEED	TYPE	GEAR	1 NO.
		MODEL NO.	TBA	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A03	14/04/2020	ISSUED FOR DETAIL DESIGN & REVISED AS MARKED (A03)	AFO	BRD	BD	BGR	
MD-216-3130-EG-ME-ELV-1003	CONV. 3130-CNV-001 -ELEVATION	A02	18/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3130-EG-ME-ELV-1002	CONV. 3130-CNV-001 -ELEVATION			COMMENTS DT. 09.03.20 & REVISED AS MARKED (A03)					
MD-216-3130-EG-ME-LPX-1001	CONV. 3130-CNV-001 -PLAN	A01	06/02/2020	ISSUED FOR APPROVAL					

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OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31382 | DOCUMENT ID: OU100673004 | REVISION: 2

DRAWN	AFO	03/02/2020
CHECKED	BRD	04/02/2020
ENGINEER	BD	04/02/2020
APPROVED	BGR	06/02/2020
EMPLOYER		
SCALE	1:75, 1:50, 1:10	A1

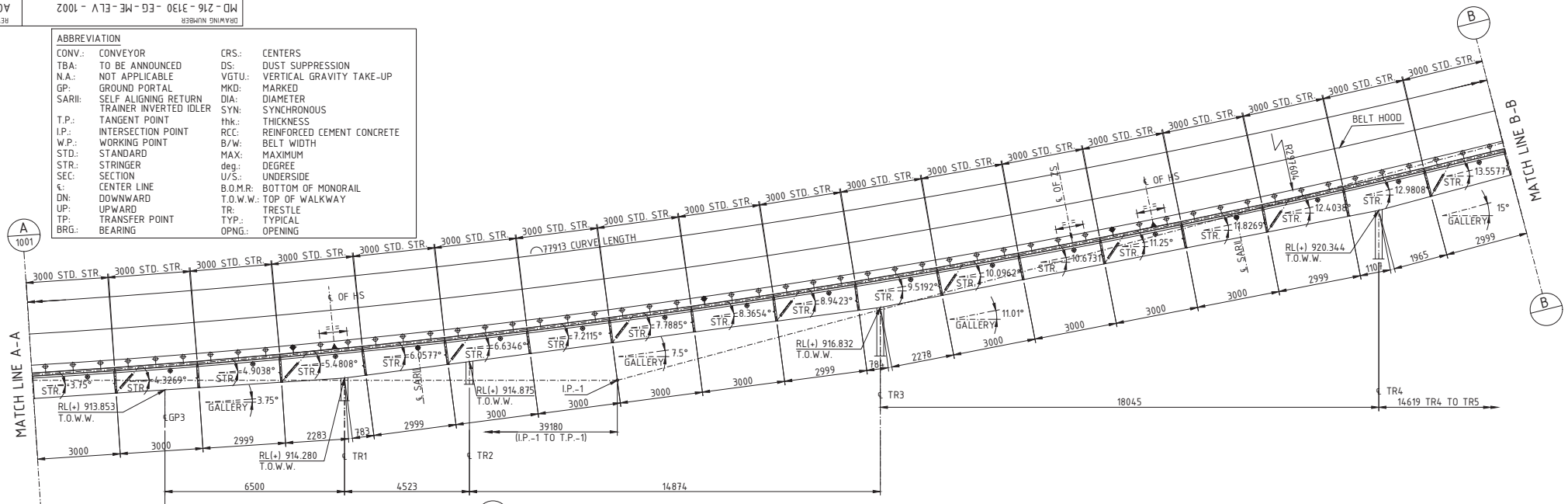
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
CONV. 3130-CNV-001 - ELEVATION

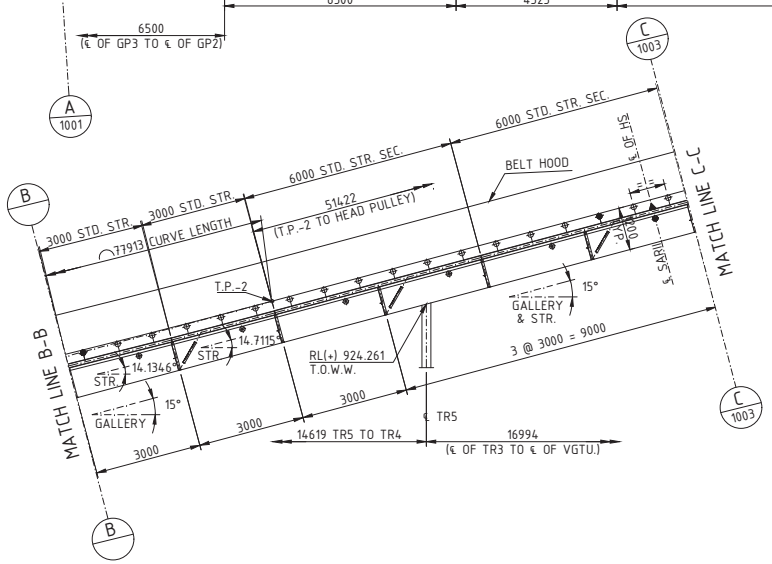
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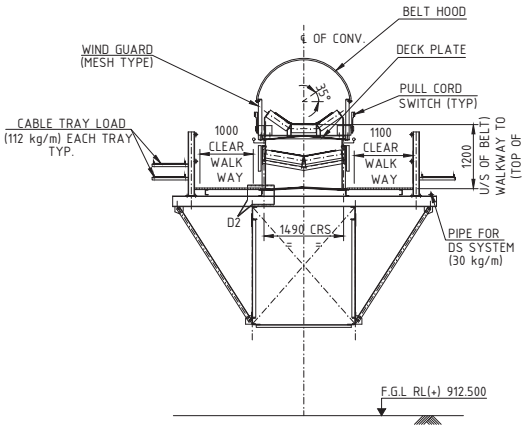
DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1002 | REV: A03

ABBREVIATION

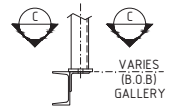
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TP:	TRANSFER POINT	TYP.:	TYPICAL
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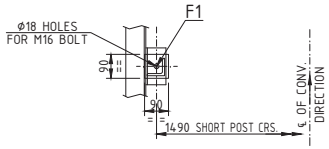




TYPICAL CONVEYOR SECTION ON GALLERY (1200 B/W)
SCALE 1:50



DETAIL-D2
SCALE 1:10



VIEW C-C
SCALE 1:10

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A03	14/04/2020	ISSUED FOR DETAIL DESIGN	AFO	BRD	BD	BGR	
MD-216-3130-EG-ME-ELV-1003	CONV. 3130-CNV-001 -ELEVATION	A02	18/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3130-EG-ME-ELV-1001	CONV. 3130-CNV-001 -ELEVATION			COMMENTS DT. 04.03.20 & REVISED AS MARKED △1					
MD-216-3130-EG-ME-LPX-1001	CONV. 3130-CNV-001 -PLAN	A01	06/02/2020	ISSUED FOR APPROVAL					

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LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16113_003		REVISION 2
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31335	DOCUMENT ID OU100669076	REVISION 2
DRAWN	AFO	03/02/2020	
CHECKED	BRD	04/02/2020	
ENGINEER	BD	04/02/2020	
APPROVED	BGR	06/02/2020	
EMPLOYER			
SCALE	1:75, 1:50, 1:10	A1	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
CONV. 3130-CNV-001 - ELEVATION

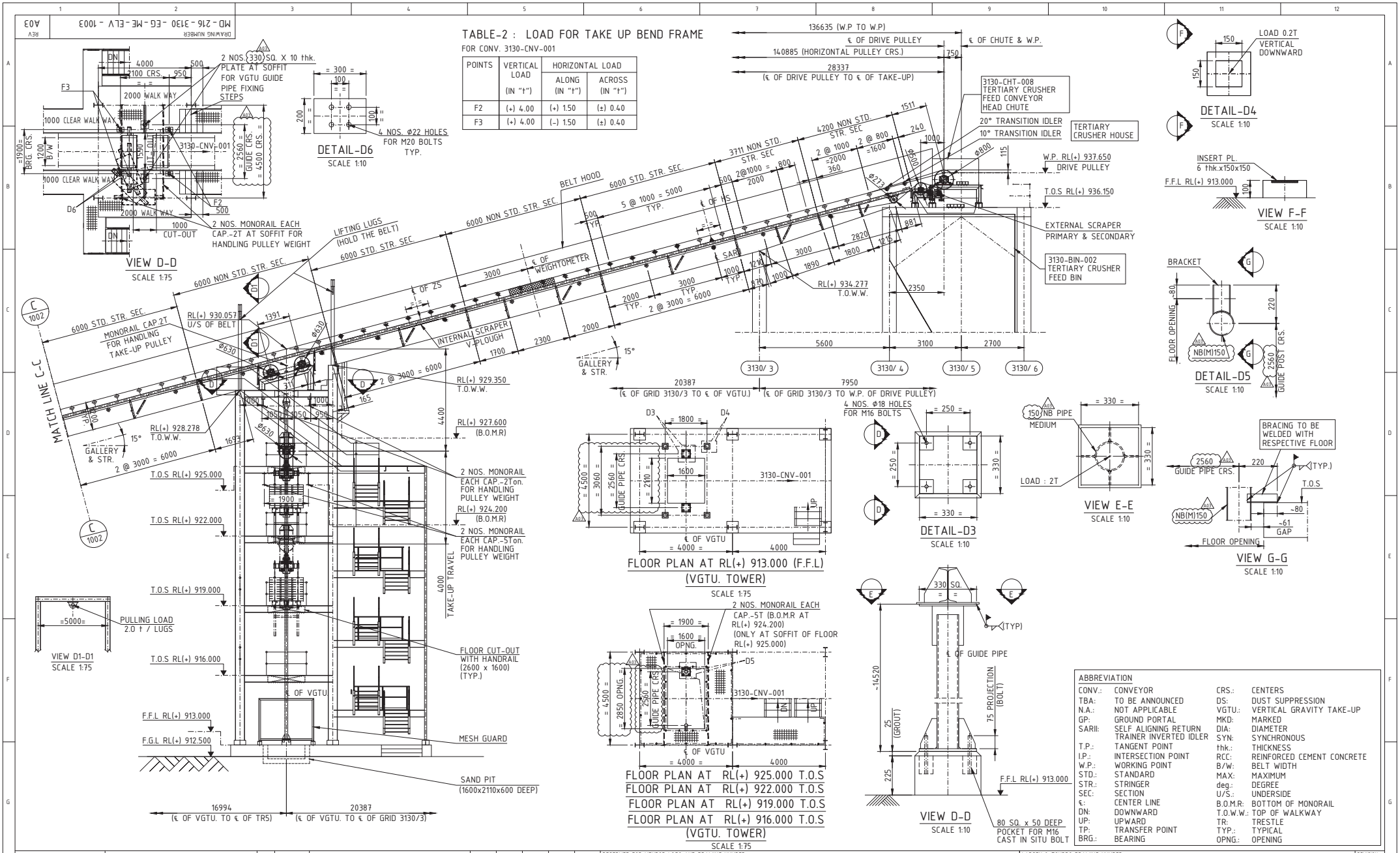
DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1002 | REV: A03

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1003 REV A03

TABLE-2 : LOAD FOR TAKE UP BEND FRAME

FOR CONV. 3130-CNV-001

POINTS	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD	
		ALONG (IN "t")	ACROSS (IN "t")
F2	(+) 4.00	(+) 1.50	(±) 0.40
F3	(+) 4.00	(–) 1.50	(±) 0.40



ABBREVIATION

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				COMMENTS DT. 08.04.2020 & REVISED AS MARKED △					
MD-216-3130-EG-ME-ELV-1002	CONV. 3130-CNV-001 -ELEVATION	A02	18/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	AFO	BRD	BD	BGR	
MD-216-3130-EG-ME-ELV-1001	CONV. 3130-CNV-001 -ELEVATION			COMMENTS DT. 04.03.20 & REVISED AS MARKED △					
MD-216-3130-EG-ME-LPX-1001	CONV. 3130-CNV-001 -PLAN	A01	06/02/2020	ISSUED FOR APPROVAL					

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16113_004 — REVISION: 2

OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31336 — DOCUMENT ID: OU100669077 — REVISION: 2

DRAWN	AFO	03/02/2020
CHECKED	BRD	04/02/2020
ENGINEER	BD	04/02/2020
APPROVED	BGR	06/02/2020
EMPLOYER		
SCALE	1:75, 1:10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
CONV. 3130-CNV-001 - ELEVATION

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1003 — REV: A03

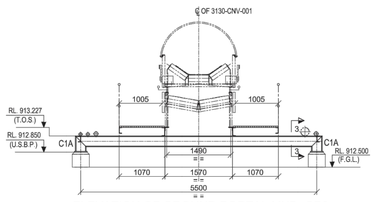
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	24-03-2020	25-04-2020	01.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

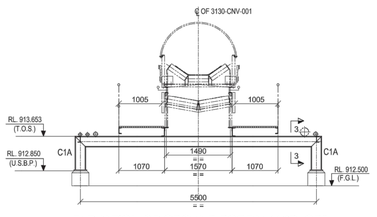
Outotec / L&T Revision No.:	0	1	2
Date	24-03-2020	25-04-2020	01.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWINGS CONTAINING BASE PLATE DETAIL**

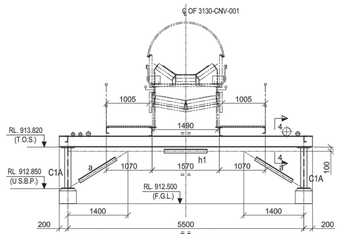
DRAWING NUMBER: MD-216-3130-EG-ST-GAD-1001 | REV: D01



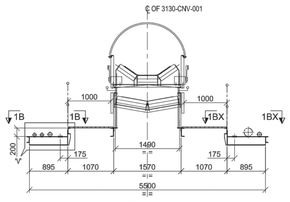
ELEVATION OF GROUND PORTAL MKD. GP2
(SCALE 1:50)



ELEVATION OF GROUND PORTAL MKD. GP3
(SCALE 1:50)



ELEVATION OF GROUND PORTAL MKD. GP4
(SCALE 1:50)

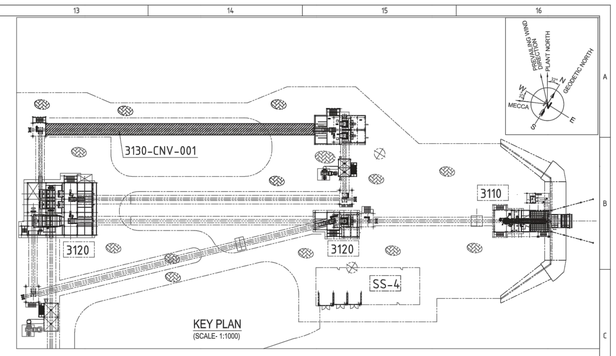


SECTION 1A-1A
(SCALE - 1:50)

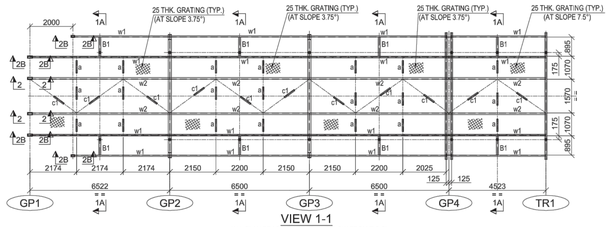
SCHEDULE OF MEMBERS FOR GP-2 TO GP4

MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1A	1 PL. 150X16 2 PL. 218X8	1 2	-	-	GUSSET PLATES 8 THK.
B1	IPE 200	I	-	-	
a	L 50X50X6	L	-	-	
c1	L 75X75X8	L	-	-	
h1	2L 75X75X8	⊥⊥	-	-	
w1	UPN 140	[	-	-	
w2	UPN 200	[	-	-	

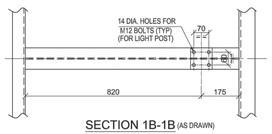
GRADE OF MATERIAL IS S275 JR
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED
FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.
ALL MEMBER CONNECTIONS SHOULD BE DESIGN FOR A MINIMUM FORCE OF 50KN.



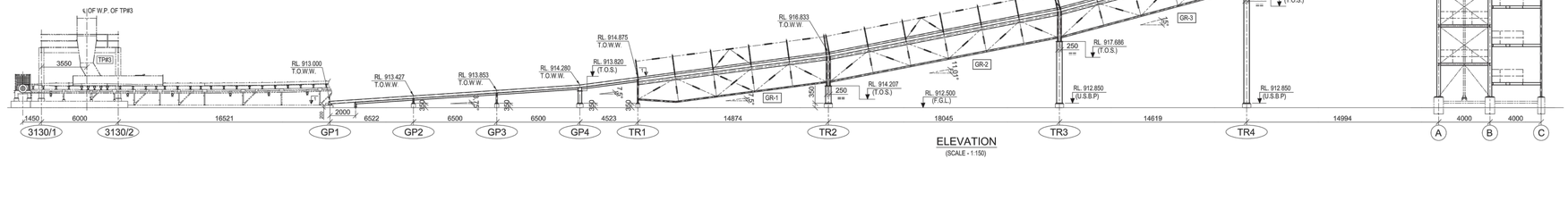
KEY PLAN
(SCALE- 1:1000)



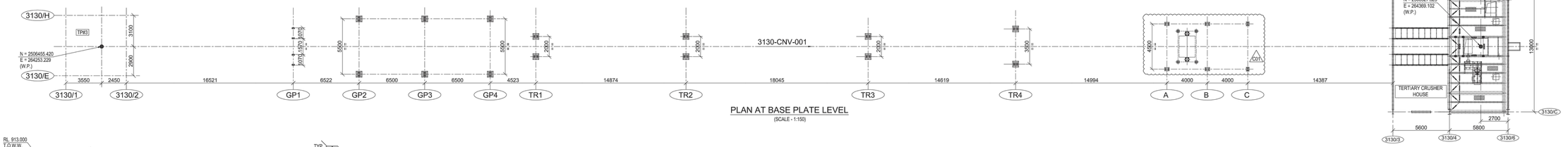
VIEW 1-1
(ALL PLAN BRACING ARE 75mm BELOW FROM T.O.S)
SCALE-(1:100)



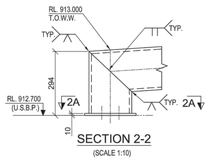
SECTION 1B-1B (AS DRAWN)
SECTION 1BX-1BX (OPP. HAND)
(SCALE - 1:10)



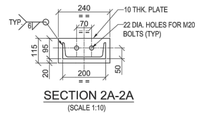
ELEVATION
(SCALE - 1:150)



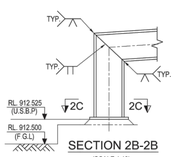
PLAN AT BASE PLATE LEVEL
(SCALE - 1:150)



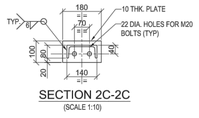
SECTION 2-2
(SCALE 1:10)



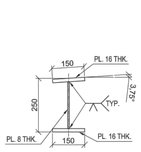
SECTION 2A-2A
(SCALE 1:10)



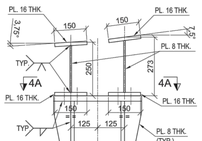
SECTION 2B-2B
(SCALE 1:10)



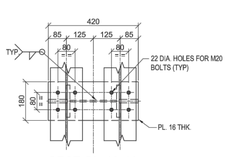
SECTION 2C-2C
(SCALE 1:10)



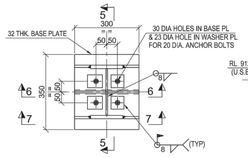
SECTION 3-3
(SCALE 1:10)



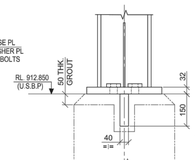
SECTION 4-4
(SCALE 1:10)



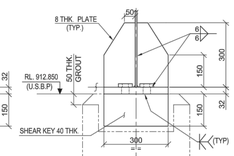
SECTION 4A-4A
(SCALE 1:10)



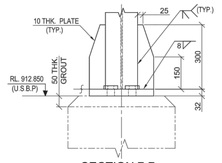
BASE PLATE DETAIL FOR GP2 & GP3
(SCALE - 1:10)



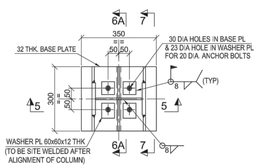
SECTION 5-5
(SCALE - 1:10)



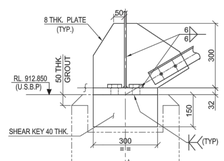
SECTION 6-6
(SCALE - 1:10)



SECTION 7-7
(SCALE - 1:10)



BASE PLATE DETAIL FOR GP4
(SCALE - 1:10)



SECTION 6A-6A
(SCALE - 1:10)

LEGENDS :-
T.O.S. - TOP OF STEEL
R.L. - REDUCED LEVEL
F/S-FAR SIDE
N/S-NEAR SIDE
TYP.-TYPICAL
W.P.-WORKING POINT
MM-MILLIMETER
PL.-PLATE
MKD.-MARKED
℄ - CENTRE LINE
T.O.C. - TOP OF CONCRETE.
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
B/S - BOTH SIDE
CHQD. PL. - CHEQUERED PLATE
U.S.B.P. - UNDER SIDE OF BASE PLATE
THK. - THICK
DN - DOWN
MC - MOMENT CONNECTION
T.O.W.W-TOP OF WALKWAY
O/P - OVER PLATE

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF STEEL USED S275JR AS PER EN10025
3. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3130-EG-ST-GAD-1002, 1003, 1004 & 1005
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1003	CONV. 3130-CNV-001 - ELEVATION								
MD-216-3130-EG-ME-ELV-1002	CONV. 3130-CNV-001 - ELEVATION	D01	01/11/2023	AS BUILT	SUS	SBY	APL	AG	
MD-216-3130-EG-ME-ELV-1001	CONV. 3130-CNV-001 - ELEVATION	C01	23/04/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	SUS	SBY	APL	AG	
MD-216-3130-EG-ME-LPX-1001	CONV. 3130-CNV-001 - PLAN	A01	20/03/2020	ISSUED FOR REFERENCE	MGH	SBY	APL	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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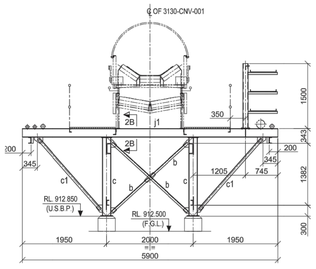
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Saudi Arabian Mining Company شركة معادن العربية السعودية
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LARSEN & TOUBRO DRAWING NUMBER	O19045_100_S_G_26013_013
REVISION	2
OUTOTEC DRAWING NUMBER	900707 __GAU01 __ZC01__MTL01 __31361
DOCUMENT ID	OU100670429
REVISION	2
DRAWN	MGH 16/03/2020
CHECKED	SBY 19/03/2020
ENGINEER	APL 18/03/2020
APPROVED	AG 20/03/2020
EMPLOYER	
SCALE	1:100,1:75,1:50,1:25,1:20,1:10
	A0

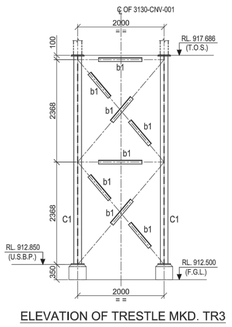
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRUCTURAL GA FOR 3130-CNV-001, Sh1
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER	REV
MD - 216 - 3130 - EG - ST - GAD - 1001	D01

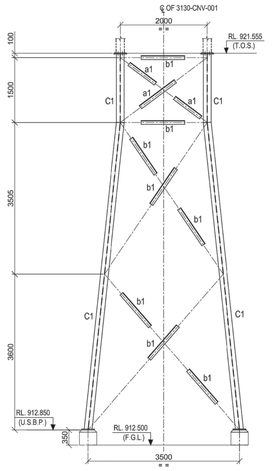
DRAWING NUMBER: MD-216-3130-EG-ST-GAD-1003 | REV: D01



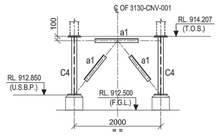
ELEVATION OF TRESTLE MKD. TR1
(SCALE 1:50)



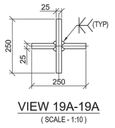
ELEVATION OF TRESTLE MKD. TR3
(SCALE 1:50)



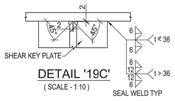
ELEVATION OF TRESTLE MKD. TR4
(SCALE 1:50)



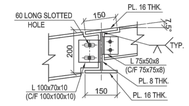
ELEVATION OF TRESTLE MKD. TR2
(SCALE 1:50)



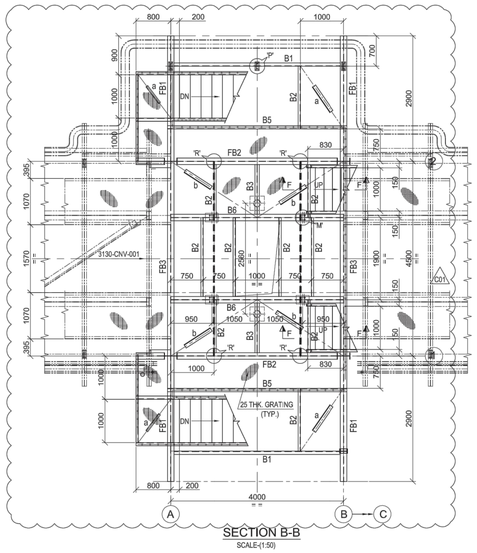
VIEW 19A-19A
(SCALE - 1:10)



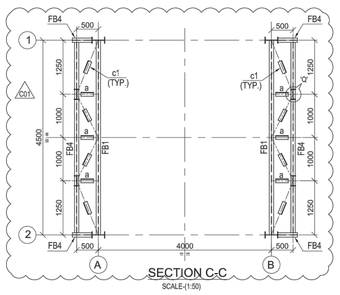
DETAIL '19C'
(SCALE - 1:10)



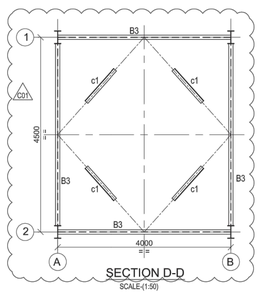
SECTION 2B-2B
(SCALE 1:10)



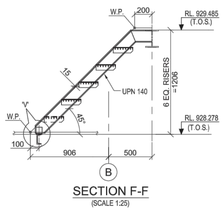
SECTION B-B
SCALE-(1:50)



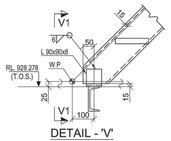
SECTION C-C
SCALE-(1:50)



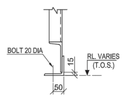
SECTION D-D
SCALE-(1:50)



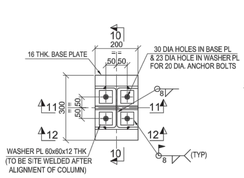
SECTION F-F
(SCALE 1:25)



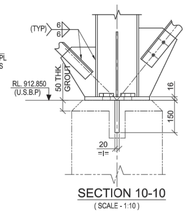
DETAIL - 'V'
(SCALE - 1:10)



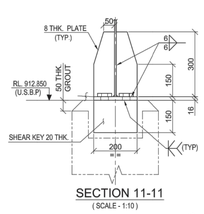
SECTION V1-V1
(SCALE - 1:10)



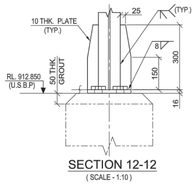
BASE PLATE DETAIL FOR TR1
(SCALE - 1:10)



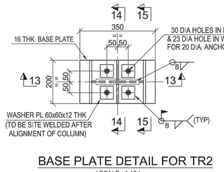
SECTION 10-10
(SCALE - 1:10)



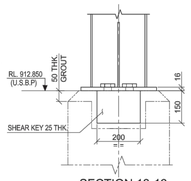
SECTION 11-11
(SCALE - 1:10)



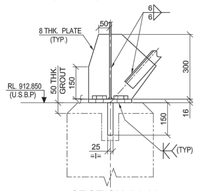
SECTION 12-12
(SCALE - 1:10)



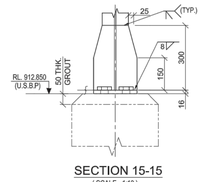
BASE PLATE DETAIL FOR TR2
(SCALE - 1:10)



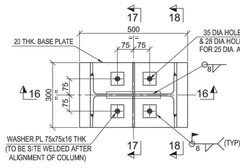
SECTION 13-13
(SCALE - 1:10)



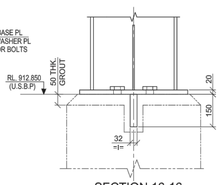
SECTION 14-14
(SCALE - 1:10)



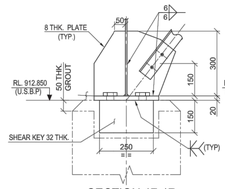
SECTION 15-15
(SCALE - 1:10)



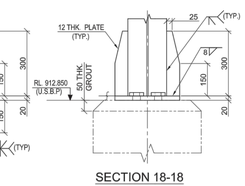
BASE PLATE DETAIL FOR TR3 & TR4
(SCALE - 1:10)



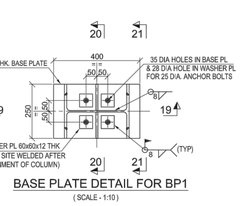
SECTION 16-16
(SCALE - 1:10)



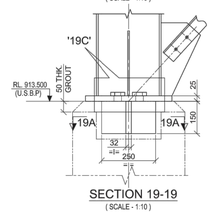
SECTION 17-17
(SCALE - 1:10)



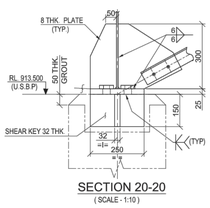
SECTION 18-18
(SCALE - 1:10)



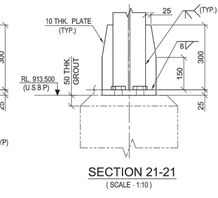
BASE PLATE DETAIL FOR BP1
(SCALE - 1:10)



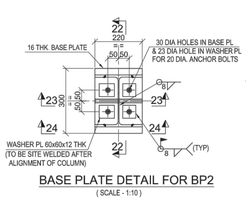
SECTION 19-19
(SCALE - 1:10)



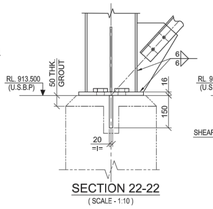
SECTION 20-20
(SCALE - 1:10)



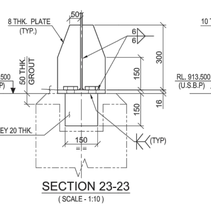
SECTION 21-21
(SCALE - 1:10)



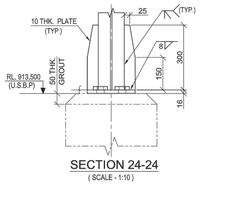
BASE PLATE DETAIL FOR BP2
(SCALE - 1:10)



SECTION 22-22
(SCALE - 1:10)



SECTION 23-23
(SCALE - 1:10)



SECTION 24-24
(SCALE - 1:10)

SCHEDULE OF MEMBERS FOR VGTU, TR1 TO TR4

MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C	IPE 200	I	-	-	
C1	IPE 400	I	-	-	
C2	IPE 300	I	-	-	
C2a (C01)	1. IPE 300 2. PL. 120X8	I-2 1	-	-	
C3	IPE 240	I	-	-	
C4	1. PL. 120X10 2. PL. 240X8	1 I-2 1	-	-	
FB1	IPE 200	I	-	-	
FB2	IPE 240	I	-	-	
FB3	IPE 300	I	-	-	
FB4	IPE 450	I	-	-	
B1	UPN 140	C	-	-	
B2	UPN 120	C	-	-	C01
B3	IPE 200	I	-	-	
B4	IPE 240	I	-	-	
B5	UPN 200	C	-	-	
B6	IPE 300	I	-	-	C01
a	L 50X50X6	L			MATERIAL GRADE S275 JR
a1	2L 50X50X6	T	100	-	
b	L 65X65X6	L	60	-	
b1	2L 65X65X6	T	120	-	
c	2L 75X75X8	T	120	-	
c1	L 75X75X8	L	140	-	
d	2L 90X90X8	T	180	-	
j1	1. PL. 150X12 2. PL. 176X8	1 I-2 1	-	-	
P1	IPE 200	I	-	-	
P2	IPE 100	I	-	-	
MR1	1. IPE 240 2. PL. 90X10	I-1 I-2	-	-	
MR2	1. IPE 450 2. PL. 140X12	I-1 I-2	-	-	

ALL GUSSET PLATE ARE 8mm THK
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED
ALL MEMBER CONNECTIONS SHOULD BE DESIGN FOR A MINIMUM FORCE OF 50KN.

LEGEND :-

R.L - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL
F.G.L - FINISHED GROUND LEVEL
F.F.L - FINISHED FLOOR LEVEL
TYP - TYPICAL
B.O.M.R. - BOTTOM OF MONORAIL
MKD - MARKED
THK - THICK
F/S - FAR SIDE
N/S - NEAR SIDE
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
U.N.O. - UNLESS NOTED OTHERWISE
T.O.W.W. - TOP OF WALKWAY

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER U.N.O.
2. GRADE OF STEEL USED IS S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD-216-3130-EG-ST-GAD-1001, 1002 ,1004 & 1005
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1003	CONV. 3130-CNV-001 - ELEVATION								
MD-216-3130-EG-ME-ELV-1002	CONV. 3130-CNV-001 - ELEVATION	D01	01/11/2023	AS BUILT	SUS	SBY	APL	AG	
MD-216-3130-EG-ME-ELV-1001	CONV. 3130-CNV-001 - ELEVATION	C01	23/04/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED △C01	SUS	SBY	APL	AG	
MD-216-3130-EG-ME-LPX-1001	CONV. 3130-CNV-001 - PLAN	A01	20/03/2020	ISSUED FOR REFERENCE	MGH	SBY	APL	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_S_G_26013_015 | REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31363 | DOCUMENT ID: OU100670431 | REVISION: 2

DRAWN	MGH	16/03/2020
CHECKED	SBY	19/03/2020
ENGINEER	APL	18/03/2020
APPROVED	AG	20/03/2020
EMPLOYER		
SCALE	1:100,50,20,10	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRUCTURAL GA FOR 3130-CNV-001, Sh3
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD-216-3130-EG-ST-GAD-1003 | REV: D01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	24-03-2020	25-04-2020	01.11.2023				Date	24-03-2020	25-04-2020	01.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR GP**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 23-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 1
JOB PART GROUND PORTAL
JOB REF CONVEYOR 3130-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0.243 0 0; 2 6.73 0 0; 3 13.217 0 0; 4 0.243 0.361 0; 5 6.73 0.77 0;
6 13.217 1.18 0; 7 0.243 0 5.5; 8 6.73 0 5.5; 9 13.217 0 5.5;
10 0.243 0.361 5.5; 11 6.73 0.77 5.5; 12 13.217 1.18 5.5; 13 0.243 0.361 0.895;
14 0.243 0.361 1.965; 15 0.243 0.361 3.535; 16 0.243 0.361 4.605;
17 6.73 0.77 0.895; 18 13.217 1.18 0.895; 23 17.767 1.85 0.895;
24 6.73 0.77 1.965; 25 13.217 1.18 1.965; 26 17.767 1.85 1.965;
27 6.73 0.77 3.535; 28 13.217 1.18 3.535; 29 17.767 1.85 3.535;
30 6.73 0.77 4.605; 31 13.217 1.18 4.605; 32 17.767 1.85 4.605;
33 -6.193 -0.0449997 0.895; 34 -6.193 -0.0449997 1.965;
35 -6.193 -0.0449997 3.535; 36 -6.193 -0.0449997 4.605; 37 13.217 1.1 0;
38 13.217 1.1 5.5; 39 13.217 1.1 1.5; 40 13.217 1.1 4;
41 2.40533 0.497333 1.965; 42 4.56767 0.633667 1.965;
43 8.89233 0.906667 1.965; 44 11.0547 1.04333 1.965; 45 2.40533 0.497333 3.535;
46 4.56767 0.633667 3.535; 47 8.89233 0.906667 3.535; 48 11.0547 1.04333 3.535;
49 -4.04767 0.0903335 1.965; 50 -1.90233 0.225667 1.965;
51 -4.04767 0.0903335 3.535; 52 -1.90233 0.225667 3.535;
53 -4.04767 0.0903333 0.895; 54 -1.90233 0.225667 0.895;
55 2.40533 0.497333 0.895; 56 4.56767 0.633667 0.895;
57 8.89233 0.906666 0.895; 58 11.0547 1.04334 0.895; 59 11.0547 1.04334 4.605;
60 8.89233 0.906666 4.605; 61 4.56767 0.633667 4.605;
62 2.40533 0.497333 4.605; 63 -1.90233 0.225667 4.605;
64 -4.04767 0.0903333 4.605; 66 -6.193 -0.0449997 0; 68 -6.193 -0.0449997 5.5;
69 -2.975 0.158 0; 70 3.4865 0.5655 0; 71 9.9735 0.975 0; 72 -2.975 0.158 5.5;
73 3.4865 0.5655 5.5; 74 9.9735 0.975 5.5; 75 -2.975 0.158 0.895;
76 -2.975 0.158 4.605; 77 3.4865 0.5655 0.895; 78 3.4865 0.5655 4.605;
79 9.9735 0.975002 0.895; 80 9.9735 0.975002 4.605; 81 -6.193 -1.35 0;
82 -6.193 -1.35 5.5; 83 0.243 -1.35 0; 84 0.243 -1.35 5.5; 85 6.73 -1.35 0;
86 6.73 -1.35 5.5; 87 13.217 -1.35 0; 88 13.217 -1.35 5.5;
89 -4.19697 0.0809152 0; 90 -4.19697 0.0809152 5.5; 91 -4.19618 -0.0310382 0;
92 -4.19618 -0.0310382 5.5;
MEMBER INCIDENCES
1 1 4; 2 2 5; 3 3 37; 4 7 10; 5 8 11; 6 9 38; 7 4 13; 8 5 17; 9 6 18; 10 13 14;
11 14 15; 12 15 16; 13 16 10; 14 17 24; 15 13 55; 16 18 25; 17 17 57; 22 18 23;
23 24 27; 24 14 41; 25 25 28; 26 24 43; 28 25 26; 29 27 30; 30 15 45; 31 28 31;
32 27 47; 34 28 29; 35 30 11; 36 16 62; 37 31 12; 38 30 60; 40 31 32; 41 33 53;
42 34 49; 43 35 51; 44 36 64; 45 3 39; 46 9 40; 47 37 6; 48 38 12; 49 37 39;
50 39 40; 51 40 38; 52 41 42; 53 42 24; 54 43 44; 55 44 25; 56 45 46; 57 46 27;
58 47 48; 59 48 28; 60 49 50; 61 50 14; 62 51 52; 63 52 15; 64 35 49; 65 49 52;
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66 52 14; 67 14 45; 68 45 42; 69 42 27; 70 27 43; 71 43 48; 72 48 25; 73 51 49;
74 52 50; 75 45 41; 76 46 42; 77 47 43; 78 48 44; 79 53 75; 80 49 53; 81 54 13;
82 50 54; 83 55 77; 84 41 55; 85 56 17; 86 42 56; 87 57 79; 88 43 57; 89 58 18;
90 44 58; 91 59 31; 92 48 59; 93 60 80; 94 47 60; 95 61 30; 96 46 61; 97 62 78;
98 45 62; 99 63 16; 100 52 63; 101 64 76; 102 51 64; 104 4 70; 105 5 71;
108 10 73; 109 11 74; 110 69 4; 111 70 5; 112 71 6; 113 72 10; 114 73 11;
115 74 12; 116 75 54; 117 69 75; 118 76 63; 119 72 76; 120 77 56; 121 70 77;
122 78 61; 123 73 78; 124 79 58; 125 71 79; 126 80 59; 127 74 80; 128 81 66;
129 82 68; 130 83 1; 131 84 7; 132 85 2; 133 86 8; 134 87 3; 135 88 9; 136 1 7;
137 2 8; 138 3 9; 139 66 33; 140 33 34; 141 34 35; 142 35 36; 143 36 68;
146 89 69; 147 90 72; 149 89 91; 151 90 92; 161 66 91; 162 91 1; 163 68 92;
164 92 7;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
1 TO 8 10 TO 14 23 29 35 47 48 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
9 16 25 31 37 TAPERED 0.25 0.008 0.25 0.2 0.016 0.2 0.016
24 26 28 30 32 34 42 43 52 TO 63 TABLE ST UPN200
15 17 22 36 38 40 41 44 79 81 83 85 87 89 91 93 95 97 99 101 116 118 120 122 -
124 126 TABLE ST UPN140
104 105 108 TO 115 146 147 149 151 TABLE ST UPN140
117 119 121 123 125 127 TABLE ST IPE200
49 TO 51 TABLE LD L90X90X8 SP 0.01
45 46 TABLE LD L65X65X6 SP 0.01
64 TO 72 TABLE ST L65X65X6
73 TO 78 80 82 84 86 88 90 92 94 96 98 100 102 TABLE ST L50X50X6
MEMBER PROPERTY EUROPEAN
128 TO 135 PRIS YD 0.6 ZD 0.6
139 TO 143 PRIS YD 0.6 ZD 0.35
136 TO 138 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY EUROPEAN
161 TO 164 PRIS YD 0.6 ZD 0.3
CONSTANTS
BETA 90 MEMB 1 TO 6 47 48
MATERIAL STEEL MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 102 104 105 108 -
109 TO 127 146 147 149 151
MATERIAL CONCRETE MEMB 128 TO 143 161 TO 164
SUPPORTS
```

```
23 26 29 32 FIXED BUT FX MX MY MZ
81 TO 88 FIXED
MEMBER TRUSS
45 46 64 TO 78 80 82 84 86 88 90 92 94 96 98 100 102
MEMBER RELEASE
9 41 TO 44 49 104 105 108 109 117 119 121 123 125 127 START MP 0.99
37 51 55 59 89 91 110 TO 115 117 119 121 123 125 127 149 151 END MP 0.99
1 TO 6 22 28 34 40 START MY MZ
*********************************************************************************
MEMBER OFFSET
139 TO 143 START 0 -0.3 0
139 TO 143 END 0 -0.3 0
136 TO 138 START 0 -0.25 0
136 TO 138 END 0 -0.25 0
161 TO 164 START 0 -0.3 0
161 TO 164 END 0 -0.3 0
LOAD 1 DL
SELFWEIGHT Y -1.1
*WALKWAY LOAD @30KG/SQM INCLUDING SEAL PLATE LOAD @ 30KG/SQM & HANDRAIL LOAD @15KG/M
*WALKWAY WIDTH 1000MM & BELT WIDTH 1200MM
MEMBER LOAD
15 17 22 36 38 40 41 44 79 81 83 85 87 89 91 93 95 97 99 101 116 118 120 122 -
124 126 UNI GY -0.32
24 26 28 30 32 34 42 43 52 TO 63 UNI GY -0.47
*PIPE DEAD LOAD
MEMBER LOAD
7 TO 9 CON GY -2.03
13 35 37 CON GY -3.57
*LIGHTING POLE LOAD = 100KG/SUPPORT POINT(ASSUMED)
7 TO 9 CON GY -1 0.695
13 35 37 CON GY -1 0.2
*CONVEYOR DEAD LOAD VERTICAL
MEMBER LOAD
42 43 CON GY -5 0.4
62 CON GY -5 0.400402
60 CON GY -5 0.400402
63 CON GY -5 1.2508
61 CON GY -5 1.2508
24 30 CON GY -5 2.07
57 CON GY -5 0.736746
53 CON GY -5 0.736746
26 32 CON GY -5 1.6
59 CON GY -5 0.266704
55 CON GY -5 0.266704
28 34 CON GY -5 1
28 34 CON GY -5 4
*CONVEYOR DEAD LOAD HORIZONTAL ALONG LENGTH
MEMBER LOAD
42 43 CON GX 1.5 0.4
62 CON GX 1.5 0.400402
60 CON GX 1.5 0.400402
63 CON GX 1.5 1.2508
61 CON GX 1.5 1.2508
56 CON GX 1.5 0.203373
```

```
52 CON GX 1.5 0.203373
57 CON GX 1.5 1.03675
53 CON GX 1.5 1.03675
26 32 CON GX 1.5 2.1
59 CON GX 1.5 0.766704
55 CON GX 1.5 0.766704
28 34 CON GX 1.5 1
28 34 CON GX 1.5 4
*CONVEYOR DEAD LOAD HORIZONTAL ACROSS LENGTH
MEMBER LOAD
42 43 CON GZ 0.8 0.4
62 CON GZ 0.8 0.400402
60 CON GZ 0.8 0.400402
63 CON GZ 0.8 1.2508
61 CON GZ 0.8 1.2508
56 CON GZ 0.8 0.203373
52 CON GZ 0.8 0.203373
57 CON GZ 0.8 1.03675
53 CON GZ 0.8 1.03675
26 32 CON GZ 0.8 2.1
59 CON GZ 0.8 0.766704
55 CON GZ 0.8 0.766704
28 34 CON GZ 0.8 1
28 34 CON GZ 0.8 4
*********************************************************************************
LOAD 2 LL
*WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
MEMBER LOAD
15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 85 87 89 91 93 -
95 97 99 101 116 118 120 122 124 126 UNI GY -2
*********************************************************************************
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
MEMBER LOAD
42 43 CON GY -5.5 0.4
62 CON GY -5.5 0.400402
60 CON GY -5.5 0.400402
63 CON GY -5.5 1.2508
61 CON GY -5.5 1.2508
24 30 CON GY -5.5 2.07
57 CON GY -5.5 0.736746
53 CON GY -5.5 0.736746
26 32 CON GY -5.5 1.6
59 CON GY -5.5 0.266704
55 CON GY -5.5 0.266704
28 34 CON GY -5.5 1
28 34 CON GY -5.5 4
** Spillage Load @75kg/ Sq. m on seal plate
24 26 28 30 32 34 42 43 52 TO 63 UNI GY -0.64
*********************************************************************************
LOAD 4 WIND LOAD IN +Z DIRN
MEMBER LOAD
42 43 CON GZ 2.58 0.4
62 CON GZ 2.58 0.400402
```

```
60 CON GZ 2.58 0.400402
63 CON GZ 2.58 1.2508
61 CON GZ 2.58 1.2508
24 30 CON GZ 2.58 2.07
57 CON GZ 2.58 0.736746
53 CON GZ 2.58 0.736746
26 32 CON GZ 2.58 1.6
59 CON GZ 2.58 0.266704
55 CON GZ 2.58 0.266704
28 34 CON GZ 2.58 1
28 34 CON GZ 2.58 4
**
42 43 CMOM GX 2.66 0.4
62 CMOM GX 2.66 0.400402
60 CMOM GX 2.66 0.400402
63 CMOM GX 2.66 1.2508
61 CMOM GX 2.66 1.2508
24 30 CMOM GX 2.66 2.07
57 CMOM GX 2.66 0.736746
53 CMOM GX 2.66 0.736746
26 32 CMOM GX 2.66 1.6
59 CMOM GX 2.66 0.266704
55 CMOM GX 2.66 0.266704
28 34 CMOM GX 2.66 1
28 34 CMOM GX 2.66 4
*42 43 CON GZ 1.35 0.4
*62 CON GZ 1.35 0.400402
*60 CON GZ 1.35 0.400402
*63 CON GZ 1.35 1.2508
*61 CON GZ 1.35 1.2508
*24 30 CON GZ 1.35 2.07
*57 CON GZ 1.35 0.736746
*53 CON GZ 1.35 0.736746
*26 32 CON GZ 1.35 1.6
*59 CON GZ 1.35 0.266704
*55 CON GZ 1.35 0.266704
*28 34 CON GZ 1.35 1
*28 34 CON GZ 1.35 4
*TRESTLE BODY WIND
1 TO 6 47 48 UNI GZ 1.2
*******************************************************************************
LOAD 5 WIND LOAD IN -Z DIRN
MEMBER LOAD
42 43 CON GZ -2.58 0.4
62 CON GZ -2.58 0.400402
60 CON GZ -2.58 0.400402
63 CON GZ -2.58 1.2508
61 CON GZ -2.58 1.2508
24 30 CON GZ -2.58 2.07
57 CON GZ -2.58 0.736746
53 CON GZ -2.58 0.736746
26 32 CON GZ -2.58 1.6
59 CON GZ -2.58 0.266704
55 CON GZ -2.58 0.266704
```

```
28 34 CON GZ -2.58 1
28 34 CON GZ -2.58 4
**
42 43 CMOM GX -2.66 0.4
62 CMOM GX -2.66 0.400402
60 CMOM GX -2.66 0.400402
63 CMOM GX -2.66 1.2508
61 CMOM GX -2.66 1.2508
24 30 CMOM GX -2.66 2.07
57 CMOM GX -2.66 0.736746
53 CMOM GX -2.66 0.736746
26 32 CMOM GX -2.66 1.6
59 CMOM GX -2.66 0.266704
55 CMOM GX -2.66 0.266704
28 34 CMOM GX -2.66 1
28 34 CMOM GX -2.66 4
**
*42 43 CON GZ -1.35 0.4
*62 CON GZ -1.35 0.400402
*60 CON GZ -1.35 0.400402
*63 CON GZ -1.35 1.2508
*61 CON GZ -1.35 1.2508
*24 30 CON GZ -1.35 2.07
*57 CON GZ -1.35 0.736746
*53 CON GZ -1.35 0.736746
*26 32 CON GZ -1.35 1.6
*59 CON GZ -1.35 0.266704
*55 CON GZ -1.35 0.266704
*28 34 CON GZ -1.35 1
*28 34 CON GZ -1.35 4
*TRESTLE BODY WIND
1 TO 6 47 48 UNI GZ -1.2
*********************************************************************************
LOAD 6 TL1
TEMPERATURE LOAD
1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 102 104 105 108 TO 127 TEMP 25
*********************************************************************************
LOAD 7 TL2
TEMPERATURE LOAD
1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 102 104 105 108 TO 127 TEMP -25
*********************************************************************************
LOAD 11 G
REPEAT LOAD
1 1.0
LOAD 12 G + QN + MN
REPEAT LOAD
1 1.0 2 1.0 3 1.0
*********************************************************************************
****LOAD COMBINATION for structural design
*LOAD COMB 201 1.4 DL
*1 1.4
*LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 6 1.2 2 1.6 3 1.6
*LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
```

```
*1 1.2 7 1.2 2 1.6 3 1.6
*LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
*1 1.2 4 1.6 2 0.5 3 0.5
*LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
*1 1.2 5 1.6 2 0.5 3 0.5
*LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
*1 0.9 4 1.6
*LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
*1 0.9 5 1.6
*******************************************************************************
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
*******************************************************************************
PERFORM ANALYSIS
*STEEL TAKE OFF LIST 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
*LOAD LIST 201 TO 207
*PARAMETER 1
*CODE LRFD
*FYLD 275000 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
*FU 410000 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
*TRACK 0 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
*BEAM 1 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
*TMAIN 300 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
*MAIN 200 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
*NSF 0.8 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
*KY 0.85 MEMB 45 46 64 TO 72
```

```
*KZ 0.85 MEMB 45 46 64 TO 72
*LZ 5.5 MEMB 7 TO 14 16 23 25 29 31 35 37
*UNT 1.57 MEMB 7 TO 14 16 23 25 29 31 35 37
*LY 1.57 MEMB 7 TO 14 16 23 25 29 31 35 37
*LX 1.57 MEMB 7 TO 14 16 23 25 29 31 35 37
*LY 5.5 MEMB 49 TO 51
*LY 2.17 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 85 -
*87 89 91 93 95 97 99 101 116 118 120 122 124 126
*LY 3 MEMB 103 TO 105 107 TO 115
*LX 2.17 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 85 -
*87 89 91 93 95 97 99 101 116 118 120 122 124 126
*LX 3 MEMB 103 TO 105 107 TO 115
*UNT 2.17 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 -
*85 87 89 91 93 95 97 99 101 116 118 120 122 124 126
*UNT 3 MEMB 103 TO 105 107 TO 115
*UNB 2.17 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 -
*85 87 89 91 93 95 97 99 101 116 118 120 122 124 126
*UNB 3 MEMB 103 TO 105 107 TO 115
*LZ 6.5 MEMB 15 17 24 26 30 32 36 38 41 TO 44 52 TO 63 79 81 83 85 87 89 91 -
*93 95 97 99 101 103 TO 105 107 TO 116 118 120 122 124 126
*LZ 4.5 MEMB 22 28 34 40
*CHECK CODE MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
LOAD LIST 401 TO 407
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 128 TO 143 161 TO 164
FYMAIN 420000 MEMB 128 TO 143 161 TO 164
TRACK 1 MEMB 128 TO 143 161 TO 164
CLB 0.05 MEMB 128 TO 143 161 TO 164
CLT 0.05 MEMB 128 TO 143 161 TO 164
CLS 0.05 MEMB 128 TO 143 161 TO 164
MAXMAIN 20 MEMB 128 TO 135
MAXMAIN 16 MEMB 136 TO 143 161 TO 164
MINMAIN 20 MEMB 128 TO 135
MINMAIN 16 MEMB 136 TO 143 161 TO 164
MINSEC 8 MEMB 128 TO 143 161 TO 164
REINF 0 MEMB 128 TO 135
DESIGN COLUMN 128 TO 135
DESIGN BEAM 136 TO 143 161 TO 164
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 24, 2020                *
*           Time=    13:55:10                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3130-CNV-001\Staad\Civil\3130-CNV-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 23-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 1
    8. JOB PART GROUND PORTAL
    9. JOB REF CONVEYOR 3130-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0.243 0 0; 2 6.73 0 0; 3 13.217 0 0; 4 0.243 0.361 0; 5 6.73 0.77 0
   18. 6 13.217 1.18 0; 7 0.243 0 5.5; 8 6.73 0 5.5; 9 13.217 0 5.5
   19. 10 0.243 0.361 5.5; 11 6.73 0.77 5.5; 12 13.217 1.18 5.5; 13 0.243 0.361 0.895
   20. 14 0.243 0.361 1.965; 15 0.243 0.361 3.535; 16 0.243 0.361 4.605
   21. 17 6.73 0.77 0.895; 18 13.217 1.18 0.895; 23 17.767 1.85 0.895
   22. 24 6.73 0.77 1.965; 25 13.217 1.18 1.965; 26 17.767 1.85 1.965
   23. 27 6.73 0.77 3.535; 28 13.217 1.18 3.535; 29 17.767 1.85 3.535
   24. 30 6.73 0.77 4.605; 31 13.217 1.18 4.605; 32 17.767 1.85 4.605
   25. 33 -6.193 -0.0449997 0.895; 34 -6.193 -0.0449997 1.965
   26. 35 -6.193 -0.0449997 3.535; 36 -6.193 -0.0449997 4.605; 37 13.217 1.1 0
   27. 38 13.217 1.1 5.5; 39 13.217 1.1 1.5; 40 13.217 1.1 4
   28. 41 2.40533 0.497333 1.965; 42 4.56767 0.633667 1.965
   29. 43 8.89233 0.906667 1.965; 44 11.0547 1.04333 1.965; 45 2.40533 0.497333 3.535
   30. 46 4.56767 0.633667 3.535; 47 8.89233 0.906667 3.535; 48 11.0547 1.04333 3.535
   31. 49 -4.04767 0.0903335 1.965; 50 -1.90233 0.225667 1.965
   32. 51 -4.04767 0.0903335 3.535; 52 -1.90233 0.225667 3.535
   33. 53 -4.04767 0.0903333 0.895; 54 -1.90233 0.225667 0.895
   34. 55 2.40533 0.497333 0.895; 56 4.56767 0.633667 0.895
   35. 57 8.89233 0.906666 0.895; 58 11.0547 1.04334 0.895; 59 11.0547 1.04334 4.605
   36. 60 8.89233 0.906666 4.605; 61 4.56767 0.633667 4.605
   37. 62 2.40533 0.497333 4.605; 63 -1.90233 0.225667 4.605
   38. 64 -4.04767 0.0903333 4.605; 66 -6.193 -0.0449997 0; 68 -6.193 -0.0449997 5.5
```

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39. 69 -2.975 0.158 0; 70 3.4865 0.5655 0; 71 9.9735 0.975 0; 72 -2.975 0.158 5.5
40. 73 3.4865 0.5655 5.5; 74 9.9735 0.975 5.5; 75 -2.975 0.158 0.895
41. 76 -2.975 0.158 4.605; 77 3.4865 0.5655 0.895; 78 3.4865 0.5655 4.605
42. 79 9.9735 0.975002 0.895; 80 9.9735 0.975002 4.605; 81 -6.193 -1.35 0
43. 82 -6.193 -1.35 5.5; 83 0.243 -1.35 0; 84 0.243 -1.35 5.5; 85 6.73 -1.35 0
44. 86 6.73 -1.35 5.5; 87 13.217 -1.35 0; 88 13.217 -1.35 5.5
45. 89 -4.19697 0.0809152 0; 90 -4.19697 0.0809152 5.5; 91 -4.19618 -0.0310382 0
46. 92 -4.19618 -0.0310382 5.5
47. MEMBER INCIDENCES
48. 1 1 4; 2 2 5; 3 3 37; 4 7 10; 5 8 11; 6 9 38; 7 4 13; 8 5 17; 9 6 18; 10 13 14
49. 11 14 15; 12 15 16; 13 16 10; 14 17 24; 15 13 55; 16 18 25; 17 17 57; 22 18 23
50. 23 24 27; 24 14 41; 25 25 28; 26 24 43; 28 25 26; 29 27 30; 30 15 45; 31 28 31
51. 32 27 47; 34 28 29; 35 30 11; 36 16 62; 37 31 12; 38 30 60; 40 31 32; 41 33 53
52. 42 34 49; 43 35 51; 44 36 64; 45 3 39; 46 9 40; 47 37 6; 48 38 12; 49 37 39
53. 50 39 40; 51 40 38; 52 41 42; 53 42 24; 54 43 44; 55 44 25; 56 45 46; 57 46 27
54. 58 47 48; 59 48 28; 60 49 50; 61 50 14; 62 51 52; 63 52 15; 64 35 49; 65 49 52
55. 66 52 14; 67 14 45; 68 45 42; 69 42 27; 70 27 43; 71 43 48; 72 48 25; 73 51 49
56. 74 52 50; 75 45 41; 76 46 42; 77 47 43; 78 48 44; 79 53 75; 80 49 53; 81 54 13
57. 82 50 54; 83 55 77; 84 41 55; 85 56 17; 86 42 56; 87 57 79; 88 43 57; 89 58 18
58. 90 44 58; 91 59 31; 92 48 59; 93 60 80; 94 47 60; 95 61 30; 96 46 61; 97 62 78
59. 98 45 62; 99 63 16; 100 52 63; 101 64 76; 102 51 64; 104 4 70; 105 5 71
60. 108 10 73; 109 11 74; 110 69 4; 111 70 5; 112 71 6; 113 72 10; 114 73 11
61. 115 74 12; 116 75 54; 117 69 75; 118 76 63; 119 72 76; 120 77 56; 121 70 77
62. 122 78 61; 123 73 78; 124 79 58; 125 71 79; 126 80 59; 127 74 80; 128 81 66
63. 129 82 68; 130 83 1; 131 84 7; 132 85 2; 133 86 8; 134 87 3; 135 88 9; 136 1 7
64. 137 2 8; 138 3 9; 139 66 33; 140 33 34; 141 34 35; 142 35 36; 143 36 68
65. 146 89 69; 147 90 72; 149 89 91; 151 90 92; 161 66 91; 162 91 1; 163 68 92
66. 164 92 7
67. DEFINE MATERIAL START
68. ISOTROPIC STEEL
69. E 2.05E+008
70. POISSON 0.3
71. DENSITY 76.8195
72. ALPHA 1.2E-005
73. DAMP 0.03
74. TYPE STEEL
75. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
76. ISOTROPIC CONCRETE
77. E 2.17185E+007
78. POISSON 0.17
79. DENSITY 23.5616
80. ALPHA 1E-005
81. DAMP 0.05
82. TYPE CONCRETE
83. STRENGTH FCU 27579
84. END DEFINE MATERIAL
85. MEMBER PROPERTY EUROPEAN
86. 1 TO 8 10 TO 14 23 29 35 47 48 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
87. 9 16 25 31 37 TAPERED 0.25 0.008 0.25 0.2 0.016 0.2 0.016
88. 24 26 28 30 32 34 42 43 52 TO 63 TABLE ST UPN200
89. 15 17 22 36 38 40 41 44 79 81 83 85 87 89 91 93 95 97 99 101 116 118 120 122 -
90. 124 126 TABLE ST UPN140
91. 104 105 108 TO 115 146 147 149 151 TABLE ST UPN140
92. 117 119 121 123 125 127 TABLE ST IPE200
93. 49 TO 51 TABLE LD L90X90X8 SP 0.01
94. 45 46 TABLE LD L65X65X6 SP 0.01
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 95. 64 TO 72 TABLE ST L65X65X6
 96. 73 TO 78 80 82 84 86 88 90 92 94 96 98 100 102 TABLE ST L50X50X6
 97. MEMBER PROPERTY EUROPEAN
 98. 128 TO 135 PRIS YD 0.6 ZD 0.6
 99. 139 TO 143 PRIS YD 0.6 ZD 0.35
100. 136 TO 138 PRIS YD 0.5 ZD 0.3
101. MEMBER PROPERTY EUROPEAN
102. 161 TO 164 PRIS YD 0.6 ZD 0.3
103. CONSTANTS
104. BETA 90 MEMB 1 TO 6 47 48
105. MATERIAL STEEL MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 102 104 105 108 -
106. 109 TO 127 146 147 149 151
107. MATERIAL CONCRETE MEMB 128 TO 143 161 TO 164
108. SUPPORTS
109. 23 26 29 32 FIXED BUT FX MX MY MZ
110. 81 TO 88 FIXED
111. MEMBER TRUSS
112. 45 46 64 TO 78 80 82 84 86 88 90 92 94 96 98 100 102
113. MEMBER RELEASE
114. 9 41 TO 44 49 104 105 108 109 117 119 121 123 125 127 START MP 0.99
115. 37 51 55 59 89 91 110 TO 115 117 119 121 123 125 127 149 151 END MP 0.99
116. 1 TO 6 22 28 34 40 START MY MZ
117. ********************************************************************************
118. MEMBER OFFSET
119. 139 TO 143 START 0 -0.3 0
120. 139 TO 143 END 0 -0.3 0
121. 136 TO 138 START 0 -0.25 0
122. 136 TO 138 END 0 -0.25 0
123. 161 TO 164 START 0 -0.3 0
124. 161 TO 164 END 0 -0.3 0
125. LOAD 1 DL
126. SELFWEIGHT Y -1.1
127. *WALKWAY LOAD @30KG/SQM INCLUDING SEAL PLATE LOAD @ 30KG/SQM _HANDRAIL LOAD @1
128. *WALKWAY WIDTH 1000MM _BELT WIDTH 1200MM
129. MEMBER LOAD
130. 15 17 22 36 38 40 41 44 79 81 83 85 87 89 91 93 95 97 99 101 116 118 120 122 -
131. 124 126 UNI GY -0.32
132. 24 26 28 30 32 34 42 43 52 TO 63 UNI GY -0.47
133. *PIPE DEAD LOAD
134. MEMBER LOAD
135. 7 TO 9 CON GY -2.03
136. 13 35 37 CON GY -3.57
137. *LIGHTING POLE LOAD = 100KG/SUPPORT POINT(ASSUMED)
138. 7 TO 9 CON GY -1 0.695
139. 13 35 37 CON GY -1 0.2
140. *CONVEYOR DEAD LOAD VERTICAL
141. MEMBER LOAD
142. 42 43 CON GY -5 0.4
143. 62 CON GY -5 0.400402
144. 60 CON GY -5 0.400402
145. 63 CON GY -5 1.2508
146. 61 CON GY -5 1.2508
147. 24 30 CON GY -5 2.07
148. 57 CON GY -5 0.736746
149. 53 CON GY -5 0.736746
150. 26 32 CON GY -5 1.6
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151. 59 CON GY -5 0.266704
152. 55 CON GY -5 0.266704
153. 28 34 CON GY -5 1
154. 28 34 CON GY -5 4
155. *CONVEYOR DEAD LOAD HORIZONTAL ALONG LENGTH
156. MEMBER LOAD
157. 42 43 CON GX 1.5 0.4
158. 62 CON GX 1.5 0.400402
159. 60 CON GX 1.5 0.400402
160. 63 CON GX 1.5 1.2508
161. 61 CON GX 1.5 1.2508
162. 56 CON GX 1.5 0.203373
163. 52 CON GX 1.5 0.203373
164. 57 CON GX 1.5 1.03675
165. 53 CON GX 1.5 1.03675
166. 26 32 CON GX 1.5 2.1
167. 59 CON GX 1.5 0.766704
168. 55 CON GX 1.5 0.766704
169. 28 34 CON GX 1.5 1
170. 28 34 CON GX 1.5 4
171. *CONVEYOR DEAD LOAD HORIZONTAL ACROSS LENGTH
172. MEMBER LOAD
173. 42 43 CON GZ 0.8 0.4
174. 62 CON GZ 0.8 0.400402
175. 60 CON GZ 0.8 0.400402
176. 63 CON GZ 0.8 1.2508
177. 61 CON GZ 0.8 1.2508
178. 56 CON GZ 0.8 0.203373
179. 52 CON GZ 0.8 0.203373
180. 57 CON GZ 0.8 1.03675
181. 53 CON GZ 0.8 1.03675
182. 26 32 CON GZ 0.8 2.1
183. 59 CON GZ 0.8 0.766704
184. 55 CON GZ 0.8 0.766704
185. 28 34 CON GZ 0.8 1
186. 28 34 CON GZ 0.8 4
187. ******************************************************************************
188. LOAD 2 LL
189. *WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
190. MEMBER LOAD
191. 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 85 87 89 91 93 -
192. 95 97 99 101 116 118 120 122 124 126 UNI GY -2
193. ******************************************************************************
194. LOAD 3 ML
195. *CONVEYOR MATERIAL LOAD
196. MEMBER LOAD
197. 42 43 CON GY -5.5 0.4
198. 62 CON GY -5.5 0.400402
199. 60 CON GY -5.5 0.400402
200. 63 CON GY -5.5 1.2508
201. 61 CON GY -5.5 1.2508
202. 24 30 CON GY -5.5 2.07
203. 57 CON GY -5.5 0.736746
204. 53 CON GY -5.5 0.736746
205. 26 32 CON GY -5.5 1.6
206. 59 CON GY -5.5 0.266704
```

```
207. 55 CON GY -5.5 0.266704
208. 28 34 CON GY -5.5 1
209. 28 34 CON GY -5.5 4
210. ** SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
211. 24 26 28 30 32 34 42 43 52 TO 63 UNI GY -0.64
212. ********************************************************************************
213. LOAD 4 WIND LOAD IN +Z DIRN
214. MEMBER LOAD
215. 42 43 CON GZ 2.58 0.4
216. 62 CON GZ 2.58 0.400402
217. 60 CON GZ 2.58 0.400402
218. 63 CON GZ 2.58 1.2508
219. 61 CON GZ 2.58 1.2508
220. 24 30 CON GZ 2.58 2.07
221. 57 CON GZ 2.58 0.736746
222. 53 CON GZ 2.58 0.736746
223. 26 32 CON GZ 2.58 1.6
224. 59 CON GZ 2.58 0.266704
225. 55 CON GZ 2.58 0.266704
226. 28 34 CON GZ 2.58 1
227. 28 34 CON GZ 2.58 4
228. **
229. 42 43 CMOM GX 2.66 0.4
230. 62 CMOM GX 2.66 0.400402
231. 60 CMOM GX 2.66 0.400402
232. 63 CMOM GX 2.66 1.2508
233. 61 CMOM GX 2.66 1.2508
234. 24 30 CMOM GX 2.66 2.07
235. 57 CMOM GX 2.66 0.736746
236. 53 CMOM GX 2.66 0.736746
237. 26 32 CMOM GX 2.66 1.6
238. 59 CMOM GX 2.66 0.266704
239. 55 CMOM GX 2.66 0.266704
240. 28 34 CMOM GX 2.66 1
241. 28 34 CMOM GX 2.66 4
242. *42 43 CON GZ 1.35 0.4
243. *62 CON GZ 1.35 0.400402
244. *60 CON GZ 1.35 0.400402
245. *63 CON GZ 1.35 1.2508
246. *61 CON GZ 1.35 1.2508
247. *24 30 CON GZ 1.35 2.07
248. *57 CON GZ 1.35 0.736746
249. *53 CON GZ 1.35 0.736746
250. *26 32 CON GZ 1.35 1.6
251. *59 CON GZ 1.35 0.266704
252. *55 CON GZ 1.35 0.266704
253. *28 34 CON GZ 1.35 1
254. *28 34 CON GZ 1.35 4
255. *TRESTLE BODY WIND
256. 1 TO 6 47 48 UNI GZ 1.2
257. ********************************************************************************
258. LOAD 5 WIND LOAD IN -Z DIRN
259. MEMBER LOAD
260. 42 43 CON GZ -2.58 0.4
261. 62 CON GZ -2.58 0.400402
262. 60 CON GZ -2.58 0.400402
```

```
263. 63 CON GZ -2.58 1.2508
264. 61 CON GZ -2.58 1.2508
265. 24 30 CON GZ -2.58 2.07
266. 57 CON GZ -2.58 0.736746
267. 53 CON GZ -2.58 0.736746
268. 26 32 CON GZ -2.58 1.6
269. 59 CON GZ -2.58 0.266704
270. 55 CON GZ -2.58 0.266704
271. 28 34 CON GZ -2.58 1
272. 28 34 CON GZ -2.58 4
273. **
274. 42 43 CMOM GX -2.66 0.4
275. 62 CMOM GX -2.66 0.400402
276. 60 CMOM GX -2.66 0.400402
277. 63 CMOM GX -2.66 1.2508
278. 61 CMOM GX -2.66 1.2508
279. 24 30 CMOM GX -2.66 2.07
280. 57 CMOM GX -2.66 0.736746
281. 53 CMOM GX -2.66 0.736746
282. 26 32 CMOM GX -2.66 1.6
283. 59 CMOM GX -2.66 0.266704
284. 55 CMOM GX -2.66 0.266704
285. 28 34 CMOM GX -2.66 1
286. 28 34 CMOM GX -2.66 4
287. **
288. *42 43 CON GZ -1.35 0.4
289. *62 CON GZ -1.35 0.400402
290. *60 CON GZ -1.35 0.400402
291. *63 CON GZ -1.35 1.2508
292. *61 CON GZ -1.35 1.2508
293. *24 30 CON GZ -1.35 2.07
294. *57 CON GZ -1.35 0.736746
295. *53 CON GZ -1.35 0.736746
296. *26 32 CON GZ -1.35 1.6
297. *59 CON GZ -1.35 0.266704
298. *55 CON GZ -1.35 0.266704
299. *28 34 CON GZ -1.35 1
300. *28 34 CON GZ -1.35 4
301. *TRESTLE BODY WIND
302. 1 TO 6 47 48 UNI GZ -1.2
303. ********************************************************************************
304. LOAD 6 TL1
305. TEMPERATURE LOAD
306. 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 102 104 105 108 TO 127 TEMP 25
307. ********************************************************************************
308. LOAD 7 TL2
309. TEMPERATURE LOAD
310. 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 102 104 105 108 TO 127 TEMP -25
311. ********************************************************************************
312. LOAD 11 G
313. REPEAT LOAD
314. 1 1.0
315. LOAD 12 G + QN + MN
316. REPEAT LOAD
317. 1 1.0 2 1.0 3 1.0
318. ********************************************************************************
```

```
****LOAD COMBINATION FOR STRUCTURAL DESIGN
*LOAD COMB 201 1.4 DL
*1 1.4
*LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 6 1.2 2 1.6 3 1.6
*LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
*1 1.2 7 1.2 2 1.6 3 1.6
*LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
*1 1.2 4 1.6 2 0.5 3 0.5
*LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
*1 1.2 5 1.6 2 0.5 3 0.5
*LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
*1 0.9 4 1.6
*LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
*1 0.9 5 1.6
********************************************************************************
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
********************************************************************************
PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 86 NUMBER OF MEMBERS 141
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 12

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 84/ 11/ 72 DOF
TOTAL PRIMARY LOAD CASES = 9, TOTAL DEGREES OF FREEDOM = 460
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 34 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.6/ 49933.6 MB

```
367. *STEEL TAKE OFF LIST 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
368. *LOAD LIST 201 TO 207
369. *PARAMETER 1
370. *CODE LRFD
371. *FYLD 275000 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
372. *FU 410000 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
373. *TRACK 0 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
374. *BEAM 1 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
375. *TMAIN 300 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
376. *MAIN 200 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
377. *NSF 0.8 MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
378. *KY 0.85 MEMB 45 46 64 TO 72
379. *KZ 0.85 MEMB 45 46 64 TO 72
380. *LZ 5.5 MEMB 7 TO 14 16 23 25 29 31 35 37
381. *UNT 1.57 MEMB 7 TO 14 16 23 25 29 31 35 37
382. *LY 1.57 MEMB 7 TO 14 16 23 25 29 31 35 37
383. *LX 1.57 MEMB 7 TO 14 16 23 25 29 31 35 37
384. *LY 5.5 MEMB 49 TO 51
385. *LY 2.17 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 85 -
386. *87 89 91 93 95 97 99 101 116 118 120 122 124 126
387. *LY 3 MEMB 103 TO 105 107 TO 115
388. *LX 2.17 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 85 -
389. *87 89 91 93 95 97 99 101 116 118 120 122 124 126
390. *LX 3 MEMB 103 TO 105 107 TO 115
391. *UNT 2.17 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 -
392. *85 87 89 91 93 95 97 99 101 116 118 120 122 124 126
393. *UNT 3 MEMB 103 TO 105 107 TO 115
394. *UNB 2.17 MEMB 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 52 TO 63 79 81 83 -
395. *85 87 89 91 93 95 97 99 101 116 118 120 122 124 126
396. *UNB 3 MEMB 103 TO 105 107 TO 115
397. *LZ 6.5 MEMB 15 17 24 26 30 32 36 38 41 TO 44 52 TO 63 79 81 83 85 87 89 91 -
398. *93 95 97 99 101 103 TO 105 107 TO 116 118 120 122 124 126
399. *LZ 4.5 MEMB 22 28 34 40
400. *CHECK CODE MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 105 107 TO 127
401. LOAD LIST 401 TO 407
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402. START CONCRETE DESIGN
403. CODE ACI 2008
404. FC 35000 MEMB 128 TO 143 161 TO 164
405. FYMAIN 420000 MEMB 128 TO 143 161 TO 164
406. TRACK 1 MEMB 128 TO 143 161 TO 164
407. CLB 0.05 MEMB 128 TO 143 161 TO 164
408. CLT 0.05 MEMB 128 TO 143 161 TO 164
409. CLS 0.05 MEMB 128 TO 143 161 TO 164
410. MAXMAIN 20 MEMB 128 TO 135
411. MAXMAIN 16 MEMB 136 TO 143 161 TO 164
412. MINMAIN 20 MEMB 128 TO 135
413. MINMAIN 16 MEMB 136 TO 143 161 TO 164
414. MINSEC 8 MEMB 128 TO 143 161 TO 164
415. REINF 0 MEMB 128 TO 135
416. DESIGN COLUMN 128 TO 135
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COLUMN NO. 128 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	85.52	37.59	0.33683

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	85.52	53.88	0.48280

==========

COLUMN NO. 129 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	86.81	37.95	0.33495

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	86.81	56.67	0.50022

==========

COLUMN NO. 130 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	114.92	44.02	0.28375

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	114.92	1.32	0.00854

==========

COLUMN NO. 131 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	118.94	43.42	0.27043

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	118.94	0.05	0.00028

==========

COLUMN NO. 132 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	81.88	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	81.88	8.77	0.07934

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COLUMN NO. 133 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	88.40	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	88.40	10.84	0.09087

==========

COLUMN NO. 134 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	68.30	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	68.30	21.67	0.23506

==========

COLUMN NO. 135 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	73.35	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	73.35	22.56	0.22788

**********END OF COLUMN DESIGN RESULTS**********

417. DESIGN BEAM 136 TO 143 161 TO 164

BEAM NO. 136 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	1105.	4166.	NO	NO

CRITICAL POS MOMENT= 4.98 KN-MET AT 2750.MM, LOAD 401
REQD STEEL= 40.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

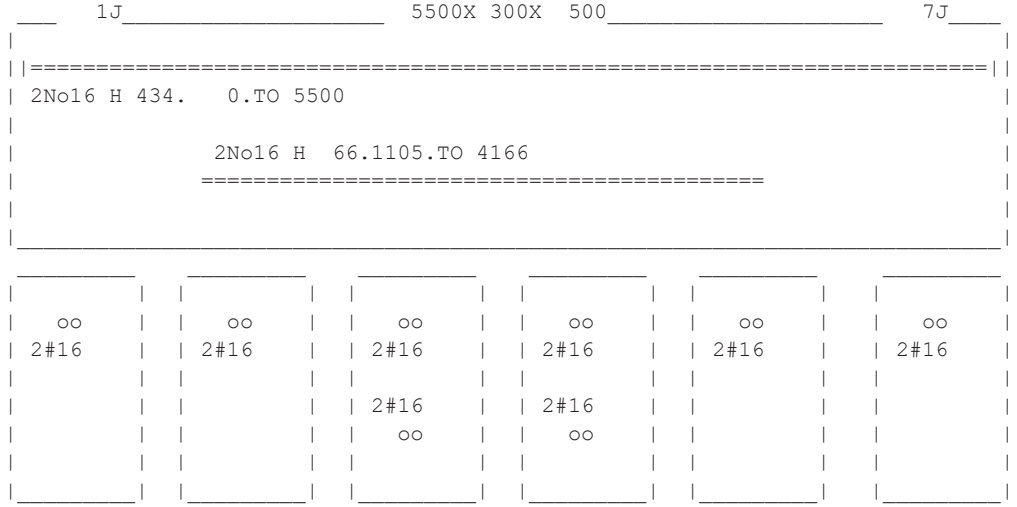
2	434.	2 - 16MM	0.	5500.	YES	YES

CRITICAL NEG MOMENT= 22.53 KN-MET AT 5500.MM, LOAD 403
REQD STEEL= 184.MM2, RHO=0.0014, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

B E A M N O. 136 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.45 KNS Vc= 110.89 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.76 KNS Vc= 110.89 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 137 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

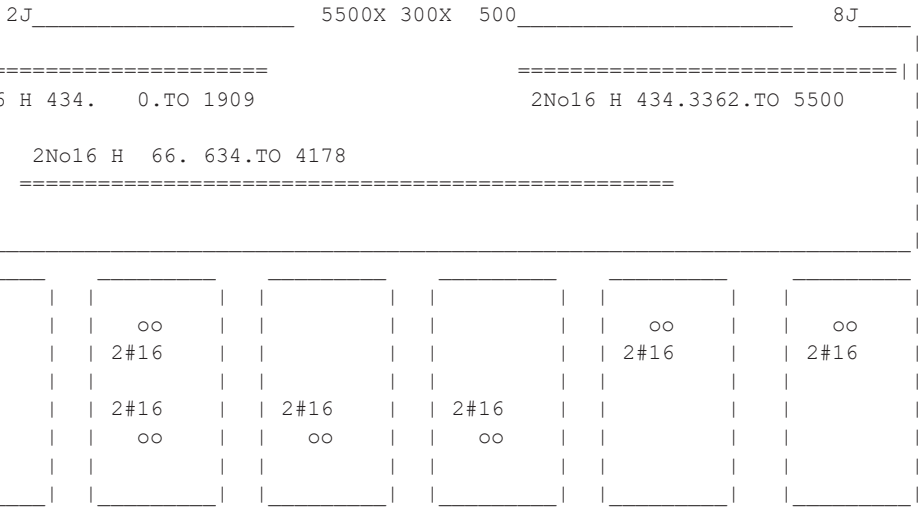
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	634.	4178.	NO	NO
	CRITICAL POS MOMENT= 6.43 KN-MET AT 2750.MM, LOAD 401					
	REQD STEEL= 52.MM2, RHO=0.0004, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	2 - 16MM	0.	1909.	YES	NO
	CRITICAL NEG MOMENT= 16.96 KN-MET AT 0.MM, LOAD 403					
	REQD STEEL= 138.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
3	434.	2 - 16MM	3362.	5500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     18.46 KN-MET  AT  5500.MM, LOAD  403|
|   REQD STEEL=   151.MM2, RHO=0.0012, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4
```

B E A M N O. 137 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   12.36 KNS  Vc=  120.22 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   12.85 KNS  Vc=  120.22 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 138 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	621.	4191.	NO	NO

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=      7.97 KN-MET  AT  2750.MM, LOAD  401|
|   REQD STEEL=    65.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

```
 2        434.     2 - 16MM         0.         1450.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     12.44 KN-MET  AT     0.MM, LOAD  403|
|   REQD STEEL=   101.MM2, RHO=0.0008, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|
```

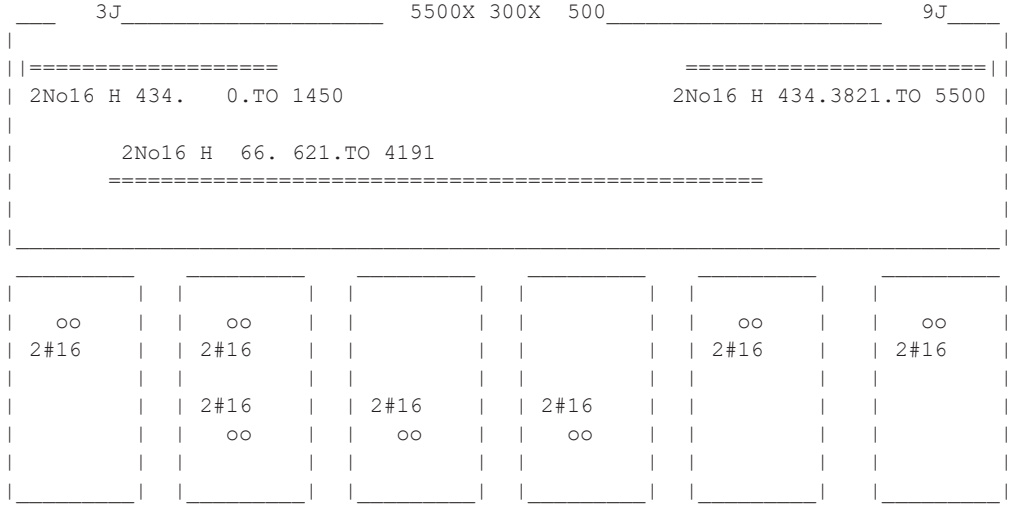
Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

```
 3        434.     2 - 16MM      3821.         5500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     12.99 KN-MET  AT  5500.MM, LOAD  403|
|   REQD STEEL=   106.MM2, RHO=0.0008, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

B E A M N O. 138 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   12.50 KNS  Vc=  122.73 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   12.71 KNS  Vc=  122.64 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 139 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 895. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	534.	3 - 16MM	0.	895.	YES	YES

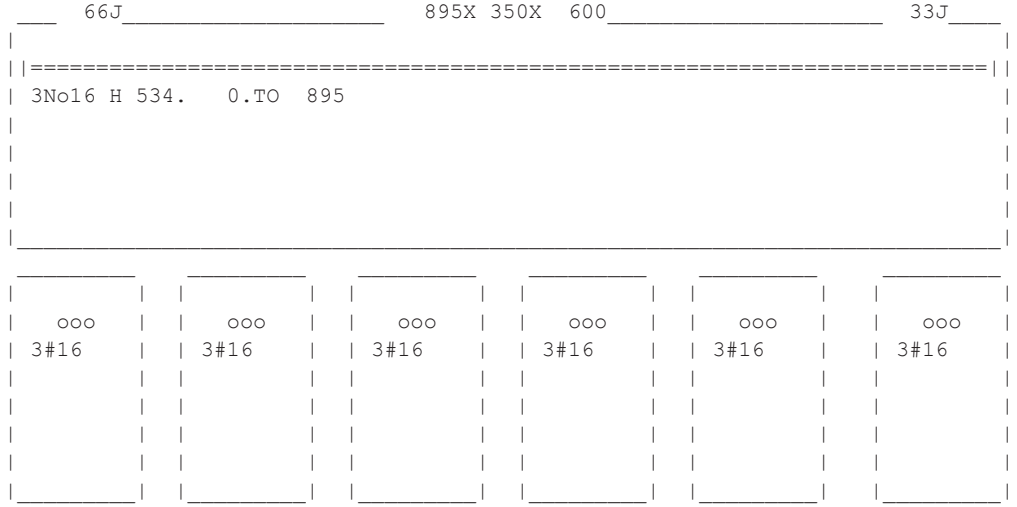
CRITICAL NEG MOMENT= 62.44 KN-MET AT 0.MM, LOAD 403
REQD STEEL= 416.MM2, RHO=0.0022, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 109. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 92778.8 cm^4

B E A M N O. 139 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 139 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 139 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



BEAM NO. 140 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1070. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	1070.	YES	YES

CRITICAL POS MOMENT= 47.30 KN-MET AT 1070.MM, LOAD 402
REQD STEEL= 314.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

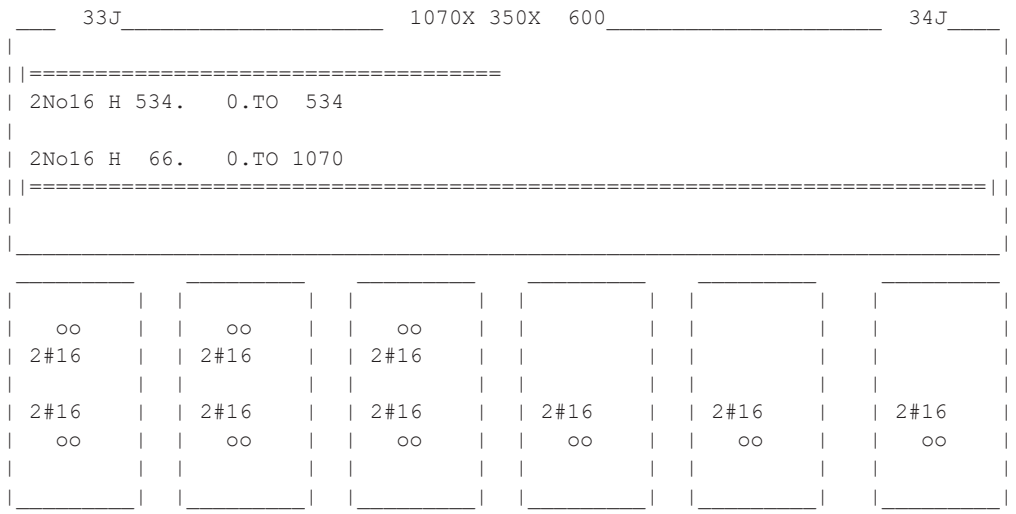
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
2	534.	2 - 16MM	0.	534.	YES	NO

CRITICAL NEG MOMENT= 3.78 KN-MET AT 0.MM, LOAD 403
REQD STEEL= 25.MM2, RHO=0.0001, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

B E A M N O. 140 D E S I G N R E S U L T S - SHEAR

```
   AT START SUPPORT - Vu=   22.20 KNS  Vc=  184.88 KNS  Vs=    0.00 KNS
   Tu=    4.88 KN-MET  Tc=    8.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                     STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=   22.18 KNS  Vc=  184.80 KNS  Vs=    0.00 KNS
   Tu=    4.88 KN-MET  Tc=    8.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                     STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 141 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1570. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

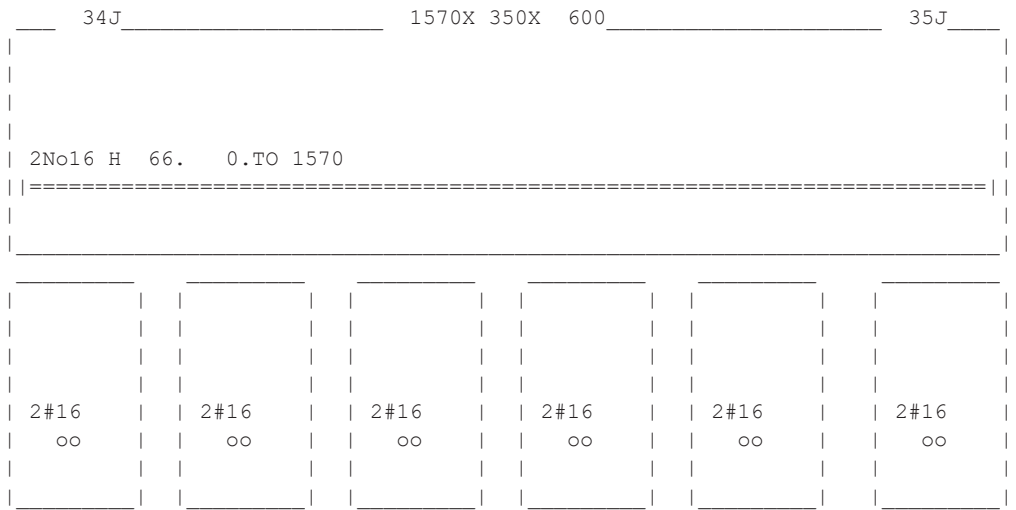
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	1570.	YES	YES

```
   CRITICAL POS MOMENT=     49.68 KN-MET  AT   785.MM, LOAD  402
   REQD STEEL=   330.MM2, RHO=0.0018, RHOMX=0.0249 RHOMN=0.0035
   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  218. MMS
   REQD. DEVELOPMENT LENGTH =   410. MMS
```

Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

B E A M N O. 141 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.67 KNS Vc= 174.71 KNS Vs= 0.00 KNS
Tu= 0.41 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 2.16 KNS Vc= 174.79 KNS Vs= 0.00 KNS
Tu= 0.41 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 142 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1070. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	1070.	YES	YES

CRITICAL POS MOMENT= 48.11 KN-MET AT 0.MM, LOAD 402
REQD STEEL= 320.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

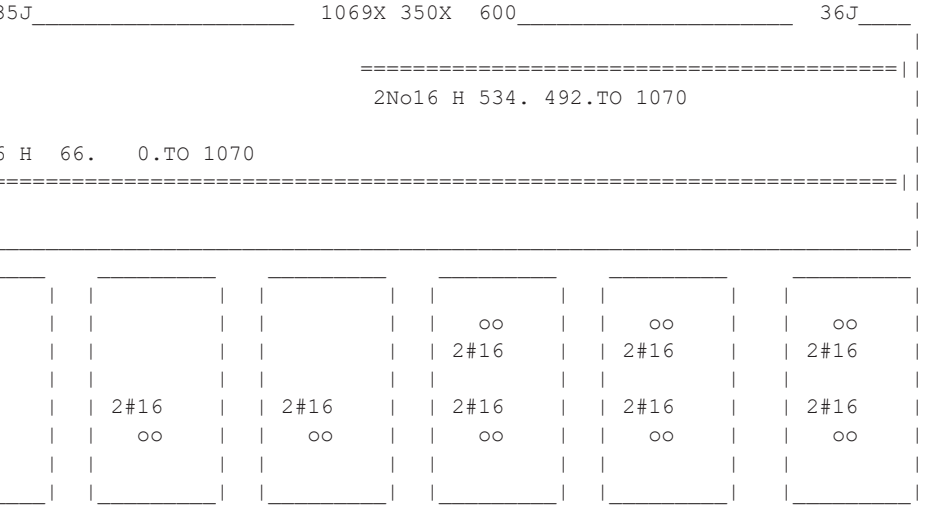
2	534.	2 - 16MM	492.	1070.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      3.93 KN-MET  AT  1070.MM, LOAD  403|
|   REQD STEEL=    26.MM2, RHO=0.0001, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  218. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    65065.9 cm^4
```

B E A M N O. 142 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   22.92 KNS  Vc=  186.38 KNS  Vs=    0.00 KNS
 Tu=    4.70 KN-MET  Tc=    8.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   22.93 KNS  Vc=  186.48 KNS  Vs=    0.00 KNS
 Tu=    4.70 KN-MET  Tc=    8.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 143 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 895. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	534.	3 - 16MM	0.	895.	YES	YES

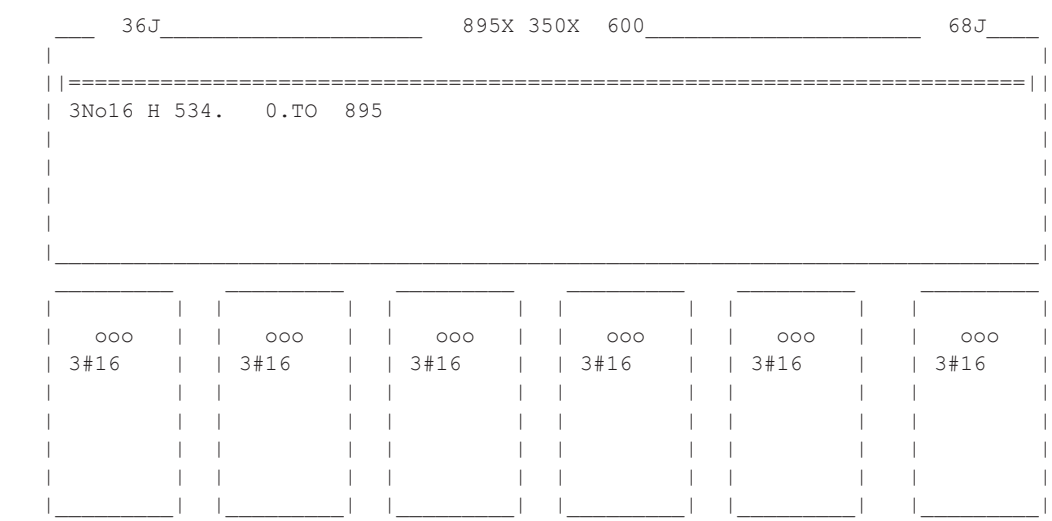
```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     63.42 KN-MET  AT   895.MM, LOAD  403|
|   REQD STEEL=   423.MM2, RHO=0.0023, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  109. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 92778.8 cm^4

B E A M N O. 143 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 143 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 143 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



BEAM NO. 161 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1997. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	837.	1997.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL POS MOMENT=     11.65 KN-MET  AT  1997.MM, LOAD  402|
|   REQD STEEL=    77.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

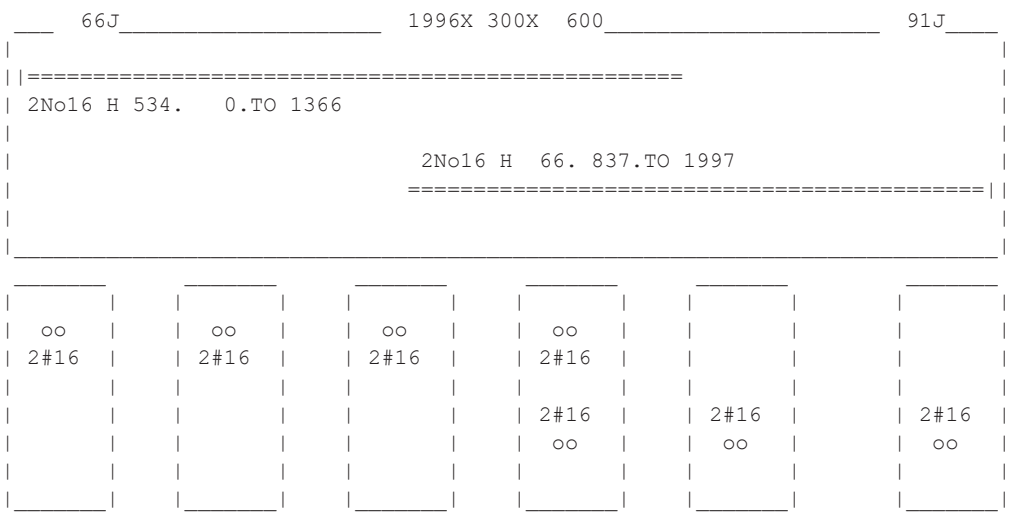
Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

```
  2        534.     2 - 16MM         0.         1366.        YES  NO
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     17.01 KN-MET  AT     0.MM, LOAD  401|
|   REQD STEEL=   113.MM2, RHO=0.0007, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

B E A M N O. 161 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   17.30 KNS  Vc=  157.91 KNS  Vs=    0.00 KNS
Tu=    1.00 KN-MET  Tc=    6.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   11.23 KNS  Vc=  157.54 KNS  Vs=    0.00 KNS
Tu=    1.00 KN-MET  Tc=    6.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 162 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4439. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	3063.	YES	NO

CRITICAL POS MOMENT= 15.39 KN-MET AT 1110.MM, LOAD 402
REQD STEEL= 102.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

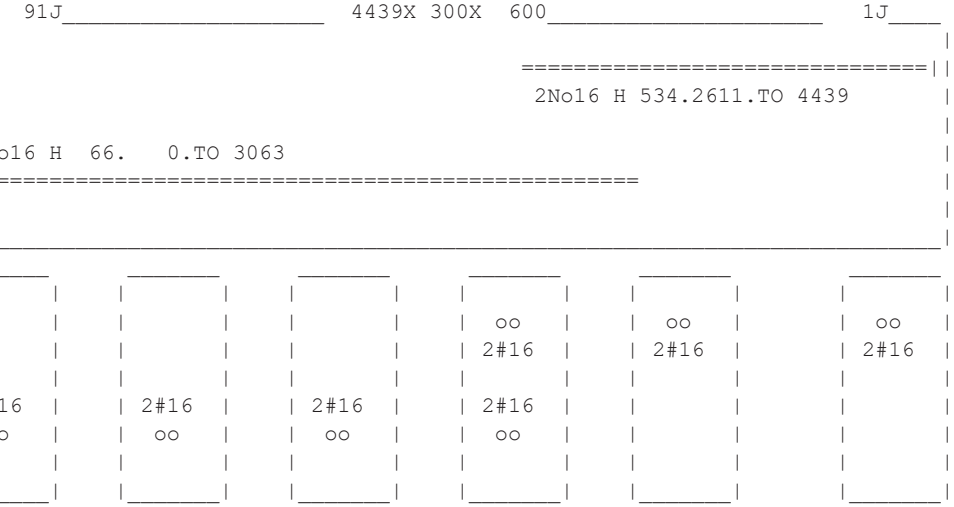
2	534.	2 - 16MM	2611.	4439.	NO	YES

CRITICAL NEG MOMENT= 22.03 KN-MET AT 4439.MM, LOAD 403
REQD STEEL= 146.MM2, RHO=0.0009, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

B E A M N O. 162 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 3.52 KNS Vc= 149.90 KNS Vs= 0.00 KNS
Tu= 1.03 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 18.50 KNS Vc= 152.51 KNS Vs= 0.00 KNS
Tu= 1.03 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 163 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1997. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	837.	1997.	NO	YES

CRITICAL POS MOMENT= 11.54 KN-MET AT 1997.MM, LOAD 402
REQD STEEL= 76.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

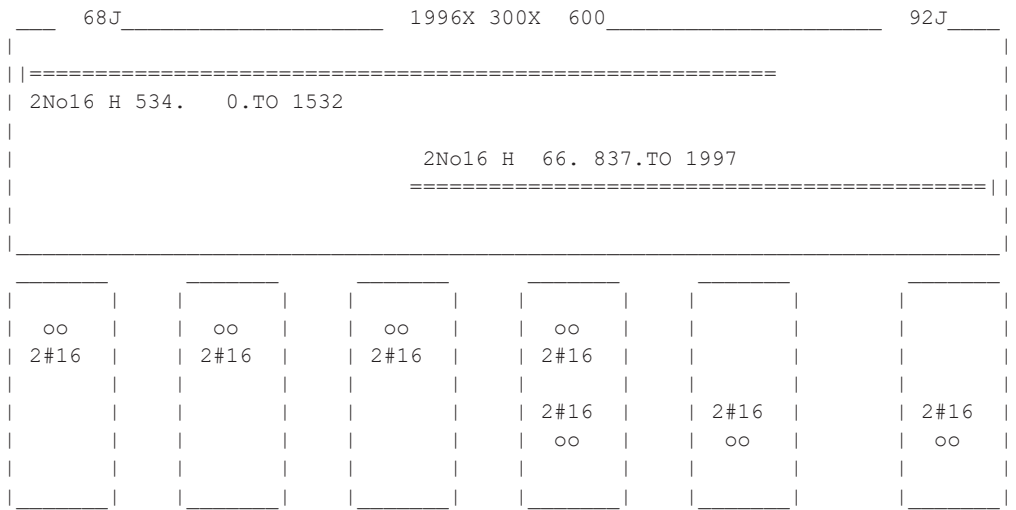
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
2	534.	2 - 16MM	0.	1532.	YES	NO

CRITICAL NEG MOMENT= 17.25 KN-MET AT 0.MM, LOAD 401
REQD STEEL= 114.MM2, RHO=0.0007, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

B E A M N O. 163 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.37 KNS Vc= 156.34 KNS Vs= 0.00 KNS
Tu= 1.06 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.31 KNS Vc= 157.42 KNS Vs= 0.00 KNS
Tu= 1.06 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 164 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4439. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	3063.	YES	NO

CRITICAL POS MOMENT= 15.31 KN-MET AT 1110.MM, LOAD 402
REQD STEEL= 101.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

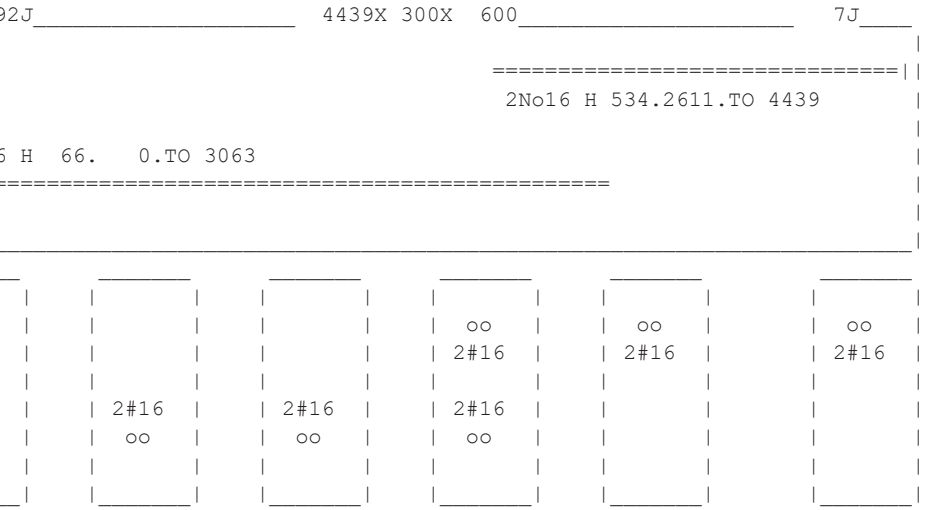
2	534.	2 - 16MM	2611.	4439.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     21.82 KN-MET  AT  4439.MM, LOAD  403|
|   REQD STEEL=   144.MM2, RHO=0.0009, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    63898.6 cm^4
```

B E A M N O. 164 D E S I G N R E S U L T S - SHEAR

```
  AT START SUPPORT - Vu=    3.59 KNS  Vc=  149.90 KNS  Vs=    0.00 KNS
  Tu=    1.08 KN-MET  Tc=    6.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                   STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=   18.43 KNS  Vc=  152.46 KNS  Vs=    0.00 KNS
  Tu=    1.08 KN-MET  Tc=    6.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                   STIRRUPS ARE NOT REQUIRED.
```




```
  ********************END OF BEAM DESIGN*************************


  418. END CONCRETE DESIGN
  419. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 24,2020 TIME= 13:55:13 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	24-03-2020	25-04-2020	01.11.2023				Date	24-03-2020	25-04-2020	01.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 4 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR TR-3**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 13-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR3 FOUNDATION
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 5 0 0.35 0;
6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7;
11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75;
15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7;
19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7;
24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75;
28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5;
33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25;
38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25;
42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25;
46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2;
51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75;
56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75;
60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5;
64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5;
69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25;
73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25;
77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25;
81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1;
86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75;
91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75;
95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75;
99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5;
103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5;
108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25;
112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25;
116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0;
121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25;
126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25;
130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25;
134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5;
138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5;
142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75;
146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75;
150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7;
154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7;
158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25;
162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25;
```

```
166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1;
171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1;
175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3;
179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25;
183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3;
188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3;
192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475;
196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475;
200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475;
204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475;
208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475;
212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75;
215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2;
219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5;
223 1.31667 -3 1.5; 224 1.03333 -3 1.25; 225 1.31667 -3 1.25; 226 1.03333 -3 1;
227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5;
231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0;
235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25;
238 1.03333 -3 -0.5; 239 1.31667 -3 -0.5; 240 1.03333 -3 -0.75;
241 1.31667 -3 -0.75; 242 1.03333 -3 -1; 243 1.31667 -3 -1;
244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25;
247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475;
251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7;
254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7;
257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5;
260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25;
263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2;
266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5;
269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25;
272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75;
275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5;
278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0;
281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25;
284 -1.31667 -3 -0.5; 285 -1.03333 -3 -0.5; 286 -1.31667 -3 -0.75;
287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1;
290 -1.31667 -3 -1.25; 291 -1.03333 -3 -1.25; 292 -1.31667 -3 3.25;
293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3;
296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7;
299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475;
302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
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84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162;
141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165;
146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168;
150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171;
162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17;
168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20;
174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24;
184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196;
187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154;
190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199;
193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156;
200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180;
204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184;
208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188;
214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44;
218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218;
222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221;
226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79;
230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226;
234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229;
238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231;
241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233;
244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235;
247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237;
250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239;
253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241;
256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243;
259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245;
262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249;
265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214;
268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251;
271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253;
274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257;
277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248;
280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38;
284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264;
288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267;
292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73;
296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272;
300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275;
304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277;
307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279;
310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281;
313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283;
316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285;

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319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287;
322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289;
325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291;
328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295;
331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260;
334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297;
337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299;
340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303;
343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX 0.387 FY -172.318 FZ -1.394
6 FX 0.468 FY -82.122 FZ 10.977
ELEMENT LOAD
```

```
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX -0.032 FY -61.148 FZ -3.382
6 FX 0.342 FY -70.854 FZ 3.277
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX 0.166 FY -42.384 FZ -2.208
6 FX 0.165 FY -43.417 FZ 2.146
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX -0.146 FY 211.746 FZ 35.206
6 FX 0.146 FY -211.746 FZ 35.106
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX 0.146 FY -211.749 FZ -35.207
6 FX -0.146 FY 211.749 FZ -35.106
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 2.902 FY 0 FZ 0
6 FX 2.902 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -2.902 FY 0 FZ 0
6 FX -2.902 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX 0 FY 0.173 FZ -18.9
6 FX 0 FY -0.173 FZ 18.94
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX 0 FY -0.173 FZ 18.9
6 FX 0 FY 0.173 FZ -18.94
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.005 FY -0.529 FZ -0.038
6 FX 0.007 FY -0.526 FZ 0.016
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX -0.009 FY 13.202 FZ 2.261
6 FX 0.009 FY -13.202 FZ 2.249
```

```
****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
```

```
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
```

```
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
****************************************
PERFORM ANALYSIS
****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 23, 2020                *
*           Time=    21:21: 5                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3130-CNV-001\Staad fdn\3130-CNV-001.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 13-MAR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR3 FOUNDATION
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 5 0 0.35 0
   18. 6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7
   19. 11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75
   20. 15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7
   21. 19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7
   22. 24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75
   23. 28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5
   24. 33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25
   25. 38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25
   26. 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25
   27. 46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2
   28. 51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75
   29. 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75
   30. 60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5
   31. 64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5
   32. 69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25
   33. 73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25
   34. 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25
   35. 81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1
   36. 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75
   37. 91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75
   38. 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75
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39. 99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5
40. 103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5
41. 108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25
42. 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25
43. 116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0
44. 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25
45. 126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25
46. 130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25
47. 134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5
48. 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5
49. 142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75
50. 146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75
51. 150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7
52. 154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7
53. 158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25
54. 162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25
55. 166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1.
56. 171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1.
57. 175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3
58. 179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25
59. 183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3
60. 188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3
61. 192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475
62. 196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475
63. 200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475
64. 204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475
65. 208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475
66. 212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75
67. 215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2
68. 219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5
69. 223 1.31667 -3 1.5; 224 1.03333 -3 1.25; 225 1.31667 -3 1.25; 226 1.03333 -3 1
70. 227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5
71. 231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0
72. 235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25
73. 238 1.03333 -3 -0.5; 239 1.31667 -3 -0.5; 240 1.03333 -3 -0.75
74. 241 1.31667 -3 -0.75; 242 1.03333 -3 -1; 243 1.31667 -3 -1.
75. 244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25
76. 247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475
77. 251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7
78. 254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7
79. 257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5
80. 260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25
81. 263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2
82. 266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5
83. 269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25
84. 272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75
85. 275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5
86. 278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0
87. 281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25
88. 284 -1.31667 -3 -0.5; 285 -1.03333 -3 -0.5; 286 -1.31667 -3 -0.75
89. 287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1.
90. 290 -1.31667 -3 -1.25; 291 -1.03333 -3 -1.25; 292 -1.31667 -3 3.25
91. 293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3
92. 296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7
93. 299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475
94. 302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475

95. MEMBER INCIDENCES
96. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
97. ELEMENT INCIDENCES SHELL
98. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
99. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
100. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
101. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
102. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
103. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
104. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
105. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
106. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
107. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
108. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
109. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
110. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
111. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
112. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
113. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
114. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
115. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
116. 119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146
117. 122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149
118. 137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162
119. 141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165
120. 146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168
121. 150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171
122. 162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17
123. 168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20
124. 174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24
125. 184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196
126. 187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154
127. 190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199
128. 193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156
129. 200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180
130. 204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184
131. 208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188
132. 214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44
133. 218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218
134. 222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221
135. 226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79
136. 230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226
137. 234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229
138. 238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231
139. 241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233
140. 244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235
141. 247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237
142. 250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239
143. 253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241
144. 256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243
145. 259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245
146. 262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249
147. 265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214
148. 268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251
149. 271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253
150. 274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257

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151. 277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248
152. 280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38
153. 284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264
154. 288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267
155. 292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73
156. 296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272
157. 300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275
158. 304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277
159. 307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279
160. 310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281
161. 313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283
162. 316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285
163. 319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287
164. 322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289
165. 325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291
166. 328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295
167. 331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260
168. 334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297
169. 337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299
170. 340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303
171. 343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294
172. START GROUP DEFINITION
173. ELEMENT
174. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
175. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
176. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
177. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
178. 214 TO 345
179. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
180. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
181. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
182. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
183. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
184. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
185. 214 TO 345
186. END GROUP DEFINITION
187. ELEMENT PROPERTY
188. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
189. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
190. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
191. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
192. DEFINE MATERIAL START
193. ISOTROPIC CONCRETE
194. E 2.17185E+007
195. POISSON 0.17
196. DENSITY 23.5616
197. ALPHA 1E-005
198. DAMP 0.05
199. TYPE CONCRETE
200. STRENGTH FCU 27579
201. END DEFINE MATERIAL
202. MEMBER PROPERTY AMERICAN
203. 1 TO 4 PRIS YD 0.7 ZD 0.5
204. 5 PRIS YD 0.5 ZD 0.3
205. CONSTANTS
206. MATERIAL CONCRETE ALL
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207. ****************************************
208. SUPPORTS
209. 1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
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JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
7	0.028125
8	0.028125
9	0.015937
10	0.015937
11	0.015937
12	0.015937
13	0.030000
14	0.066666
15	0.035416
16	0.028125
17	0.062500
18	0.028125
19	0.062500
20	0.062500
21	0.028125
22	0.062500
23	0.028125
24	0.062500
25	0.030000
26	0.066666
27	0.035416
28	0.066666
29	0.035416
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.066666
36	0.035416
37	0.066666
38	0.035416
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.066666
45	0.035416
46	0.066666
47	0.035416
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.066666
53	0.035416

54	0.066666
55	0.035416
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.066666
62	0.035416
63	0.066666
64	0.035416
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.066666
71	0.035416
72	0.066666
73	0.035416
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.066666
80	0.035416
81	0.066666
82	0.035416
83	0.062500
84	0.062500
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86	0.062500
87	0.062500
88	0.066666
89	0.035416
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91	0.035416
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94	0.062500
95	0.062500
96	0.062500
97	0.066666
98	0.035416
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100	0.035416
101	0.062500
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105	0.062500
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107	0.035416
108	0.066666
109	0.035416

110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.066666
116	0.035416
117	0.066666
118	0.035416
119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.066666
124	0.035416
125	0.066666
126	0.035416
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.066666
133	0.035416
134	0.066666
135	0.035416
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.066666
142	0.035416
143	0.066666
144	0.035416
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.066666
151	0.035416
152	0.030000
153	0.028125
154	0.028125
155	0.028125
156	0.028125
157	0.030000
158	0.066666
159	0.035416
160	0.063333
161	0.033645
162	0.062500
163	0.059375
164	0.062500
165	0.059375

166	0.062500
167	0.059375
168	0.062500
169	0.059375
170	0.062500
171	0.059375
172	0.066666
173	0.063333
174	0.035416
175	0.033645
176	0.063333
177	0.033645
178	0.066666
179	0.035416
180	0.059375
181	0.062500
182	0.059375
183	0.062500
184	0.059375
185	0.062500
186	0.059375
187	0.062500
188	0.059375
189	0.062500
190	0.063333
191	0.066666
192	0.033645
193	0.035416
194	0.060000
195	0.031875
196	0.056250
197	0.056250
198	0.056250
199	0.056250
200	0.056250
201	0.060000
202	0.031875
203	0.060000
204	0.031875
205	0.056250
206	0.056250
207	0.056250
208	0.056250
209	0.056250
210	0.060000
211	0.031875
212	0.070834
213	0.070834
214	0.070834
215	0.070834
216	0.070834
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218	0.070834
219	0.070834
220	0.070834
221	0.070834

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239	0.070834
240	0.070834
241	0.070834
242	0.070834
243	0.070834
244	0.067292
245	0.067292
246	0.067292
247	0.070834
248	0.067292
249	0.070834
250	0.063750
251	0.063750
252	0.031875
253	0.031875
254	0.031875
255	0.063750
256	0.031875
257	0.063750
258	0.070834
259	0.070834
260	0.070834
261	0.070834
262	0.070834
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264	0.070834
265	0.070834
266	0.070834
267	0.070834
268	0.070834
269	0.070834
270	0.070834
271	0.070834
272	0.070834
273	0.070834
274	0.070834
275	0.070834
276	0.070834
277	0.070834

278	0.070834
279	0.070834
280	0.070834
281	0.070834
282	0.070834
283	0.070834
284	0.070834
285	0.070834
286	0.070834
287	0.070834
288	0.070834
289	0.070834
290	0.067292
291	0.067292
292	0.067292
293	0.070834
294	0.067292
295	0.070834
296	0.063750
297	0.063750
298	0.031875
299	0.031875
300	0.031875
301	0.063750
302	0.031875
303	0.063750
TOTAL	17.280014

```
210. ****************************************
211. LOAD 1 DEAD LOAD DL
212. SELFWEIGHT Y -1
213. JOINT LOAD
214. 5 FX 0.387 FY -172.318 FZ -1.394
215. 6 FX 0.468 FY -82.122 FZ 10.977
216. ELEMENT LOAD
217. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
218. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
219. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
220. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
221. ****************************************
222. LOAD 2 LIVE LOAD LL
223. JOINT LOAD
224. 5 FX -0.032 FY -61.148 FZ -3.382
225. 6 FX 0.342 FY -70.854 FZ 3.277
226. ****************************************
227. LOAD 3 MATERIAL LOAD LL
228. JOINT LOAD
229. 5 FX 0.166 FY -42.384 FZ -2.208
230. 6 FX 0.165 FY -43.417 FZ 2.146
231. ****************************************
232. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
233. JOINT LOAD
234. 5 FX -0.146 FY 211.746 FZ 35.206
235. 6 FX 0.146 FY -211.746 FZ 35.106
236. ****************************************
237. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
238. JOINT LOAD
```

```
239. 5 FX 0.146 FY -211.749 FZ -35.207
240. 6 FX -0.146 FY 211.749 FZ -35.106
241. ****************************************
242. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
243. JOINT LOAD
244. 5 FX 2.902 FY 0 FZ 0
245. 6 FX 2.902 FY 0 FZ 0
246. ****************************************
247. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
248. JOINT LOAD
249. 5 FX -2.902 FY 0 FZ 0
250. 6 FX -2.902 FY 0 FZ 0
251. ****************************************
252. LOAD 8 TEMP LOAD   TL1
253. JOINT LOAD
254. 5 FX 0 FY 0.173 FZ -18.9
255. 6 FX 0 FY -0.173 FZ 18.94
256. ****************************************
257. LOAD 9 TEMP LOAD   TL2
258. JOINT LOAD
259. 5 FX 0 FY -0.173 FZ 18.9
260. 6 FX 0 FY 0.173 FZ -18.94
261. ****************************************
262. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
263. JOINT LOAD
264. 5 FX 0.005 FY -0.529 FZ -0.038
265. 6 FX 0.007 FY -0.526 FZ 0.016
266. ****************************************
267. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
268. JOINT LOAD
269. 5 FX -0.009 FY 13.202 FZ 2.261
270. 6 FX 0.009 FY -13.202 FZ 2.249
271. ****************************************
272. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
273. LOAD 301 1.0 DL
274. REPEAT LOAD
275. 1 1.0
276. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
277. REPEAT LOAD
278. 1 1.0 8 1.0 2 1.0 3 1.0
279. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
280. REPEAT LOAD
281. 1 1.0 9 1.0 2 1.0 3 1.0
282. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
283. REPEAT LOAD
284. 1 1.0 4 0.75 2 0.75 3 0.75
285. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
286. REPEAT LOAD
287. 1 1.0 5 0.75 2 0.75 3 0.75
288. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
289. REPEAT LOAD
290. 1 1.0 6 0.75 2 0.75 3 0.75
291. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
292. REPEAT LOAD
293. 1 1.0 7 0.75 2 0.75 3 0.75
294. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
```

```
295. REPEAT LOAD
296. 1 1.0 10 0.525 2 0.75 3 0.75
297. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
298. REPEAT LOAD
299. 1 1.0 10 -0.525 2 0.75 3 0.75
300. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
301. REPEAT LOAD
302. 1 1.0 11 0.525 2 0.75 3 0.75
303. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
304. REPEAT LOAD
305. 1 1.0 11 -0.525 2 0.75 3 0.75
306. LOAD 312 0.6 DL + 1.0 WL(+Z)
307. REPEAT LOAD
308. 1 0.6 4 1.0
309. LOAD 313 0.6 DL + 1.0 WL(-Z)
310. REPEAT LOAD
311. 1 0.6 5 1.0
312. LOAD 314 0.6 DL + 1.0 WL(+X)
313. REPEAT LOAD
314. 1 0.6 6 1.0
315. LOAD 315 0.6 DL + 1.0 WL(-X)
316. REPEAT LOAD
317. 1 0.6 7 1.0
318. LOAD 316 0.6 DL + 0.7 EL(+X)
319. REPEAT LOAD
320. 1 0.6 10 0.7
321. LOAD 317 0.6 DL + 0.7 EL(-X)
322. REPEAT LOAD
323. 1 0.6 10 -0.7
324. LOAD 318 0.6 DL + 0.7 EL(+Z)
325. REPEAT LOAD
326. 1 0.6 11 0.7
327. LOAD 319 0.6 DL + 0.7 EL(-Z)
328. REPEAT LOAD
329. 1 0.6 11 -0.7
330. ****************************************
331. *****LOAD COMBINATION FOR FOUNDATION DESIGN
332. LOAD 401 1.4 DL
333. REPEAT LOAD
334. 1 1.4
335. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
336. REPEAT LOAD
337. 1 1.4 8 1.4 2 1.7 3 1.7
338. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
339. REPEAT LOAD
340. 1 1.4 9 1.4 2 1.7 3 1.7
341. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
342. REPEAT LOAD
343. 1 1.2 4 1.6 2 0.5 3 0.5
344. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
345. REPEAT LOAD
346. 1 1.2 5 1.6 2 0.5 3 0.5
347. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
348. REPEAT LOAD
349. 1 1.2 6 1.6 2 0.5 3 0.5
350. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
```

```
351. REPEAT LOAD
352. 1 1.2 7 1.6 2 0.5 3 0.5
353. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
354. REPEAT LOAD
355. 1 1.2 10 1.0 2 0.5 3 0.5
356. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
357. REPEAT LOAD
358. 1 1.2 10 -1.0 2 0.5 3 0.5
359. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
360. REPEAT LOAD
361. 1 1.2 11 1.0 2 0.5 3 0.5
362. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
363. REPEAT LOAD
364. 1 1.2 11 -1.0 2 0.5 3 0.5
365. LOAD 412 0.9 DL + 1.6 WL(+Z)
366. REPEAT LOAD
367. 1 0.9 4 1.6
368. LOAD 413 0.9 DL + 1.6 WL(-Z)
369. REPEAT LOAD
370. 1 0.9 5 1.6
371. LOAD 414 0.9 DL + 1.6 WL(+X)
372. REPEAT LOAD
373. 1 0.9 6 1.6
374. LOAD 415 0.9 DL + 1.6 WL(-X)
375. REPEAT LOAD
376. 1 0.9 7 1.6
377. LOAD 416 0.9 DL + 1.0 EL(+X)
378. REPEAT LOAD
379. 1 0.9 10 1.0
380. LOAD 417 0.9 DL + 1.0 EL(-X)
381. REPEAT LOAD
382. 1 0.9 10 -1.0
383. LOAD 418 0.9 DL + 1.0 EL(+Z)
384. REPEAT LOAD
385. 1 0.9 11 1.0
386. LOAD 419 0.9 DL + 1.0 EL(-Z)
387. REPEAT LOAD
388. 1 0.9 11 -1.0
389. ****************************************
390. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**303**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**264**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**299**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

```
  ORIGINAL/FINAL BAND-WIDTH=   290/    30/     93 DOF
TOTAL      PRIMARY LOAD CASES =     49, TOTAL DEGREES OF FREEDOM =     921
TOTAL LOAD COMBINATION  CASES =      0  SO FAR.
  SIZE OF STIFFNESS MATRIX =       86 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     14.9/  50943.3 MB
```

```
391. ****************************************
392. LOAD LIST 401 TO 419
393. START CONCRETE DESIGN
```

```
394. CODE ACI 2008
395. FC 35000 MEMB 1 TO 5
396. FYMAIN 420000 MEMB 1 TO 5
397. TRACK 1 ALL
398. CLB 0.05 MEMB 1 TO 5
399. CLT 0.05 MEMB 1 TO 5
400. CLS 0.05 ALL
401. MAXMAIN 25 ALL
402. MINMAIN 16 ALL
403. MINSEC 8 ALL
404. REINF 0 MEMB 1 TO 4
405. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	366.82	0.31	0.00028

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	366.82	14.38	0.01307

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	371.15	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	371.15	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	208.86	0.33	0.00053

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	208.86	14.70	0.02347

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	176.88	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	176.88	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

406. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

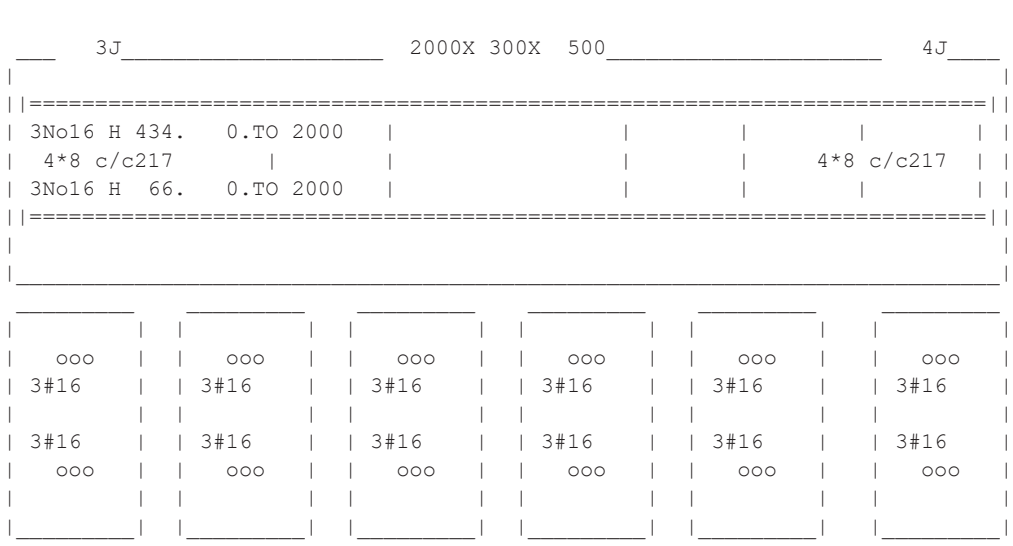
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	2000.	YES	YES
	CRITICAL POS MOMENT= 82.70 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 519.MM2, RHO=0.0040, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	434.	3 - 16MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 89.14 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 560.MM2, RHO=0.0043, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 83.52 KNS Vc= 126.04 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM

AT END SUPPORT - Vu= 88.32 KNS Vc= 126.04 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM



**********END OF BEAM DESIGN**********

407. END CONCRETE DESIGN
408. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 23,2020 TIME= 21:21:13 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.        *
*               http://www.bentley.com                      *
*************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	24-03-2020	25-04-2020	01.11.2023				Date	24-03-2020	25-04-2020	01.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 5 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR TR-4**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 23-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR4 FOUNDATION
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 3.5; 3 0 1.49012e-008 0; 4 0 1.49012e-008 3.5; 5 0 0.35 0;
6 0 0.35 3.5;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 2 FIXED
*****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX -0.042 FY -132.305 FZ -9.049
6 FX 0.044 FY -95.934 FZ 17.049
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX 0.01 FY -61.654 FZ -6.866
6 FX -0.01 FY -58.346 FZ 6.866
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX 0.008 FY -36.458 FZ -4.182
6 FX -0.008 FY -36.542 FZ 4.182
*****************************************
```

```
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX -0.026 FY 197.421 FZ 40.019
6 FX 0.026 FY -197.421 FZ 41.033
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX 0.026 FY -197.421 FZ -40.019
6 FX -0.026 FY 197.421 FZ -41.033
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 7.004 FY 0 FZ 0
6 FX 6.999 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -7.004 FY 0 FZ 0
6 FX -6.999 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX 0.002 FY -0.022 FZ 5.56
6 FX -0.002 FY 0.022 FZ -5.559
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX -0.002 FY 0.022 FZ -5.56
6 FX 0.002 FY -0.022 FZ 5.559
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.002 FY -0.048 FZ 0
6 FX -0.001 FY 0.048 FZ 0
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX 0.004 FY 9.338 FZ 1.658
6 FX -0.004 FY -9.338 FZ 1.707
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
```

```
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
```

```
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 23, 2020                *
*           Time=    21:23:10                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3130-CNV-001\Staad fdn\3130-CNV-001.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 23-MAR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR4 FOUNDATION
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 3.5; 3 0 1.49012E-008 0; 4 0 1.49012E-008 3.5; 5 0 0.35 0
   18. 6 0 0.35 3.5
   19. MEMBER INCIDENCES
   20. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
   21. DEFINE MATERIAL START
   22. ISOTROPIC CONCRETE
   23. E 2.17185E+007
   24. POISSON 0.17
   25. DENSITY 23.5616
   26. ALPHA 1E-005
   27. DAMP 0.05
   28. TYPE CONCRETE
   29. STRENGTH FCU 27579
   30. END DEFINE MATERIAL
   31. MEMBER PROPERTY AMERICAN
   32. 1 TO 4 PRIS YD 0.7 ZD 0.5
   33. 5 PRIS YD 0.5 ZD 0.3
   34. CONSTANTS
   35. MATERIAL CONCRETE ALL
   36. SUPPORTS
   37. 1 2 FIXED
   38. ****************************************
```

```
39. LOAD 1 DEAD LOAD DL
40. SELFWEIGHT Y -1
41. JOINT LOAD
42. 5 FX -0.042 FY -132.305 FZ -9.049
43. 6 FX 0.044 FY -95.934 FZ 17.049
44. ********************************************
45. LOAD 2 LIVE LOAD LL
46. JOINT LOAD
47. 5 FX 0.01 FY -61.654 FZ -6.866
48. 6 FX -0.01 FY -58.346 FZ 6.866
49. ********************************************
50. LOAD 3 MATERIAL LOAD LL
51. JOINT LOAD
52. 5 FX 0.008 FY -36.458 FZ -4.182
53. 6 FX -0.008 FY -36.542 FZ 4.182
54. ********************************************
55. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
56. JOINT LOAD
57. 5 FX -0.026 FY 197.421 FZ 40.019
58. 6 FX 0.026 FY -197.421 FZ 41.033
59. ********************************************
60. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
61. JOINT LOAD
62. 5 FX 0.026 FY -197.421 FZ -40.019
63. 6 FX -0.026 FY 197.421 FZ -41.033
64. ********************************************
65. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
66. JOINT LOAD
67. 5 FX 7.004 FY 0 FZ 0
68. 6 FX 6.999 FY 0 FZ 0
69. ********************************************
70. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
71. JOINT LOAD
72. 5 FX -7.004 FY 0 FZ 0
73. 6 FX -6.999 FY 0 FZ 0
74. ********************************************
75. LOAD 8 TEMP LOAD   TL1
76. JOINT LOAD
77. 5 FX 0.002 FY -0.022 FZ 5.56
78. 6 FX -0.002 FY 0.022 FZ -5.559
79. ********************************************
80. LOAD 9 TEMP LOAD   TL2
81. JOINT LOAD
82. 5 FX -0.002 FY 0.022 FZ -5.56
83. 6 FX 0.002 FY -0.022 FZ 5.559
84. ********************************************
85. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
86. JOINT LOAD
87. 5 FX 0.002 FY -0.048 FZ 0
88. 6 FX -0.001 FY 0.048 FZ 0
89. ********************************************
90. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
91. JOINT LOAD
92. 5 FX 0.004 FY 9.338 FZ 1.658
93. 6 FX -0.004 FY -9.338 FZ 1.707
94. ********************************************
```

```
 95. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
 96. LOAD COMB 301 1.0 DL
 97. 1 1.0
 98. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
 99. 1 1.0 8 1.0 2 1.0 3 1.0
100. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
101. 1 1.0 9 1.0 2 1.0 3 1.0
102. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
103. 1 1.0 4 0.75 2 0.75 3 0.75
104. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
105. 1 1.0 5 0.75 2 0.75 3 0.75
106. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
107. 1 1.0 6 0.75 2 0.75 3 0.75
108. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
109. 1 1.0 7 0.75 2 0.75 3 0.75
110. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
111. 1 1.0 10 0.525 2 0.75 3 0.75
112. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
113. 1 1.0 10 -0.525 2 0.75 3 0.75
114. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
115. 1 1.0 11 0.525 2 0.75 3 0.75
116. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
117. 1 1.0 11 -0.525 2 0.75 3 0.75
118. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
119. 1 0.6 4 1.0
120. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
121. 1 0.6 5 1.0
122. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
123. 1 0.6 6 1.0
124. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
125. 1 0.6 7 1.0
126. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
127. 1 0.6 10 0.7
128. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
129. 1 0.6 10 -0.7
130. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
131. 1 0.6 11 0.7
132. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
133. 1 0.6 11 -0.7
134. ****************************************
135. *****LOAD COMBINATION FOR FOUNDATION DESIGN
136. LOAD COMB 401 1.4 DL
137. 1 1.4
138. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
139. 1 1.4 8 1.4 2 1.7 3 1.7
140. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
141. 1 1.4 9 1.4 2 1.7 3 1.7
142. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
143. 1 1.2 4 1.6 2 0.5 3 0.5
144. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
145. 1 1.2 5 1.6 2 0.5 3 0.5
146. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
147. 1 1.2 6 1.6 2 0.5 3 0.5
148. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
149. 1 1.2 7 1.6 2 0.5 3 0.5
150. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
```

```
151. 1 1.2 10 1.0 2 0.5 3 0.5
152. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
153. 1 1.2 10 -1.0 2 0.5 3 0.5
154. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
155. 1 1.2 11 1.0 2 0.5 3 0.5
156. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
157. 1 1.2 11 -1.0 2 0.5 3 0.5
158. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
159. 1 0.9 4 1.6
160. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
161. 1 0.9 5 1.6
162. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
163. 1 0.9 6 1.6
164. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
165. 1 0.9 7 1.6
166. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
167. 1 0.9 10 1.0
168. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
169. 1 0.9 10 -1.0
170. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
171. 1 0.9 11 1.0
172. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
173. 1 0.9 11 -1.0
174. ****************************************
175. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

```
  ORIGINAL/FINAL BAND-WIDTH=     2/     2/     18 DOF
TOTAL      PRIMARY LOAD CASES =    11, TOTAL DEGREES OF FREEDOM =      24
TOTAL LOAD COMBINATION  CASES =    38  SO FAR.
  SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.0/  50937.4 MB
```

```
176. LOAD LIST 301 TO 319
177. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	0.03	163.19	-4.22	-8.23	-0.01	-0.11
	302	0.02	261.31	-4.92	-8.78	-0.01	-0.06
	303	0.02	261.27	-6.35	-9.92	-0.01	-0.07
	304	0.04	67.75	-35.72	-74.28	-0.02	-0.12
	305	0.01	405.78	25.16	56.14	-0.00	-0.02
	306	-5.23	236.77	-5.28	-9.07	-0.01	17.52
	307	5.28	236.77	-5.28	-9.07	-0.01	-17.67
	308	0.02	236.79	-5.28	-9.07	-0.01	-0.07
	309	0.02	236.74	-5.28	-9.07	-0.01	-0.08
	310	0.02	231.26	-6.17	-10.97	-0.01	-0.07
	311	0.02	242.27	-4.40	-7.17	-0.01	-0.08
	312	0.04	-127.44	-43.12	-91.88	-0.02	-0.13
	313	-0.00	323.26	38.06	82.00	0.00	0.00
	314	-6.98	97.91	-2.53	-4.94	-0.01	23.40
	315	7.02	97.91	-2.53	-4.94	-0.01	-23.53
	316	0.02	97.95	-2.53	-4.94	-0.01	-0.06
	317	0.02	97.88	-2.53	-4.94	-0.01	-0.07
	318	0.02	90.57	-3.71	-7.47	-0.01	-0.06
	319	0.02	105.26	-1.35	-2.41	-0.01	-0.07
2	301	-0.04	132.67	-3.78	-8.32	-0.01	0.11
	302	-0.02	227.55	-3.08	-7.74	-0.01	0.06
	303	-0.02	227.59	-1.65	-6.60	-0.01	0.07
	304	-0.04	372.86	-33.06	-72.59	-0.02	0.13
	305	-0.01	34.83	27.63	57.67	-0.00	0.03
	306	-5.27	203.85	-2.72	-7.46	-0.01	17.67
	307	5.22	203.85	-2.72	-7.46	-0.01	-17.51
	308	-0.02	203.82	-2.72	-7.46	-0.01	0.08
	309	-0.03	203.87	-2.72	-7.46	-0.01	0.08
	310	-0.02	209.35	-3.60	-9.36	-0.01	0.07
	311	-0.03	198.34	-1.84	-5.56	-0.01	0.08
	312	-0.04	304.95	-42.73	-91.83	-0.02	0.13
	313	-0.00	-145.74	38.19	81.85	0.00	0.00
	314	-7.02	79.60	-2.27	-4.99	-0.01	23.52
	315	6.98	79.60	-2.27	-4.99	-0.01	-23.38
	316	-0.02	79.57	-2.27	-4.99	-0.01	0.07
	317	-0.02	79.64	-2.27	-4.99	-0.01	0.07
	318	-0.02	86.95	-3.45	-7.52	-0.01	0.06
	319	-0.02	72.26	-1.10	-2.46	-0.01	0.08

********** END OF LATEST ANALYSIS RESULT **********

```
178. ****************************************
179. LOAD LIST 401 TO 419
180. START CONCRETE DESIGN
181. CODE ACI 2008
182. FC 35000 ALL
183. FYMAIN 420000 ALL
184. TRACK 1 ALL
185. CLB 0.05 ALL
186. CLT 0.05 ALL
187. CLS 0.05 ALL
188. MAXMAIN 25 ALL
189. MINMAIN 16 ALL
190. MINSEC 8 ALL
191. REINF 0 MEMB 1 TO 4
192. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	298.19	0.01	0.00002

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	298.19	9.51	0.01063

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	284.96	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	284.96	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	232.47	0.01	0.00002

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	232.47	6.54	0.00938

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	206.63	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	206.63	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

193. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

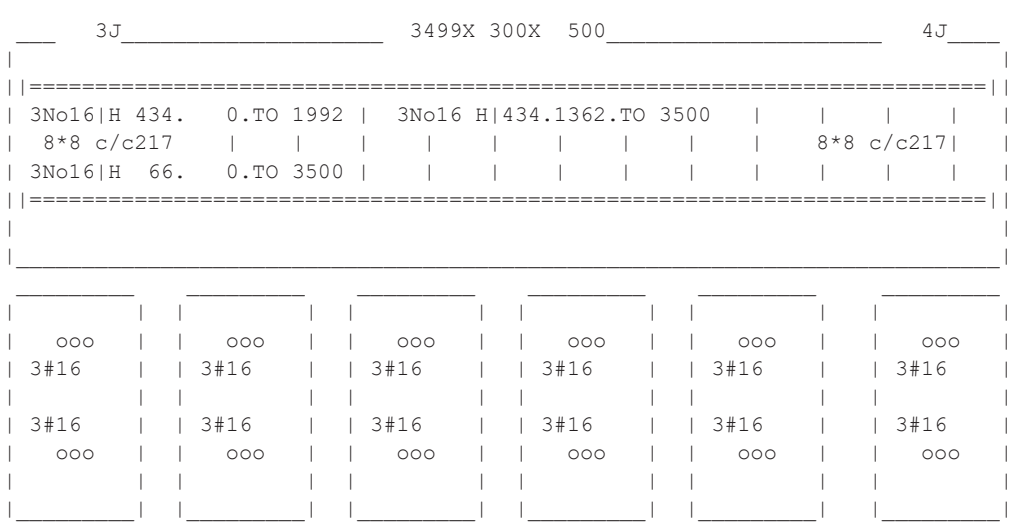
LEN - 3500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	3500.	YES	YES
	CRITICAL POS MOMENT= 79.28 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 497.MM2, RHO=0.0038, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	434.	3 - 16MM	0.	1992.	YES	NO
	CRITICAL NEG MOMENT= 77.02 KN-MET AT 0.MM, LOAD 413					
	REQD STEEL= 482.MM2, RHO=0.0037, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
3	434.	3 - 16MM	1362.	3500.	NO	YES
	CRITICAL NEG MOMENT= 89.43 KN-MET AT 3500.MM, LOAD 404					
	REQD STEEL= 562.MM2, RHO=0.0043, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 46.75 KNS Vc= 122.67 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 405
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 1316. MM

AT END SUPPORT - Vu= 53.78 KNS Vc= 122.21 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 1316. MM



**********END OF BEAM DESIGN**********

194. END CONCRETE DESIGN
195. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 23,2020 TIME= 21:23:10 ****

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************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	24-03-2020	25-04-2020	01.11.2023				Date	24-03-2020	25-04-2020	01.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 6 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR VGTU**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 22-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 1
JOB PART VGTU FDN
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 4.5; 3 4 -3 0; 4 4 -3 4.5; 5 8 -3 0; 6 8 -3 4.5; 7 0 0.35 0;
8 0 0.35 4.5; 9 4 0.35 0; 10 4 0.35 4.5; 11 8 0.35 0; 12 8 0.35 4.5; 13 0 1 0;
14 0 1 4.5; 15 4 1 0; 16 4 1 4.5; 17 8 1 0; 18 8 1 4.5;
MEMBER INCIDENCES
1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17;
11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10;
19 10 12;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
9 TO 12 PRIS YD 0.6 ZD 0.6
13 TO 15 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY AMERICAN
16 TO 19 PRIS YD 0.6 ZD 0.3
MEMBER PROPERTY AMERICAN
1 TO 8 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 TO 6 FIXED
*****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
13  FX  -17.302 FY  -292.376    FZ  -12.756
14  FX  -18.26  FY  -245.674    FZ  16.001
15  FX  -14.626 FY  -119.938    FZ  2.1
16  FX  -15.819 FY  -88.543 FZ  5.968
17  FX  0.007   FY  -30.141 FZ  -1.204
18  FX  0   FY  -22.224 FZ  -0.493
```

```
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
13  FX  -1.14   FY  -208.123    FZ  -11.282
14  FX  -3.27   FY  -179.66 FZ  11.541
15  FX  3.312   FY  -279.969    FZ  -5.278
16  FX  1.116   FY  -292.848    FZ  6.911
17  FX  -0.006  FY  -69.16  FZ  -0.954
18  FX  -0.011  FY  -54.457 FZ  -0.952

*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
13  FX  -0.01   FY  -17.625 FZ  -1.124
14  FX  -0.107  FY  -17.275 FZ  0.919
15  FX  0.108   FY  -24.85  FZ  0.125
16  FX  0.011   FY  -26.048 FZ  0.233
17  FX  -0.001  FY  -0.443  FZ  -0.087
18  FX  0   FY  0.443   FZ  -0.087

*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
13  FX  -1.594  FY  162.605 FZ  33.388
14  FX  1.597   FY  -162.605    FZ  33.597
15  FX  -2.338  FY  162.036 FZ  28.538
16  FX  2.336   FY  -162.036    FZ  28.536
17  FX  -0.002  FY  52.838  FZ  15.971
18  FX  0.001   FY  -52.839 FZ  15.974

*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
13  FX  1.595   FY  -162.605    FZ  -33.388
14  FX  -1.598  FY  162.605 FZ  -33.598
15  FX  2.338   FY  -162.037    FZ  -28.538
16  FX  -2.336  FY  162.036 FZ  -28.536
17  FX  0.002   FY  -52.838 FZ  -15.971
18  FX  -0.001  FY  52.839  FZ  -15.974

*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
13  FX  50.722  FY  263.588 FZ  11.419
14  FX  50.438  FY  265.501 FZ  -11.997
15  FX  50.77   FY  -261.31 FZ  0.307
16  FX  51.021  FY  -267.775    FZ  1.163
17  FX  1.098   FY  -2.279  FZ  -0.446
18  FX  0.571   FY  2.274   FZ  -0.446

*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
```

```
13  FX  -50.722 FY  -263.588    FZ  -11.419
14  FX  -50.438 FY  -265.501    FZ  11.997
15  FX  -50.77  FY  261.31  FZ  -0.307
16  FX  -51.021 FY  267.775 FZ  -1.163
17  FX  -1.098  FY  2.279   FZ  0.446
18  FX  -0.571  FY  -2.274  FZ  0.446

****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
13  FX  -1.464  FY  -0.018  FZ  -37.278
14  FX  -1.431  FY  0.018   FZ  37.312
15  FX  1.325   FY  0.061   FZ  -0.644
16  FX  1.178   FY  -0.061  FZ  0.633
17  FX  0.095   FY  -0.029  FZ  -0.497
18  FX  0.297   FY  0.029   FZ  0.486

****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
13  FX  1.464   FY  0.018   FZ  37.278
14  FX  1.431   FY  -0.018  FZ  -37.312
15  FX  -1.325  FY  -0.061  FZ  0.644
16  FX  -1.178  FY  0.061   FZ  -0.633
17  FX  -0.095  FY  0.029   FZ  0.497
18  FX  -0.297  FY  -0.029  FZ  -0.486

****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
13  FX  2.798   FY  16.845  FZ  0.613
14  FX  2.43    FY  16.637  FZ  -0.937
15  FX  2.801   FY  -16.517 FZ  0.169
16  FX  2.434   FY  -16.84  FZ  0.225
17  FX  -0.002  FY  0.121   FZ  0.024
18  FX  -0.001  FY  -0.122  FZ  0.024

****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
13  FX  -0.237  FY  13.452  FZ  2.442
14  FX  0.237   FY  -13.452 FZ  2.458
15  FX  -0.3    FY  13.375  FZ  1.959
16  FX  0.299   FY  -13.375 FZ  1.961
17  FX  0   FY  3.261   FZ  0.678
18  FX  0   FY  -3.261  FZ  0.679

****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
```

```
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
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```
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
**************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
********************************************************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 12
DESIGN COLUMN 1 TO 12
DESIGN BEAM 13 TO 19
END CONCRETE DESIGN
FINISH
```

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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 25, 2020                *
*           Time=     0:40:55                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
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```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3130-CNV-001\Staad\Civil\3130-CNV-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 22-MAR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 1
    8. JOB PART VGTU FDN
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-001
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 4.5; 3 4 -3 0; 4 4 -3 4.5; 5 8 -3 0; 6 8 -3 4.5; 7 0 0.35 0
   18. 8 0 0.35 4.5; 9 4 0.35 0; 10 4 0.35 4.5; 11 8 0.35 0; 12 8 0.35 4.5; 13 0 1 0
   19. 14 0 1 4.5; 15 4 1 0; 16 4 1 4.5; 17 8 1 0; 18 8 1 4.5
   20. MEMBER INCIDENCES
   21. 1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17
   22. 11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10
   23. 19 10 12
   24. DEFINE MATERIAL START
   25. ISOTROPIC CONCRETE
   26. E 2.17185E+007
   27. POISSON 0.17
   28. DENSITY 23.5616
   29. ALPHA 1E-005
   30. DAMP 0.05
   31. TYPE CONCRETE
   32. STRENGTH FCU 27579
   33. END DEFINE MATERIAL
   34. MEMBER PROPERTY AMERICAN
   35. 9 TO 12 PRIS YD 0.6 ZD 0.6
   36. 13 TO 15 PRIS YD 0.5 ZD 0.3
   37. MEMBER PROPERTY AMERICAN
   38. 16 TO 19 PRIS YD 0.6 ZD 0.3
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39. MEMBER PROPERTY AMERICAN
40. 1 TO 8 PRIS YD 0.6 ZD 0.6
41. CONSTANTS
42. MATERIAL CONCRETE ALL
43. SUPPORTS
44. 1 TO 6 FIXED
45. ******************************************
46. LOAD 1 DEAD LOAD DL
47. SELFWEIGHT Y -1
48. JOINT LOAD
49. 13 FX -17.302 FY -292.376 FZ -12.756
50. 14 FX -18.26 FY -245.674 FZ 16.001
51. 15 FX -14.626 FY -119.938 FZ 2.1
52. 16 FX -15.819 FY -88.543 FZ 5.968
53. 17 FX 0.007 FY -30.141 FZ -1.204
54. 18 FX 0 FY -22.224 FZ -0.493
56. ******************************************
57. LOAD 2 LIVE LOAD LL
58. JOINT LOAD
59. 13 FX -1.14 FY -208.123 FZ -11.282
60. 14 FX -3.27 FY -179.66 FZ 11.541
61. 15 FX 3.312 FY -279.969 FZ -5.278
62. 16 FX 1.116 FY -292.848 FZ 6.911
63. 17 FX -0.006 FY -69.16 FZ -0.954
64. 18 FX -0.011 FY -54.457 FZ -0.952
66. ******************************************
67. LOAD 3 MATERIAL LOAD LL
68. JOINT LOAD
69. 13 FX -0.01 FY -17.625 FZ -1.124
70. 14 FX -0.107 FY -17.275 FZ 0.919
71. 15 FX 0.108 FY -24.85 FZ 0.125
72. 16 FX 0.011 FY -26.048 FZ 0.233
73. 17 FX -0.001 FY -0.443 FZ -0.087
74. 18 FX 0 FY 0.443 FZ -0.087
76. ******************************************
77. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
78. JOINT LOAD
79. 13 FX -1.594 FY 162.605 FZ 33.388
80. 14 FX 1.597 FY -162.605 FZ 33.597
81. 15 FX -2.338 FY 162.036 FZ 28.538
82. 16 FX 2.336 FY -162.036 FZ 28.536
83. 17 FX -0.002 FY 52.838 FZ 15.971
84. 18 FX 0.001 FY -52.839 FZ 15.974
86. ******************************************
87. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
88. JOINT LOAD
89. 13 FX 1.595 FY -162.605 FZ -33.388
90. 14 FX -1.598 FY 162.605 FZ -33.598
91. 15 FX 2.338 FY -162.037 FZ -28.538
92. 16 FX -2.336 FY 162.036 FZ -28.536
93. 17 FX 0.002 FY -52.838 FZ -15.971
94. 18 FX -0.001 FY 52.839 FZ -15.974
96. ******************************************
97. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
98. JOINT LOAD
99. 13 FX 50.722 FY 263.588 FZ 11.419
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100. 14 FX 50.438 FY 265.501 FZ -11.997
101. 15 FX 50.77 FY -261.31 FZ 0.307
102. 16 FX 51.021 FY -267.775 FZ 1.163
103. 17 FX 1.098 FY -2.279 FZ -0.446
104. 18 FX 0.571 FY 2.274 FZ -0.446
106. ****************************************
107. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
108. JOINT LOAD
109. 13 FX -50.722 FY -263.588 FZ -11.419
110. 14 FX -50.438 FY -265.501 FZ 11.997
111. 15 FX -50.77 FY 261.31 FZ -0.307
112. 16 FX -51.021 FY 267.775 FZ -1.163
113. 17 FX -1.098 FY 2.279 FZ 0.446
114. 18 FX -0.571 FY -2.274 FZ 0.446
116. ****************************************
117. LOAD 8 TEMP LOAD   TL1
118. JOINT LOAD
119. 13 FX -1.464 FY -0.018 FZ -37.278
120. 14 FX -1.431 FY 0.018 FZ 37.312
121. 15 FX 1.325 FY 0.061 FZ -0.644
122. 16 FX 1.178 FY -0.061 FZ 0.633
123. 17 FX 0.095 FY -0.029 FZ -0.497
124. 18 FX 0.297 FY 0.029 FZ 0.486
126. ****************************************
127. LOAD 9 TEMP LOAD   TL2
128. JOINT LOAD
129. 13 FX 1.464 FY 0.018 FZ 37.278
130. 14 FX 1.431 FY -0.018 FZ -37.312
131. 15 FX -1.325 FY -0.061 FZ 0.644
132. 16 FX -1.178 FY 0.061 FZ -0.633
133. 17 FX -0.095 FY 0.029 FZ 0.497
134. 18 FX -0.297 FY -0.029 FZ -0.486
136. ****************************************
137. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
138. JOINT LOAD
139. 13 FX 2.798 FY 16.845 FZ 0.613
140. 14 FX 2.43 FY 16.637 FZ -0.937
141. 15 FX 2.801 FY -16.517 FZ 0.169
142. 16 FX 2.434 FY -16.84 FZ 0.225
143. 17 FX -0.002 FY 0.121 FZ 0.024
144. 18 FX -0.001 FY -0.122 FZ 0.024
146. ****************************************
147. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
148. JOINT LOAD
149. 13 FX -0.237 FY 13.452 FZ 2.442
150. 14 FX 0.237 FY -13.452 FZ 2.458
151. 15 FX -0.3 FY 13.375 FZ 1.959
152. 16 FX 0.299 FY -13.375 FZ 1.961
153. 17 FX 0 FY 3.261 FZ 0.678
154. 18 FX 0 FY -3.261 FZ 0.679
156. ****************************************
157. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
158. LOAD COMB 301 1.0 DL
159. 1 1.0
160. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
161. 1 1.0 8 1.0 2 1.0 3 1.0
```

```
162. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
163. 1 1.0 9 1.0 2 1.0 3 1.0
164. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
165. 1 1.0 4 0.75 2 0.75 3 0.75
166. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
167. 1 1.0 5 0.75 2 0.75 3 0.75
168. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
169. 1 1.0 6 0.75 2 0.75 3 0.75
170. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
171. 1 1.0 7 0.75 2 0.75 3 0.75
172. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
173. 1 1.0 10 0.525 2 0.75 3 0.75
174. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
175. 1 1.0 10 -0.525 2 0.75 3 0.75
176. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
177. 1 1.0 11 0.525 2 0.75 3 0.75
178. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
179. 1 1.0 11 -0.525 2 0.75 3 0.75
180. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
181. 1 0.6 4 1.0
182. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
183. 1 0.6 5 1.0
184. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
185. 1 0.6 6 1.0
186. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
187. 1 0.6 7 1.0
188. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
189. 1 0.6 10 0.7
190. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
191. 1 0.6 10 -0.7
192. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
193. 1 0.6 11 0.7
194. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
195. 1 0.6 11 -0.7
196. ****************************************
197. *****LOAD COMBINATION FOR FOUNDATION DESIGN
198. LOAD COMB 401 1.4 DL
199. 1 1.4
200. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
201. 1 1.4 8 1.4 2 1.7 3 1.7
202. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
203. 1 1.4 9 1.4 2 1.7 3 1.7
204. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
205. 1 1.2 4 1.6 2 0.5 3 0.5
206. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
207. 1 1.2 5 1.6 2 0.5 3 0.5
208. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
209. 1 1.2 6 1.6 2 0.5 3 0.5
210. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
211. 1 1.2 7 1.6 2 0.5 3 0.5
212. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
213. 1 1.2 10 1.0 2 0.5 3 0.5
214. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
215. 1 1.2 10 -1.0 2 0.5 3 0.5
216. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
217. 1 1.2 11 1.0 2 0.5 3 0.5
```

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218. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
219. 1 1.2 11 -1.0 2 0.5 3 0.5
220. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
221. 1 0.9 4 1.6
222. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
223. 1 0.9 5 1.6
224. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
225. 1 0.9 6 1.6
226. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
227. 1 0.9 7 1.6
228. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
229. 1 0.9 10 1.0
230. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
231. 1 0.9 10 -1.0
232. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
233. 1 0.9 11 1.0
234. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
235. 1 0.9 11 -1.0
236. ****************************************
237. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**18**	**NUMBER OF MEMBERS**	**19**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 6/ 6/ 42 DOF
TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 72
TOTAL LOAD COMBINATION CASES = 38 SO FAR.
SIZE OF STIFFNESS MATRIX = 4 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.1/ 49841.9 MB

```
238. LOAD LIST 301 TO 319
239. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	9.79	348.17	-2.87	-5.56	0.39	-24.33
	302	9.47	574.36	-14.36	-16.19	0.82	-23.83
	303	9.97	574.22	2.64	-0.62	0.50	-24.16
	304	10.52	382.67	-29.92	-75.73	-0.13	-26.06
	305	8.95	652.84	19.70	60.34	1.32	-22.10
	306	-9.05	303.99	-2.98	-5.52	0.66	28.01
	307	28.52	731.52	-7.24	-9.87	0.53	-76.17
	308	9.03	508.35	-4.95	-7.43	0.63	-22.11
	309	10.44	527.16	-5.27	-7.96	0.56	-26.05
	310	9.81	510.06	-6.38	-11.17	0.54	-24.26
	311	9.66	525.46	-3.85	-4.22	0.65	-23.89
	312	6.92	28.78	-34.80	-94.05	-0.73	-17.24
	313	4.82	389.02	31.36	87.38	1.20	-11.96
	314	-19.17	-76.12	1.11	-0.43	0.33	54.85
	315	30.92	493.92	-4.56	-6.24	0.15	-84.05
	316	4.94	196.36	-1.51	-2.98	0.28	-11.97
	317	6.81	221.44	-1.94	-3.69	0.19	-17.22
	318	5.97	198.63	-3.41	-7.97	0.16	-14.84
	319	5.77	219.17	-0.03	1.30	0.31	-14.35
2	301	10.21	304.22	-0.91	-4.36	0.28	-25.37
	302	11.08	503.02	10.32	5.75	0.33	-27.65
	303	11.61	502.93	-6.62	-9.65	0.67	-28.08
	304	10.28	588.37	-23.62	-70.55	-0.28	-25.26
	305	11.85	318.20	25.93	65.45	1.17	-29.22
	306	-7.60	238.06	-0.70	-4.01	0.59	24.56
	307	29.72	668.51	3.02	-1.09	0.29	-79.04
	308	10.46	444.00	1.13	-2.45	0.48	-25.51
	309	11.67	462.57	1.18	-2.64	0.41	-28.97
	310	10.99	460.99	-0.11	-6.02	0.39	-27.06
	311	11.14	445.59	2.42	0.92	0.50	-27.43
	312	5.08	362.65	-33.58	-93.28	-0.80	-12.58
	313	7.18	2.42	32.49	88.06	1.14	-17.86
	314	-18.75	-104.43	-3.03	-4.56	0.37	53.85
	315	31.01	469.50	1.94	-0.67	-0.03	-84.29
	316	5.32	170.15	-0.58	-2.49	0.22	-12.92
	317	6.94	194.92	-0.51	-2.74	0.12	-17.53
	318	6.03	192.80	-2.23	-7.24	0.10	-14.97
	319	6.23	172.27	1.14	2.02	0.24	-15.47
3	301	12.84	177.16	-1.45	-6.69	-0.21	-27.27
	302	12.75	479.26	-3.97	-10.20	-0.10	-26.88
	303	12.19	479.64	-3.11	-9.36	-0.34	-26.23
	304	13.94	271.38	-23.74	-66.07	-1.96	-29.33
	305	11.18	536.38	17.70	48.06	1.53	-24.14
	306	-17.74	598.89	-3.38	-9.99	-0.25	36.82
	307	42.86	208.86	-2.65	-8.02	-0.18	-90.29
	308	11.42	412.46	-3.09	-9.20	-0.20	-24.32

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	309	13.71	395.30	-2.94	-8.81	-0.23	-29.15
	310	12.69	396.35	-4.00	-11.72	-0.34	-26.97
	311	12.44	411.41	-2.04	-6.30	-0.09	-26.50
	312	9.55	-70.37	-28.50	-80.10	-2.46	-19.82
	313	5.86	282.96	26.76	72.08	2.20	-12.90
	314	-32.70	366.32	-1.36	-5.32	-0.18	68.38
	315	48.10	-153.73	-0.38	-2.70	-0.08	-101.10
	316	6.17	117.74	-0.97	-4.27	-0.10	-13.14
	317	9.23	94.86	-0.77	-3.75	-0.15	-19.58
	318	7.87	96.26	-2.18	-7.62	-0.30	-16.68
	319	7.54	116.34	0.44	-0.40	0.04	-16.04
4	301	13.46	149.45	-4.98	-10.47	-0.31	-28.49
	302	14.80	465.70	-3.74	-10.48	-0.53	-31.10
	303	14.37	465.88	-4.61	-11.35	-0.27	-30.66
	304	12.92	519.21	-25.10	-67.87	-2.12	-27.69
	305	15.68	254.21	16.34	46.26	1.37	-32.88
	306	-15.79	586.87	-4.74	-11.80	-0.35	32.90
	307	44.39	186.55	-4.02	-9.81	-0.40	-93.46
	308	13.29	395.52	-4.45	-11.00	-0.35	-28.16
	309	15.31	377.89	-4.31	-10.61	-0.39	-32.41
	310	14.18	394.24	-5.36	-13.51	-0.50	-30.05
	311	14.43	379.18	-3.40	-8.10	-0.25	-30.52
	312	6.23	266.33	-30.62	-82.37	-2.52	-13.63
	313	9.92	-86.99	24.64	69.80	2.15	-20.55
	314	-32.04	356.55	-3.47	-7.61	-0.15	67.15
	315	48.19	-177.21	-2.51	-4.95	-0.21	-101.33
	316	6.73	101.42	-3.09	-6.54	-0.16	-14.26
	317	9.42	77.92	-2.89	-6.02	-0.21	-19.93
	318	7.91	99.71	-4.30	-9.89	-0.35	-16.78
	319	8.24	79.63	-1.68	-2.67	-0.02	-17.41
5	301	9.57	73.81	2.51	3.11	-0.59	-23.37
	302	8.08	145.63	3.11	5.11	-0.75	-21.56
	303	8.07	145.43	3.35	5.31	-0.71	-21.50
	304	9.84	79.53	-9.94	-31.01	-2.17	-24.55
	305	7.05	175.67	16.04	40.38	0.79	-19.43
	306	-19.38	146.35	3.28	5.32	-0.79	37.76
	307	36.27	108.85	2.82	4.06	-0.59	-81.75
	308	7.37	128.17	3.02	4.62	-0.69	-19.70
	309	9.52	127.03	3.08	4.76	-0.69	-24.28
	310	8.57	125.56	2.62	3.50	-0.80	-22.22
	311	8.33	129.64	3.48	5.87	-0.58	-21.76
	312	7.60	-19.81	-15.81	-45.73	-2.32	-17.44
	313	3.88	108.37	18.82	49.46	1.62	-10.61
	314	-31.36	69.28	1.81	2.71	-0.48	65.65
	315	42.84	19.29	1.20	1.03	-0.22	-93.69
	316	4.31	45.05	1.47	1.77	-0.35	-10.97

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	317	7.17	43.52	1.54	1.96	-0.35	-17.07
	318	5.90	41.56	0.94	0.29	-0.50	-14.33
	319	5.58	47.01	2.07	3.45	-0.21	-13.72
6	301	10.14	65.23	-1.91	-1.49	-0.59	-24.50
	302	9.82	120.08	-0.95	0.84	-0.76	-25.34
	303	9.79	119.96	-1.22	0.58	-0.79	-25.35
	304	8.49	154.39	-14.28	-35.54	-2.21	-22.57
	305	11.28	58.26	11.70	35.86	0.75	-27.69
	306	-17.92	121.39	-1.06	0.79	-0.84	34.44
	307	37.70	91.26	-1.52	-0.47	-0.62	-84.71
	308	8.94	106.99	-1.32	0.09	-0.73	-23.11
	309	10.84	105.66	-1.26	0.23	-0.73	-27.16
	310	9.77	108.37	-1.72	-1.02	-0.84	-24.90
	311	10.01	104.28	-0.86	1.34	-0.62	-25.36
	312	4.22	103.23	-18.46	-48.49	-2.32	-11.28
	313	7.94	-24.95	16.17	46.70	1.62	-18.11
	314	-31.00	59.22	-0.84	-0.05	-0.50	64.74
	315	43.16	19.06	-1.45	-1.74	-0.21	-94.14
	316	4.81	40.02	-1.18	-0.99	-0.35	-12.00
	317	7.35	38.25	-1.11	-0.80	-0.35	-17.40
	318	5.92	41.86	-1.71	-2.47	-0.50	-14.39
	319	6.24	36.42	-0.58	0.68	-0.21	-15.01

********** END OF LATEST ANALYSIS RESULT **********

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240. ********************************************************************************
241. LOAD LIST 401 TO 419
242. START CONCRETE DESIGN
243. CODE ACI 2008
244. FC 35000 ALL
245. FYMAIN 420000 ALL
246. TRACK 1 ALL
247. CLB 0.05 ALL
248. CLT 0.05 ALL
249. CLS 0.05 ALL
250. MAXMAIN 25 ALL
251. MINMAIN 16 ALL
252. MINSEC 8 ALL
253. REINF 0 MEMB 1 TO 12
254. DESIGN COLUMN 1 TO 12
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	688.69	18.22	0.00790

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	688.69	8.73	0.00378

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COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	629.73	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	629.73	0.00	0.00000

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COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	594.05	19.05	0.00957

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	594.05	2.80	0.00141

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COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	529.14	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	529.14	0.00	0.00000

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COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	320.38	33.91	0.03160

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	320.38	3.94	0.00367

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COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	258.33	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	258.33	0.00	0.00000

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COLUMN NO. 7 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	260.69	35.75	0.04094

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	260.69	13.41	0.01536

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COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	190.71	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	190.71	0.00	0.00000

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COLUMN NO. 9 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	97.77	18.68	0.05705

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	97.77	11.38	0.03476

==========

COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     64.92      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     64.92      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 11 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     79.29     20.37    0.07667
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 411.95  -1583.37     79.29     10.55    0.03972
 ---------------------------------------------------------
```

==========

COLUMN NO. 12 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	47.87	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	47.87	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

255. DESIGN BEAM 13 TO 19

BEAM NO. 13 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4500.	YES	YES

CRITICAL POS MOMENT=	62.40 KN-MET AT 0.MM, LOAD 412
REQD STEEL=	457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH =	410. MMS

Cracked Moment of Inertia Iz at above location = 58050.0 cm^4

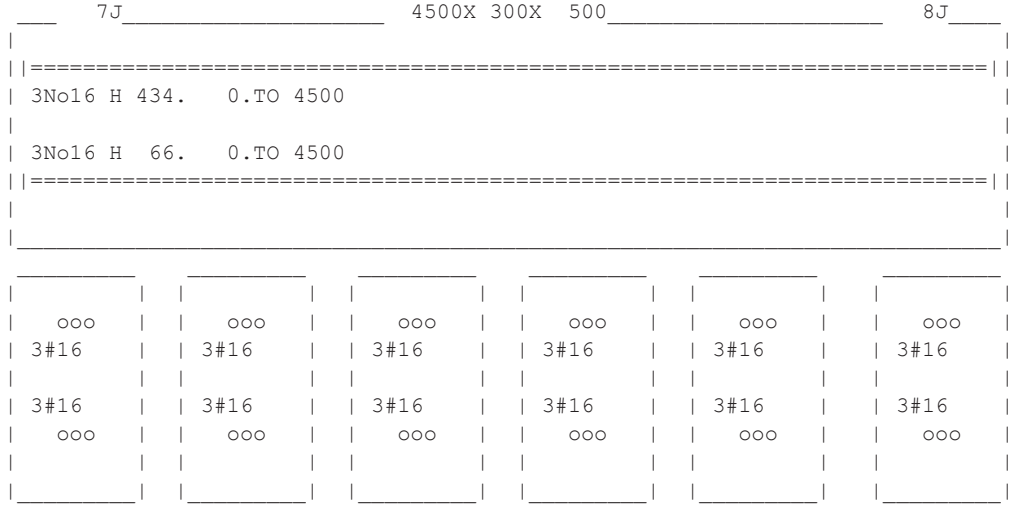
2	434.	3 - 16MM	0.	4500.	YES	YES

CRITICAL NEG MOMENT=	77.27 KN-MET AT 4500.MM, LOAD 404
REQD STEEL=	484.MM2, RHO=0.0037, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH =	410. MMS

Cracked Moment of Inertia Iz at above location = 58050.0 cm^4

B E A M N O. 13 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.37 KNS Vc= 122.61 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.61 KNS Vc= 122.61 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 14 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

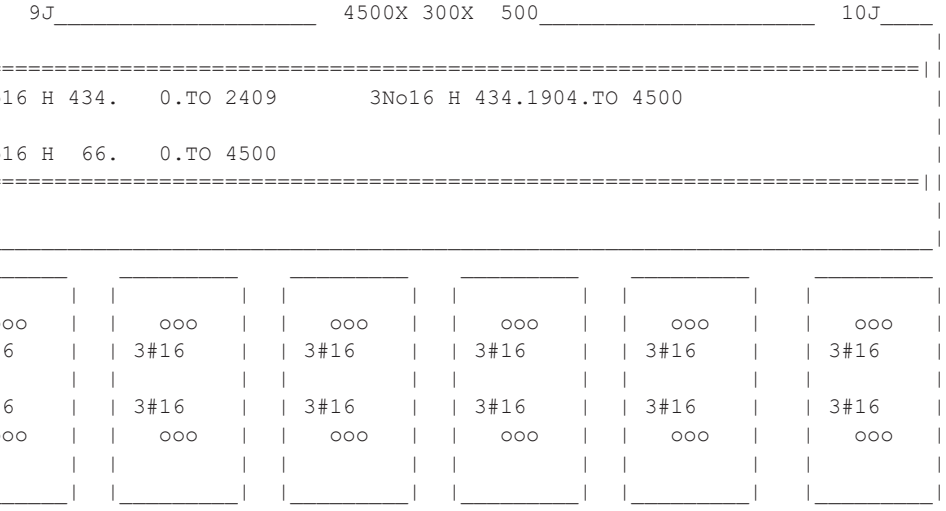
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4500.	YES	YES
	CRITICAL POS MOMENT= 53.75 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 444.MM2, RHO=0.0034, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	434.	3 - 16MM	0.	2409.	YES	NO
	CRITICAL NEG MOMENT= 56.31 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
3	434.	3 - 16MM	1904.	4500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     67.19 KN-MET  AT  4500.MM, LOAD  404|
|   REQD STEEL=   457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    58050.0 cm^4
```

B E A M N O. 14 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    6.36 KNS  Vc=  127.85 KNS  Vs=    0.00 KNS
 Tu=    0.02 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   11.61 KNS  Vc=  127.85 KNS  Vs=    0.00 KNS
 Tu=    0.02 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 15 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	4500.	YES	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=     30.93 KN-MET  AT  4500.MM, LOAD  413|
|   REQD STEEL=   253.MM2, RHO=0.0019, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4

  2        434.     2 - 16MM         0.         2409.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     42.76 KN-MET  AT     0.MM, LOAD  405|
|   REQD STEEL=   352.MM2, RHO=0.0027, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|

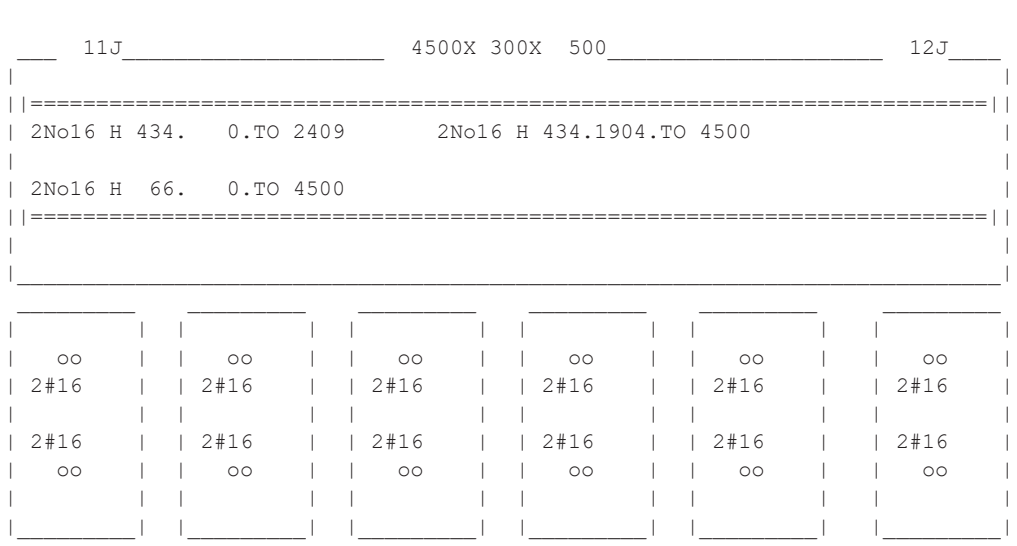
  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4

  3        434.     2 - 16MM      1904.         4500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     41.08 KN-MET  AT  4500.MM, LOAD  404|
|   REQD STEEL=   338.MM2, RHO=0.0026, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4
```

B E A M N O. 15 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    9.18 KNS  Vc=  127.25 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    8.79 KNS  Vc=  128.69 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 16 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES

CRITICAL POS MOMENT= 70.33 KN-MET AT 4000.MM, LOAD 407
REQD STEEL= 471.MM2, RHO=0.0029, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

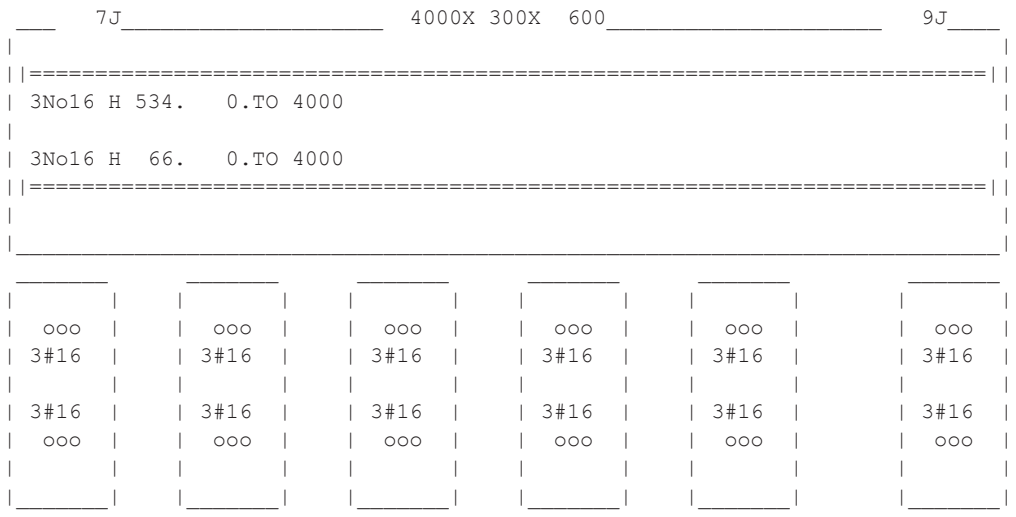
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
2	534.	3 - 16MM	0.	4000.	YES	YES

CRITICAL NEG MOMENT= 100.01 KN-MET AT 0.MM, LOAD 407
REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

B E A M N O. 16 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.92 KNS Vc= 154.74 KNS Vs= 0.00 KNS
Tu= 0.60 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 0.51 KNS Vc= 154.74 KNS Vs= 0.00 KNS
Tu= 0.60 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 17 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES

CRITICAL POS MOMENT= 81.35 KN-MET AT 4000.MM, LOAD 407
REQD STEEL= 546.MM2, RHO=0.0034, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

2	534.	3 - 16MM	0.	2200.	YES	NO

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     92.39 KN-MET  AT     0.MM, LOAD  407|
|   REQD STEEL=   562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

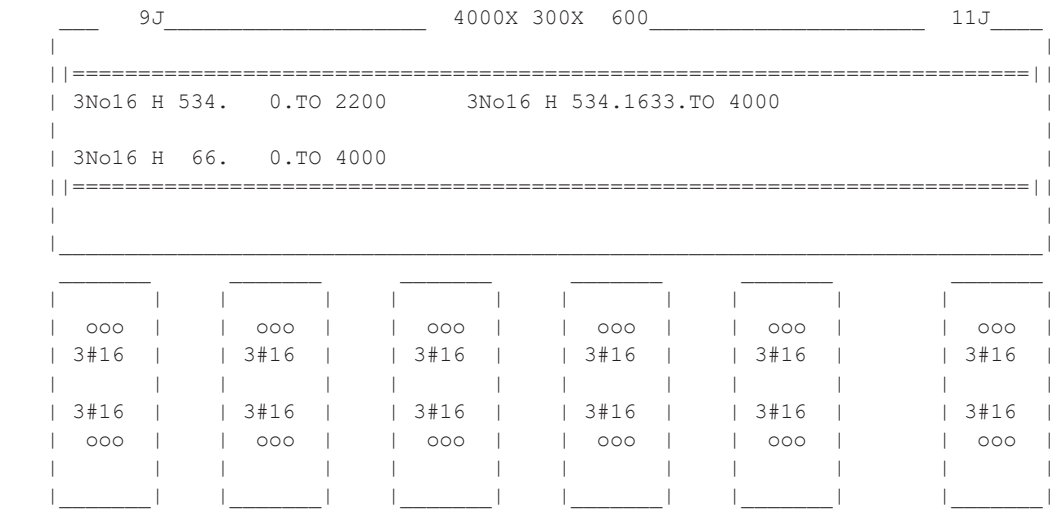
Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

```
  3        534.     3 - 16MM      1633.         4000.        NO   YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     64.66 KN-MET  AT  4000.MM, LOAD  414|
|   REQD STEEL=   432.MM2, RHO=0.0027, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

B E A M N O. 17 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   18.28 KNS  Vc=  154.07 KNS  Vs=    0.00 KNS
Tu=    0.55 KN-MET  Tc=    6.5 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    0.87 KNS  Vc=  154.07 KNS  Vs=    0.00 KNS
Tu=    0.55 KN-MET  Tc=    6.5 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 18 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES

CRITICAL POS MOMENT= 72.05 KN-MET AT 4000.MM, LOAD 407
REQD STEEL= 482.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

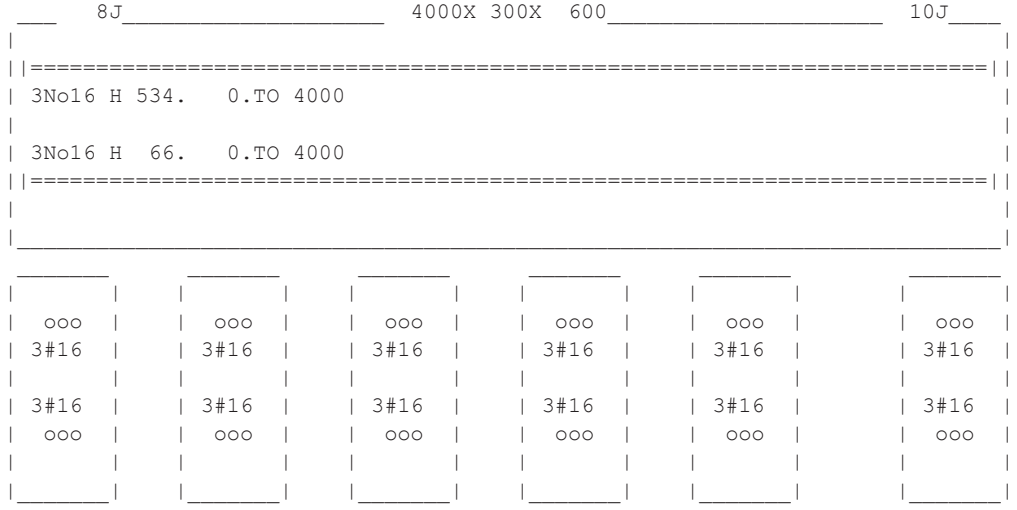
2	534.	3 - 16MM	0.	4000.	YES	YES

CRITICAL NEG MOMENT= 101.86 KN-MET AT 0.MM, LOAD 407
REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

B E A M N O. 18 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 18.55 KNS Vc= 154.52 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 1.14 KNS Vc= 154.52 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 19 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

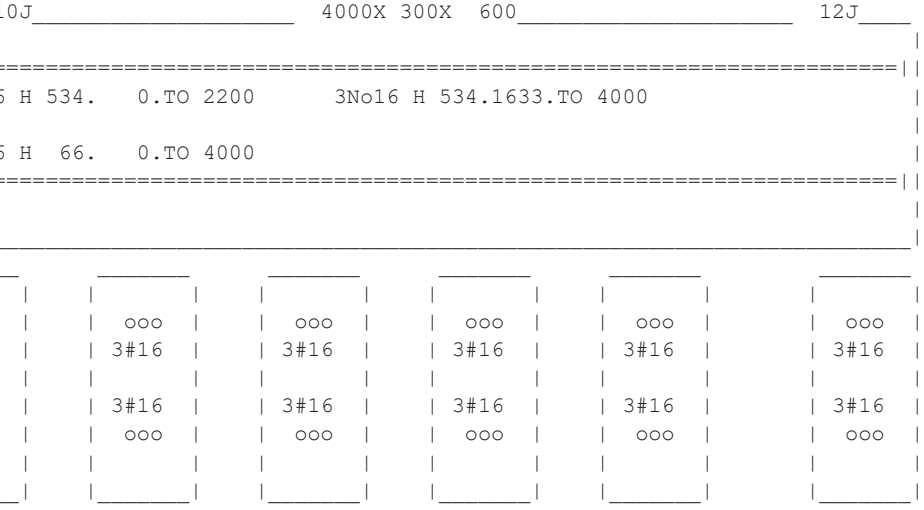
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 82.58 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 554.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	2200.	YES	NO
	CRITICAL NEG MOMENT= 93.81 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
3	534.	3 - 16MM	1633.	4000.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     63.76 KN-MET  AT  4000.MM, LOAD  414|
|   REQD STEEL=   426.MM2, RHO=0.0027, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                      |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    90761.5 cm^4
```

B E A M N O. 19 D E S I G N R E S U L T S - SHEAR

```
  AT START SUPPORT - Vu=   18.81 KNS  Vc=  153.72 KNS  Vs=    0.00 KNS
  Tu=    0.66 KN-MET  Tc=    6.5 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                   STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=    1.40 KNS  Vc=  153.72 KNS  Vs=    0.00 KNS
  Tu=    0.66 KN-MET  Tc=    6.5 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                   STIRRUPS ARE NOT REQUIRED.
```




```
  **********END OF BEAM DESIGN**********

  256. END CONCRETE DESIGN
  257. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 25,2020 TIME= 0:40:56 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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